

EXHIBIT 45

UNITED STATES INTERNATIONAL TRADE COMMISSION
Washington, D.C.

In the Matter of

**CERTAIN LIQUID CRYSTAL
DISPLAY DEVICES, INCLUDING
MONITORS, TELEVISIONS, AND
MODULES, AND COMPONENTS
THEREOF**

Investigation No. 337-TA-749
Investigation No. 337-TA-741

COMMISSION OPINION

On March 26, 2012, the Commission determined to review a portion of the presiding administrative law judge's ("ALJ") final initial determination ("ID") issued on January 12, 2012. The ALJ found a violation of Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337 ("Section 337"), by respondents Chimei InnoLux Corporation of Taiwan, Chi Mei Optoelectronics USA, Inc. of San Jose, California, and InnoLux Corporation of Austin, Texas (collectively, "CMI"); Qisda Corporation of Taoyuan, Taiwan, Qisda (Suzhou) Co. Ltd. of China, and Qisda America Corporation of Irvine, California (collectively, "Qisda"); and BenQ Corporation of Taipei, Taiwan, BenQ America Corporation of Irvine, California, and BenQ Latin America Corporation of Miami, Florida (collectively "BenQ"). The ALJ found that the asserted claims of U.S. Patent No. 5,648,674 ("the '674 patent") are infringed by the CMI accused products including the "Type 2 Array Circuitry" and any Qisda or BenQ accused product incorporating these CMI accused products. The ALJ found that no other accused products infringe the asserted claims of the '674 patent. The ALJ also found that no accused products infringe the asserted claims of United States Patent Nos. 5,978,063 ("the '063 patent"), 5,375,006 ("the '006 patent"), 5,621,556 ("the '556 patent"), and 6,121,941 ("the '941 patent").

The ALJ also found that claims 1, 2, 3, 4, 8, 11, 12, 14, and 18 of the '063 patent and claims 4 and 14 of the '006 patent are invalid. The ALJ concluded that a domestic industry exists in the United States that exploits the asserted patents as required by 19 U.S.C. § 1337(a)(2).

The Commission determined to review the following portions of the ALJ's ID: (1) claim construction of the term "layer" of the asserted claims of the '006 patent; (2) infringement of the asserted claims of the '006 patent; (3) anticipation of claims 4 and 7 of the '006 patent by Scheuble; (4) claim construction of the limitations "mechanically rubbing" / "mechanically rubbed," "a plurality of spacing elements," and "an affixing layer" of the asserted claims of the '063 patent; (5) infringement of the asserted claims of the '063 patent; (6) obviousness of the asserted claims of the '063 patent in view of Sugata and Tsuboyama; (7) whether Lowe and Miyazaki are prior art to the asserted claims of the '063 patent; (8) anticipation of the asserted claims of the '063 patent by Lowe; (9) anticipation of the asserted claims of the '063 patent by Miyazaki; (10) obviousness of the asserted claim of the '556 patent in view of Takizawa and Possin; (11) anticipation and obviousness of the asserted claims of the '674 patent in view of Fujitsu; (12) claim construction of the "second rate" "determined by" limitation of the asserted claims of the '941 patent and the "input video signal" limitation of claim 4 of the '941 patent; (13) infringement of the asserted claims of the '941 patent; (14) anticipation of the asserted claims of the '941 patent by Baba; (15) exclusion of evidence of the ViewFrame II+2 LCD Panel; and (16) the economic prong of the domestic industry requirement. The Commission also determined to review and to take no position on the claim construction of the terms "drain electrodes" and "source electrodes" of the '556 patent. The Commission determined not to review any other issues.

On review, the Commission has determined to reverse the ALJ's finding of a violation of section 337 with respect to the '674 patent. The Commission has determined to affirm, with modifications, the ALJ's finding of no violation of section 337 as to the remaining asserted patents. Specifically, the Commission finds that the asserted claims of the '674 patent are infringed by the accused products of respondents CMI, Qisda, and BenQ, but that claims 1, 7, 8, 14, 16, 17, and 18 of the '674 patent are anticipated by Fujitsu and that claims 9, 11, and 13 are obvious in view of Fujitsu and the knowledge of one of ordinary skill in the art, . The Commission also finds that (a) respondents' accused products do not infringe the asserted claims of the '006 patent; (b) Scheuble does not anticipate claims 4 and 7 of the '006 patent; (c) the accused products of respondents AU Optronics Corp., AU Optronics Corp. America, Qisda, and BenQ infringe claims 11, 12, 14, 17, and 18, but not the remaining asserted claims of the '063 patent; (d) respondent CMI's accused products do not infringe the asserted claims of the '063 patent; (e) the asserted claims of the '063 patent are obvious in view of Sugata and Tsuboyama; (f) Lowe and Miyazaki are prior art to claims 1-4 and 8 of the '063 patent, but not the remaining asserted claims of the '063 patent; (g) respondents' have not shown that Lowe anticipates the asserted claims of the '063 patent; (h) Miyazaki anticipates claims 11, 12, 14, 17, and 18 of the '063 patent, but not any of the remaining asserted claims of the '063 patent; (i) respondents have not shown that claim 3 of the '556 patent is obvious in view of Takizawa and Possin; (j) respondents' accused products do not infringe the asserted claims of the '941 patent; (k) respondents have not shown that the asserted claims of the '941 patent are obvious in view of Baba; and (l) Thomson has established that a domestic industry exists under 19 U.S.C. § 1337(a)(3)(C). The Commission also determined to remand to the ALJ to decide whether

respondents have shown that the asserted claims of the '941 patent are anticipated by the ViewFrame II+2 LCD Panel.

I. BACKGROUND

On October 18, 2010, the Commission instituted a first investigation, No. 337-TA-741, based on a complaint filed by Thomson Licensing SAS of France and Thomson Licensing LLC of Princeton, New Jersey (collectively, "Thomson"). 75 *Fed. Reg.* 63856 (Oct. 18, 2010). The complaint alleges violations of Section 337 by reason of infringement of certain claims of the '941, '063, '674, '556, and '006 patents. The named respondents are CMI, MStar Semiconductor Inc. ("MStar") of Taiwan, Qisda, BenQ, and Realtek Semicondustor Corp. ("Realtek") of Taiwan.

On November 23, 2010, the Commission instituted a second investigation, No. 337-TA-749, based on another complaint filed by Thomson. 75 *Fed. Reg.* 74080 (Nov. 23, 2010). The complaint alleges violations of Section 337 by reason of infringement of certain claims of the '556, '063, and '006 patents. The named respondents are Qisda, BenQ, and AU Optronics Corp. of Taiwan and AU Optronics Corp. America of Houston, Texas (collectively "AUO").

On December 16, 2010, the ALJ consolidated Investigation Nos. 337-TA-741 and 337-TA-749. Investigation No. 337-TA-749 was designated as the lead case for the consolidated investigation.

The products accused of infringing the '063, '006, and '556 patents are CMI, Qisda, and BenQ LCD monitors that contain certain AUO and CMI LCD modules. The products accused of infringing the '674 patent are CMI, Qisda, and BenQ monitors that contain certain CMI LCD modules. The products accused of infringing the '941 patent are CMI, Qisda, and BenQ displays that include an MStar or Realtek LCD controller (also referred to as a scaler chip).

The ALJ held an evidentiary hearing from September 13, 2011 through September 19, 2011, and received post-hearing briefs from the parties thereafter. On January 12, 2012, the ALJ issued a final ID finding a violation of Section 337 by CMI, Qisda, and BenQ. The ALJ found that the CMI accused products that include the Type 2 Array Circuitry and any Qisda or BenQ accused products incorporating these CMI accused products infringe the asserted claims of the '674 patent. The ALJ found that no other accused products infringe the '674 patent. The ALJ also found that no accused products infringe the asserted claims of the '063, '006, '556 patent, or '941 patents. The ALJ also found that claims 1, 2, 3, 4, 8, 11, 12, 14, and 18 of the '063 patent are invalid for obviousness under 35 U.S.C. § 103, and that claims 4 and 14 of the '006 patent are invalid as being anticipated under 35 U.S.C. § 102. Finally, the ALJ concluded that a domestic industry exists in the United States that exploits the asserted patents as required by 19 U.S.C. § 1337(a)(2).

With respect to remedy, the ALJ recommended that the Commission issue a limited exclusion order directed to products that infringe the asserted claims of the '674 patent and that the order should contain a certification provision. The ALJ recommended that the Commission also issue a cease and desist order against Qisda America only. The ALJ also recommended that the bond during the Presidential review period be set at 0%.

On January 25, 2012, Thomson filed a petition for review challenging the ALJ's finding with respect to claim construction, infringement, and invalidity for the '063 and '006 patents, and challenging the ALJ's finding with respect to claim construction and infringement for the '556 and '941 patents. Complainant Thomson's Petition for Review of Initial Determination (Jan. 25, 2012) ("Thom. Pet."). On the same day, AUO filed a petition for review challenging the ALJ's finding with respect to claim construction, infringement, and invalidity for the '063,

'006, and '556 patents. Respondent AUO's Petition for Review of Initial Determination (Jan. 25, 2012) ("AUO Pet."). CMI also filed a petition for review challenging the ALJ's finding with respect to claim construction, infringement, and invalidity for the '063, '006, '556, and '941 patents, and challenging the ALJ's findings with respect to infringement and invalidity for the '674 patent. Respondent CMI's Petition for Review of Initial Determination (Jan. 25, 2012) ("CMI Pet."). Mstar also filed a petition for review challenging the ALJ's finding with respect to claim construction, infringement, and invalidity of the '941 patent. Respondent MStar's Petition for Review of Initial Determination (Jan. 25, 2012) ("MStar Pet."). Realtek also filed a petition for review challenging the ALJ's finding with respect to claim construction, infringement, and invalidity of the '941 patent. Respondent Realtek's Petition for Review of Initial Determination (Jan. 25, 2012) ("Realtek Pet."). Finally, Qisda and BenQ also filed a petition for review incorporating all of the respondents' arguments by reference.

On February 2, 2012, Thomson filed a reply to each of the respondents' petitions for review. Complainant Thomson's Reply to Respondent AUO's Petition for Review of Initial Determination (Feb. 2, 2012); Complainant Thomson's Reply to Respondent CMI's Petition for Review of Initial Determination (Feb. 2, 2012); Complainant Thomson's Reply to Respondent MStar's Petition for Review of Initial Determination (Feb. 2, 2012); Complainant Thomson's Reply to Respondent Realtek's Petition for Review of Initial Determination (Feb. 2, 2012) ("Thom. Rep. Realtek Pet."); Complainant Thomson's Reply to Respondents Qisda and BenQ's Petition for Review of Initial Determination (Feb. 2, 2012); On the same day, each of the respondents filed a reply to Thomson's petition for review. Respondent AUO's Reply to Complainant Thomson's Petition for Review (Feb. 2, 2012); Respondent CMI's Reply to Complainant Thomson's Petition for Review (Feb. 2, 2012); Respondent MStar's Reply to

Complainant Thomson's Petition for Review (Feb. 2, 2012) ("MStar Rep. Thom. Pet.");
Respondent Realtek's Reply to Complainant Thomson's Petition for Review (Feb. 2, 2012);
Respondents Qisda and BenQ's Reply to Complainant Thomson's Petition for Review (Feb. 2, 2012);

On March 26, 2012, the Commission determined to review the following portions of the ALJ's ID: (1) claim construction of the term "layer" of the asserted claims of the '006 patent; (2) infringement of the asserted claims of the '006 patent; (3) anticipation of claims 4 and 7 of the '006 patent by Scheuble; (4) claim construction of the limitations "mechanically rubbing" / "mechanically rubbed," "a plurality of spacing elements," and "an affixing layer" of the asserted claims of the '063 patent; (5) infringement of the asserted claims of the '063 patent; (6) obviousness of the asserted claims of the '063 patent in view of Sugata and Tsuboyama; (7) whether Lowe and Miyazaki are prior art to the asserted claims of the '063 patent; (8) anticipation of the asserted claims of the '063 patent by Lowe; (9) anticipation of the asserted claims of the '063 patent by Miyazaki; (10) obviousness of the asserted claim of the '556 patent in view of Takizawa and Possin; (11) anticipation and obviousness of the asserted claims of the '674 patent in view of Fujitsu; (12) claim construction of the "second rate" "determined by" limitation of the asserted claims of the '941 patent and the "input video signal" limitation of claim 4 of the '941 patent; (13) infringement of the asserted claims of the '941 patent; (14) anticipation of the asserted claims of the '941 patent by Baba; (15) exclusion of evidence of the ViewFrame II+2 LCD Panel; and (16) the economic prong of the domestic industry requirement. The Commission also determined to review and to take no position on the claim construction of the terms "drain electrodes" and "source electrodes" of the '556 patent. The Commission

determined not to review any other issues. The Commission requested briefing from the parties on the issues on review, as well as on remedy, the public interest, and bonding.

On April 9, 2012, Thomson and each of the respondents filed a response to the Commission's request for written submissions. Written Submission of Complainant Thomson In Response to the Commission's Determination to Review-in-Part a Final Determination (Apr. 9, 2012) ("Thom. Resp."); Written Submission of Respondent AUO In Response to the Commission's Determination to Review-in-Part a Final Determination (Apr. 9, 2012); Written Submission of Respondent CMI In Response to the Commission's Determination to Review-in-Part a Final Determination (Apr. 9, 2012) ("CMI Resp."); Written Submission of Respondent MStar In Response to the Commission's Determination to Review-in-Part a Final Determination (Apr. 9, 2012) ("MStar Resp."); Written Submission of Respondent Realtek In Response to the Commission's Determination to Review-in-Part a Final Determination (Apr. 9, 2012) ("Realtek Resp."); Written Submission of Respondents Qisda and BenQ In Response to the Commission's Determination to Review-in-Part a Final Determination (Apr. 9, 2012). On April 16, 2012, Thomson and each of the respondents filed a reply submission.

II. ANALYSIS

A. The '006 Patent

1. The Invention of the '006 Patent

The basic components of an LCD consist of a thin layer of liquid crystal sandwiched between a pair of glass substrates, consisting of a first substrate with electrodes to control the pixels of the display and a second substrate that contains a color filter. The first substrate is associated with a first polarizer that polarizes light in one direction, while the second substrate is associated with a second polarizer that is perpendicular to the first polarizer. The first polarizer selects one linear polarization of light and allows that polarization to shine through. The second

polarizer, in the absence of any manipulation of the polarization, will block the light because it is perpendicular. The liquid crystal material changes the polarization of the light depending on the voltage applied to the liquid crystal.

The '006 patent is directed to a particular type of LCD known as “twisted nematic display,” in which the alignment direction of the liquid crystal molecules at one substrate is approximately perpendicular to the alignment of the liquid crystal molecules at the other substrate when no voltage is applied (*i.e.*, ON state). This twisted alignment allows the polarization of the polarized light to rotate by 90 degrees when the LCD is in the ON state, thus allowing the light to shine through the second polarizer.

When a voltage is applied, the electrical field created by the voltage untwists the liquid crystal molecules so that the polarization of the light will ideally be unchanged, thus causing the light to be blocked by the second polarizer. This corresponds to the OFF state. When voltage is applied, however, a property of nematic liquid crystals known as “birefringence” causes unwanted changes to the polarization of light. As a consequence, the polarization state of an incoming beam of polarized light can be modified, especially at viewing angles other than the normal or perpendicular direction of observation. This unwanted polarization change in turn causes some light to be leaked through the second polarizer in the OFF state. As the viewing angle moves away from the normal or perpendicular direction, the polarization change increases and the amount of light leakage increases. As a result, the contrast ratio of an image displayed on a twisted-nematic LCD is poorer at large viewing angles.

The invention of the '006 patent addresses this contrast ratio problem by adding a compensating plate formed by “a uniaxial birefringent optical” medium to the twisted-nematic liquid crystal layer of the LCD. The plate counters the “birefringent” properties of the nematic

liquid crystal to create a more homogenous contrast ratio for the display device in a wider angle of observation. The asserted claims are claims 4, 7, and 14.

2. Construction of the Term “Layer” of the ’006 Patent

The Commission determined to review the ALJ’s construction of the term “layer,” which appears in independent claim 14 as part of the limitations “a first birefringent layer” and a “liquid crystal layer.” The term also appears in independent claim 1 as part of the limitation “a layer of twisted nematic liquid crystal.” The ALJ found that the term means “one or more thicknesses,” as proposed by Thomson, as opposed to the “the entire thickness” of the material, as proposed by respondents. *Id.* at 75. The ALJ found that there is no clear intrinsic evidence indicating that the inventors intended to limit the meaning of “layer” to an “entire thickness.” *Id.* at 77. The ALJ cited the general rule that the terms “a” or “an” in a patent claim carries the meaning of “one or more.” *Id.* Because the term “layer” appears in two different contexts in the asserted claims: “a layer of twisted nematic liquid crystal” and “a first birefringent layer,” the ALJ reasoned that the proper construction must apply to both instances. *Id.* at 76. Thus, the ALJ concluded that the inclusion of the “one or more” language as proposed by Thomson is proper for both the limitation “a layer of twisted nematic liquid crystal” as well as the limitation “a first birefringent layer.” *Id.*

We begin our analysis with independent claim 14, which recites both a “twisted nematic liquid crystal *layer*” and a “birefringent *layer*.” The language of claim 14 which includes the limitation “liquid crystal layer” refers to the entire thickness of the liquid crystal. In particular, claim 14 requires “providing a voltage across the liquid crystal layer” by placing “a pair of transparent electrodes opposing one another across the liquid crystal layer.” In other words, the electrode must be placed on each side of the entire thickness of the liquid crystal layer. Similarly, the term “birefringent layer” must refer to the entire thickness of the birefringent material. It is

well-settled that “the same terms appearing in different portions of the claims should be given the same meaning.” *Fin Control Sys. Pty, Ltd. v. OAM, Inc.*, 265 F.3d 1311, 1318 (Fed. Cir. 2001). Thus, the term “layer” in claim 14 means “the entire thickness.”

The term “layer” also appears in claim 1 as part of the limitation “a layer of twisted nematic liquid crystal.” Unlike claim 14, the language of claim 1 does not recite details regarding the structure surrounding the liquid crystal layer. Nevertheless, because claim terms are normally used consistently throughout the patent, the usage of a term in one claim can often illuminate the meaning of the same term in other claims. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1314 (Fed. Cir. 2005). The patent provides no indication that the inventors intended the term “layer” to have different meanings in the claims. Because claim 14 uses the term “layer” to refer to the entire thickness of the liquid crystal, the term “layer” should likewise have the same meaning in claim 1.

The specification supports construing the term layer to mean the entire thickness of a material. The specification explains that “the layer of twisted nematic liquid crystal is a layer of uniform thickness, imprisoned between two transparent plates 3 and 4 with parallel faces.” ’006 patent at 5:28-32. In addition, although the specification does not use the term “layer” to describe the negatively birefringent compensator, there is no indication that the structure of the compensator can be considered as having one or more thicknesses. Rather, the specification describes the structure of the negatively birefringent compensator as either a “blade” or a “plate.” *Id.* at 1:22-23. There may be more than one blade or plate that accomplishes the compensation, but each blade or plate refers to the entire thickness of the material. *Id.* at 1:21-25.

For the foregoing reasons, we reverse the ALJ's construction of the term "layer" appearing in claims 1 and 14 of the '006 patent. We find that the proper construction of the term layer is the entire thickness of the material.

3. Direct Infringement of the '006 Patent

The products accused of infringing the asserted claims of the '006 patent are CMI, Qisda, and BenQ monitors and televisions that contain twisted nematic LCD modules,¹ including AUO and CMI LCD modules, that use [] compensation film []. The AUO and CMI modules are identified in the direct witness statement of Thomson's expert Dr. Escuti. CX-4241C at Q.273-277, 279-280. The CMI, Qisda, and BenQ LCD displays that use such AUO and CMI modules are identified in CMI's, Qisda's, and BenQ's interrogatory responses. CX-4241C at Q.278; CX-459C; CX-367C, and CX-379C.

Before the ALJ, the primary claim construction and infringement disputes were the meaning of "uniaxial" and whether the [] in the accused products meets the limitation "uniaxial compensating means with negative birefringence" of claim 1 and the limitation "a first birefringent layer . . . wherein the first birefringent layer has the property that it provides uniaxial negative birefringence" of claim 14. Thomson argued that the term "uniaxial" should be construed as "a refractive index along one axis (n_1) that is less than the refractive indices along the orthogonal axes (n_2 , n_3), where n_2 and n_3 are substantially the same ($n_1 < n_2 \approx n_3$)." Respondents argued that the term "uniaxial" means "having a single optical axis," where an "optical axis" is a "direction in a doubly-refracting (birefringent) material along which the two refracted rays travel at the same speed – *i.e.*, without double refraction." Respondents' construction requires the refractive indices n_2 and n_3 to be the same, and not just substantially

¹ An LCD module, or LCD panel, is the part of an LCD that includes the polarizers, two substrates, and liquid crystal. The LCD module is housed within the LCD casing and is electronically connected to the control circuitry of the display.

the same. The ALJ adopted respondents' construction of "uniaxial." The Commission determined not to review the ALJ's construction.

With respect to infringement, Thomson advanced two theories explaining why the [] serves as the recited uniaxial negatively birefringent compensator recited in the claim. First, Thomson argued that each layer of the [] portion² of the [] constitutes the uniaxial compensator with negative birefringence, and that [] in the [] compensates the positive birefringence of each liquid crystal layer in the LCD. For this "sublayer theory," Thomson uses the ALJ's construction of the term "uniaxial," which requires the two refracted rays in the birefringent material to travel at the same speed, *i.e.*, $n_2=n_3$. Second, Thomson argued that the [] as a whole is a uniaxial compensator with negative birefringence. Under this "entire layer theory," Thomson uses its own proposed construction of the term "uniaxial," which requires only that the refractive indices n_2 and n_3 be *substantially* equal.

Even though the ALJ did not adopt Thomson's construction of "uniaxial," the ALJ nevertheless analyzed each theory. The ALJ did not find either theory persuasive. We address each of the two theories in turn. Because we construe the term "layer" to mean "the entire thickness," we address Thomson's entire layer theory first.

a. Thomson's Entire Layer Theory

The ALJ rejected Thomson's theory that the [] as a whole is a uniaxial compensator with negative birefringence. The ALJ found that testing performed by AUO's expert shows that the [] does not have a single optical axis or a non-zero birefringence, as required under the ALJ's construction of "uniaxial." ID at 356. The ALJ found on the other hand that testing performed by Thomson's expert showing that the film is uniaxial

² The [].

was based on flawed methodology. *Id.* at 357. The ALJ further found that the two technical articles by Yamahara (“Yamahara articles”) discussing the [], on which Thomson rests its infringement argument, not only contained incorrect assumptions but did not fully describe the complex nature of the []. *Id.* at 355.

We agree with the ALJ’s finding. Thomson argues against the ALJ’s construction of “uniaxial” by challenging that it requires n_2 and n_3 to be equal to an infinite amount of decimal places and thus makes it impossible to prove that a compensator is uniaxial in the real world. We find this argument to be unpersuasive for two reasons. First, it improperly conflates the definition of uniaxial with the evidence required to prove that a material is uniaxial. The possibility of measurement errors does not change the definition of “uniaxial.” It merely impacts the evidence one would need to prove that a material is “uniaxial.” Second, and more importantly, Thomson incorrectly applies the ALJ’s construction of “uniaxial.” The ALJ did not construe the term “uniaxial” to merely mean that refractive indices n_2 and n_3 are equal. Rather, the ALJ construed the term “uniaxial” as “having a single optical axis” and the term “optical axis” in turn as a direction in a birefringent material in which n_2 and n_3 are equal. *Id.* at 71-75. In other words, the term “uniaxial” means that there is *only one* direction in the material in which n_2 and n_3 are equal. Thus, one cannot simply prove infringement by finding *a* direction in which n_2 and n_3 are equal. Rather, to satisfy the “uniaxial” limitation, there must *be only one* such direction.

Because a proper analysis of Thomson’s infringement evidence with respect to the “uniaxial” limitation requires an understanding of how birefringent materials are tested and described in practice, we begin with an explanation of these concepts. It is undisputed that one cannot measure n_2 and n_3 of a birefringent material directly, but must rely on other measured

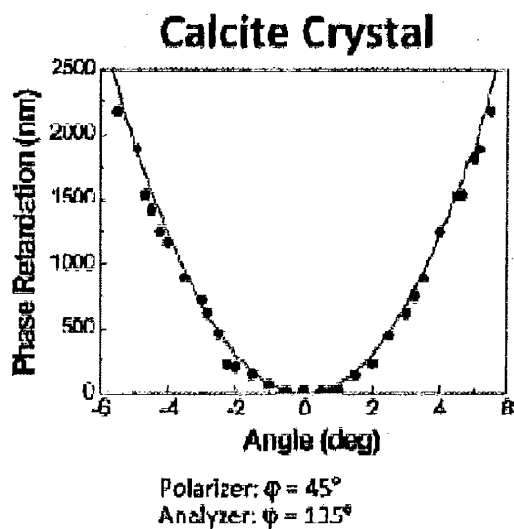
parameters of the system. RX-556C at Q.58-64, 99. One way of testing whether a birefringent material is “uniaxial” is to measure the “phase retardation” of the material under a limited set of conditions. RX-556C at Q.58-64, 99; RX-557 at Q.22-24; RX-558C at Q.27. Specifically, the “phase retardation” of a birefringent material quantifies the amount of polarization change of light passing through the material, and is defined as the product of the birefringence of the material, Δn (*i.e.*, the difference between n_2 and n_3), and the thickness of the particular birefringent material, d . RX-556C at Q.58-64, 72; CX-4241C at Q.40. Because a compensation material always has a certain thickness (*i.e.*, d cannot be zero), retardation is only zero when birefringence is zero, *i.e.*, $n_2 - n_3 = 0$. RX-157C at Q.135; RX-556C at Q.196. The phase retardation value may not be exactly zero, but the value must be within the experimental error of the test for that material, which differs depending on the particular material. Tr. at 1440:15-1442:1. When there is only a single zero-point phase retardation value, within the experimental error of the test for that material, then the material has only a single optical axis and is thus “uniaxial.” RX-557 at Q.64-74. Thus, the ALJ’s construction does not impose an impossible standard for demonstrating that a material is uniaxial.

Thomson did not present its own testing results of the [], but attempts to show infringement by using the retardation test results of the [] by AUO’s expert Dr. Wu, as well as the retardation test results of calcite by Dr. Wu. Thomson argues that the [] must be uniaxial because the lowest retardation value found by Dr. Wu for the [] is 13nm, which is lower than the retardation value of 15nm found by Dr. Wu for calcite. Thom. Resp. at 28; Thom. Pet. at 55. According to Thomson, Dr. Wu’s tests confirmed that n_2 and n_3 in the [] are “equal” to three decimal places, because 13 nanometers divided by the thickness of the [], which is 80 microns, is 0.0001625. Thom. Pet. at 55.

We find Thomson's self-performed calculations to be inaccurate. First, Thomson does not take into account that the experimental error for phase retardation measurements differs based on the particular material being tested. Tr. at 1440:15-1442:1. For a relatively thick material such as calcite, the experimental error is 20 nm, while for a relatively thin material such as the accused [], the experimental error is 5 nm. *Id.*; RX-557 at Q.66. Thus, a retardation value of 15 nm for calcite is within the 20 nm experimental error for that material, while a retardation value of 13nm for the [] is much higher than the 5 nm experimental error for the film. RX-556C at Q.123; RX-557 at Q.68, 71-73. In other words, as observed by the ALJ, Dr. Wu's testing shows that there is a single direction in calcite in which there is no birefringence, *i.e.*, where n_2 and n_3 in calcite are equal; while there is no direction in the [] where there is no birefringence, *i.e.*, where n_2 and n_3 in the film are equal. ID at 356.

In addition, Thomson based its calculation for the [] on only one low phase retardation value measured in the [] and ignores that Dr. Wu also found additional low phase retardation values in the film. RX-557 at Q.66-70. As shown below in RX-480, Dr. Wu measured *three* distinct low phase retardation points -- two measuring approximately 13 nm in the horizontal direction and one measuring approximately 30 nm in the vertical direction. RX-480; RX-557 at Q.66-70. The ALJ's claim construction, however, requires that there be only one optical axis. By contrast, the uniaxial calcite measurements, shown below in RX-481, show a symmetric curve around a *single* low point of zero.

[



]

RX-480; RX-481; RX-556 at Q.126. Thus, even assuming that the three distinct low phase retardation values for the [] are close to zero, one can only conclude from Dr. Wu's test results that the [] has more than one direction in which $n_2=n_3$, and thus has more than one optical axes. Thus, it cannot be uniaxial under the ALJ's construction of the term as adopted by the Commission.

Thomson also relies on the description of the [] in the Yamahara articles, which described the [] as having "uniaxial negative birefringence," with n_2 and n_3 being 1.5999 nm and 1.6 nm and thus substantially the same. CX-4241C at Q.424-35; CX-48; CX-65. We find that the ALJ properly found the Yamahara articles to be unreliable evidence. ID at 355. AUO's expert Dr. Drzaic testified in depth about the Yamahara articles, explaining that they were an early analysis of the [] in which the author acknowledged using a simplified model with a limited set of data that did not fully describe the complex nature of the []

[. RX-556 at Q.100-104. Moreover, as pointed out by the ALJ, Dr. Drzaic explained that the model used in the Yamahara articles contained out-of-date assumptions, and that there is currently no consensus in the industry as to how the [] should be modeled. *Id.* at

Q.108-110. AUO's expert Dr. Yeh also provided similar testimony regarding the Yamahara papers. RX-558 at Q.257-58.

Accordingly, we agree with the ALJ that Thomson has not shown that the entire accused [] meets the "uniaxial" limitation of the asserted claims.

b. Thomson's Sublayer Theory

With respect to Thomson's sublayer theory, the ALJ found that Thomson has offered no evidence that the [] portion in the [] consists of distinct sublayers other than the fact that []. *Id.* at 358-59. The ALJ found that such computer modeling does not reflect the actual physical composition of the []. *Id.* at 359. The ALJ thus concluded that Thomson has failed to demonstrate that the accused products meet the "uniaxial compensating means with birefringence" limitation of claims 4 and 7 and the "first birefringent layer . . . wherein the first birefringent layer has the property that it provides uniaxial negative birefringence" limitation of claim 14. *Id.* at 360.

We agree with the ALJ that Thomson's sublayer theory is unpersuasive. As found by the ALJ, Thomson did not use any test results or printed publications to support its assertion that each layer of the [] portion of the [] is "uniaxial," but instead relied on computer modeling data acquired by []:

[

]

CX-39C at AUO-THO 0307904; JX-49C at 48-51. Thomson did not explain how this modeling evidence equates to the actual physical composition of the []. Rather, as pointed out by the ALJ, AUO's expert Dr. Yeh testified that the fact that the [] may be modeled as [] does not mean that such sublayers actually exist. RX-558C at Q.273-75. Even Thomson's expert Dr. Escuti admitted that the alleged sublayers do not have distinct boundaries. CX-4241C at Q.420. In addition, AUO's expert Dr. Drzaic credibly explained that "[t]he layer-by-layer method is a mathematical convenience used in many types of optical modeling merely as a way to facilitate the calculation" and "[i]t does not follow that the simplified model is an accurate description of the structure or phenomenon." RX-556C at Q.80-81. Thus, even if the alleged sublayers exist, Thomson has not provided any actual evidence showing that the values listed in the table above are the actual n_2 and n_3 values for the sublayers. Accordingly, we agree with the ALJ's finding that even under Thomson's sublayer theory, which uses the ALJ's construction of the term

“uniaxial,” Thomson has not shown that the accused products meet the uniaxial negatively birefringent compensator limitation of the asserted claims.

Moreover, as discussed in the claim construction portion of this opinion, the term “layer” means “the entire thickness of a material.” This means that Thomson must establish that the accused products meet the “uniaxial” limitation by considering the accused [] compensation film as a whole, rather than sublayers of the [] portion for the film. Thomson cannot show infringement by simply showing that each sublayer of the [] meets the “uniaxial” limitation.

For the foregoing reasons, the Commission affirms the ALJ’s finding that Thomson has not shown that the accused devices directly infringe the asserted claims of the ’006 patent under either Thomson’s entire layer theory or under Thomson’s sublayer theory.

4. Infringement of the ’006 Patent under the Doctrine of Equivalents

An element of an accused product is equivalent to a claim limitation if the differences between the two are insubstantial, a question that turns on whether the element of the accused product performs substantially the same function in substantially the same way to obtain the same result as the claim limitation. *Absolute Software, Inc. v. Stelath Signal, Inc.*, 659 F.3d 1121, 1139-40 (Fed. Cir. 2011). The patentee must provide particularized testimony and linking argument with respect to the function, way, result test when such evidence is presented to support a finding of infringement under the doctrine of equivalents. *Texas Instruments Inc. v. Cypress Semiconductor Corp.*, 90 F.3d 1558, 1567 (Fed. Cir. 1996).

Before the ALJ, Thomson argued that the uniaxial negatively birefringent compensator limitation is satisfied by the [] under the doctrine of equivalents, under both its sublayer theory and its entire layer theory. With respect to Thomson’s sublayer theory, the ALJ found that the sublayers of the [] do not perform the same function because AUO’s

expert Dr. Drzaic testified that an arbitrarily thin sublayer of the film would be too thin to provide sufficient retardation to effectively serve as a compensator in an LCD. ID at 361 (citing RX-556C at Q.156).

With respect to Thomson's entire layer theory, although not explicitly stated, the ALJ's finding implied that the [] as a whole does perform the same compensation function and produces the same compensation result. ID at 361-62. The ALJ found, however, that the way the [] compensates as a whole is substantially different. *Id.* at 361. The ALJ found that Thomson uses the same evidence that the ALJ rejected in his direct infringement analysis. *Id.* In addition, the ALJ found that accepting Thomson's position that a material where n_2 is approximately equal to n_3 can meet the "uniaxial" limitation would entirely vitiate the limitation. *Id.* at 361-362. Thus, the ALJ concluded that Thomson failed to demonstrate that the [] meets the "uniaxial" claim limitation under the doctrine of equivalents. *Id.* at 362.

Thomson argues that the [] as whole compensates in the same way as the compensator of the asserted claims of the '006 patent and that the ALJ erred in applying the doctrine of claim vitiation. We disagree with Thomson and find that the term "uniaxial" is not entitled to a numerical range of equivalents. The ALJ construed the term "uniaxial" as "having a *single* optical axis." As found by the ALJ, a birefringent material may have zero, one, or two optical axes." ID at 70-71. Thus, the number of axes in a "uniaxial" material is qualitatively different from a value that is defined by a range of degrees or values. Given the very small discrete set of possible values for the number of optical axes for a birefringent material – zero to two – the term "uniaxial" cannot readily be assigned a range of numerical values.

Thomson cites three cases purporting to show that the doctrine of equivalents is applicable where claims require an exact numerical value. We find these cases to be

distinguishable because unlike the situation here, the facts of each of these cases involve a parameter that is defined by a *range* of values rather than by a discrete value. *See Adams Respiratory Therapeutics, Inc. v. Perrigo Co.*, 616 F.3d 1283, 1292 (Fed. Cir. 2010) (discussing the limitation “at least 3500hr*ng/mL”); *Pozen Inc. v. Par Pharm., Inc.*, 800 F. Supp. 2d 789, 809-13 (E.D. Tex. 2011) (discussing a limitation construed to require “at least 90% of the naproxen”); *Phillips Corp. v. Iwasaki Elec. Co.*, 505 F.3d 1371, 1376 (Fed. Cir. 2007) (discussing a limitation requiring between 10^{-6} and 10^{-4} $\mu\text{mol}/\text{mm}^3$ halogen).

We find that the facts here are akin to those in *Moore U.S.A., Inc. v. Standard Register Co.*, 229 F.3d 1091, 1119 (Fed Cir. 2000), where the Court found that the term “majority” is not entitled to a range of equivalents covering 47.8%. According to *Phillips Corp. v. Iwasaki Elec. Co.* (one of the three cases cited by Thomson), discussing *Moore*, “[e]ven though the term ‘majority’ means a quantity greater than 50% and less than or equal to 100%, holding 47.8% to be equivalent to a majority *changes the character* of the claim limitation in a way that merely broadening a range need not.” 505 F.3d at 1379 (emphasis added). Here, allowing $n_2 \approx n_3$ to be equivalent to $n_2 = n_3$ would also change the character of the claim limitation because, as Thomson’s expert Dr. Escuti admitted, the scope of materials where $n_2 \approx n_3$ “would include materials that fall under the textbook definition of biaxial.” Tr. 360:5-9. This would vitiate the requirement that the compensator have a single optical axis.

Additionally, we note that Thomson makes the same arguments and cites to the same evidence in support of application of the doctrine of equivalents as it did with respect to direct infringement. Specifically, Thomson cites to the same retardation measurements of the [

] taken by AUO’s expert Dr. Wu to argue that n_2 and n_3 are substantially equal. Thomson also cites to the same two Yamahara articles discussing the [] to argue that n_2 and n_3

do not have to be strictly equal for a material to be considered uniaxial. We find that the ALJ properly rejected these arguments, stating that they are “just a repeat assertion from the literal infringement analysis.” ID at 361. As held by the Federal Circuit, “[t]he evidence and argument on the doctrine of equivalents cannot merely be subsumed in plaintiff’s case of literal infringement.” *Lear Siegler, Inc. v. Sealy Mattress*, 873 F.2d 1422, 1425 (Fed. Cir. 1989). Accordingly, Thomson has not shown that the [] as a whole meets the “uniaxial” limitation under the doctrine of equivalents.

Lastly, we turn to Thomson’s argument that the sublayers of [] accomplish the same compensation *function* as the claimed compensator because each sublayer of the film compensates a sublayer of the liquid crystal. Thom. Pet. at 64. In other words, Thomson argues that the asserted claims do not require the recited compensator to compensate the birefringence of the entire liquid crystal layer, but merely a portion of the birefringence of the crystal layer. We find Thomson’s argument to be unpersuasive. It is illogical to urge that a sublayer contributing a small amount of compensation is performing the function of the associated compensator. To the contrary, the language of claim 1 states that the compensator is “associated with” the “layer of twisted nematic liquid crystal *placed between two polarizers*,” thus indicating that the compensator must compensate for the entire layer of twisted nematic liquid crystal that is placed between two polarizers, not just a portion of the layer. Thus, we agree with the ALJ that a sublayer of the [] would not perform the compensation function because it would be far too thin to provide sufficient retardation to effectively serve as a compensator in a liquid crystal display. ID at 361.

In addition, we note that Thomson’s argument is based on an incorrect construction of the term “layer.” As discussed above, the proper construction of the term “layer” is the entire

thickness of a material, and therefore Thomson must show infringement of the “uniaxial” limitation by considering the entire thickness of the compensator as well as the entire thickness of the liquid crystal layer. Thus, Thomson has not shown that each of the sublayers of the accused [] meet the “uniaxial” limitation under of the doctrine of equivalents.

For the forgoing reasons, the Commission affirms the ALJ’s finding that Thomson has not shown that the accused devices infringe the asserted claims of the ’006 patent under the doctrine of equivalents, using either Thomson’s entire layer theory or its sublayer theory.

5. Anticipation of Claims 4 and 7 of the ’006 Patent by Scheuble

The ALJ found that respondents failed to establish by clear and convincing evidence that U.S. Patent No. 6,327,010 to Scheuble (“Scheuble”) anticipates asserted claims 4 and 7. Before the ALJ, Thomson argued that Scheuble does not disclose several limitations of claims 4 and 7. The ALJ addressed only one of these limitations, finding that Scheuble does not disclose the limitation “a birefringent plate” of claim 3, from which asserted claims 4 depends, and “a pair of uniaxial birefringent plates” of claim 7.³ ID at 219. Specifically, the ALJ found that although he did not explicitly construe the term “plate” in his claim construction analysis, under both parties’ proffered constructions of the term, a “plate” must be solid and not liquid. ID at 220. The ALJ analyzed expert testimony from both sides and found that respondents have not shown by clear and convincing evidence that the liquid-crystalline compensation layer disclosed in Scheuble is a “plate” as required by claims 4 and 7. *Id.*

Respondents argue that the “birefringent plate” limitation of claims 4 and 7 encompasses the liquid compensation layer sealed between two glass substrates disclosed in Scheuble. We find that neither the intrinsic evidence nor extrinsic evidence supports respondents’ argument.

³ The parties agree that for purposes of the anticipation analysis, the terms “plate” and “plates” should be construed the same.

The language “said compensating means comprises a birefringent plate” of claim 3 and “said compensating means comprises a pair of uniaxial birefringent plates” of claim 7 suggests that the *compensating means itself* be a plate, not the compensating means and surrounding glass structures. Also, the specification describes compensating element 11 as either “a plate with parallel faces cut out of a uniaxial birefringent medium” or a “blade” formed by such a medium, and does not suggest that the compensating element 11 can be a liquid layer sealed between two substrates. ’006 patent at 2:66-3:4, 1:21-25. Moreover, AUO’s expert Dr. Yeh never opined that Scheuble discloses the “birefringent plate” limitation of claims 4 and 7 by disclosing a liquid compensation layer sealed between two glass substrates. Rather, Dr. Yeh testified that the glass substrates themselves are plates, not the combination of the glass substrates and the liquid compensator.

Scheuble also discloses that *glass substrates* may be used at the end of column 8. A person of ordinary skill in the art would understand that a glass substrate is a smooth, flat, relatively thin, rigid body of uniform thickness meeting this limitation under Respondents’ construction of plate.

RX-157C at Q.381 (emphasis added). Thus, we agree with Thomson that a compensating liquid enclosed by two glass plates such as that disclosed in Scheuble is not a “birefringent plate” required by claims 4 and 7.

We turn to respondents’ argument that even if the enclosed liquid compensation structure of Scheuble is not a “birefringent plate,” other portions of Scheuble disclose a solid compensator that satisfies the limitation. As pointed out by AUO, the ALJ’s analysis focused only on the liquid crystal compensator disclosed in column 8 of Scheuble and did not address Scheuble’s disclosure of compensating “films” in column 9:

However, the object of FIG. 17 is merely to illustrate the principle of a compensation layer of this type, and a wide range of variants are possible. Thus, for example, the layers of monomeric nematic liquid crystal can be replaced or combined with

stacks of films comprising liquid-crystalline polymers whose mesogenic groups are aligned correspondingly, and/or with *films* comprising isotropic polymer material which are axially stretched correspondingly.

RX-75 at 9:1-9 (emphasis added). Also, as pointed out by CMI, the ALJ did not consider Scheuble's disclosure that the compensation layer of its invention may be made of a "thermoplastic polymer" that is stretched two-dimensionally:

In addition to this liquid-crystal layer, the electrooptical systems according to the invention may contain one or more, preferably not more than 2 and in particular one, compensation layer. *The compensation layers may be based on low-molecular weight liquid crystals, liquid-crystalline polymers or thermoplastic polymers, which are, for example, stretched 2-dimensionally and are thus made optically uniaxial.*

RX-75 at 3:38-42 (emphasis added). We find, however, that respondents did not provide clear support showing that the "films" or "polymers" referred in these passages are indeed solid material within the meaning of the "birefringent plate" limitation of claims 4 and 7.

AUO's expert Dr. Yeh did not rely on these passages when opining on whether Scheuble discloses "plates," but cited only to the particular portion of Scheuble discussing the glass substrates that enclose the liquid compensator. RX-157C at Q.356.

Because we find that respondents have not shown by clear and convincing evidence that Scheuble discloses the "birefringent plate" limitation of claims 4 and 7, we affirm the ALJ's finding that Scheuble does not anticipate claims 4 and 7 of the '006 patent.

B. The '063 Patent

1. The Invention of the '063 Patent

As discussed in the background section of the '006 patent, the basic components of an LCD consist of a thin layer of liquid crystal sandwiched between a pair of glass substrates, each substrate having a polarizer and a set of electrodes cemented to its surfaces. The liquid crystal thus sandwiched between two substrates will not maintain a uniform thickness by itself. Rather,

the thickness of the liquid crystal layer is kept uniform by using “spacers” that are made of glass fibers or plastic microspheres.

The invention of the '063 patent is directed to an improved liquid crystal display cell assembly in which the spacers are engineered to be highly “anisotropic” in shape so that they can be compatible with the aggressive mechanical rubbing process of LCD assembly. The patent defines the term “anisotropic” as a shape in which one side of the spacer is longer than the other (as opposed to the prior art shape where all sides of the spacer are equal in length). In addition to allowing the spacers to be compatible with the aggressive mechanical rubbing process of LCD assembly, the anisotropic shape of the spacers also allows the spacers to be situated within the non-active areas of the LCD cell, *i.e.*, areas that do not transmit light. Thus the spacers will not interfere with the active areas, *i.e.*, areas that transmit light.

Figure 6 shows a bottom substrate of a four-pixel liquid crystal display cell with active areas 34 and non-active areas 36:

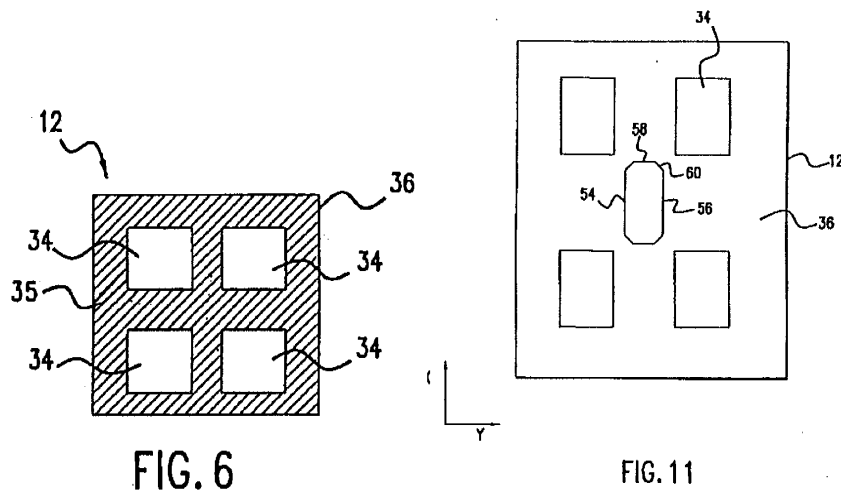


Figure 11 shows an anisotropic spacer 54 formed in non-active area 36 having a first side 56 along the X direction that is longer than a second side 58 along the Y direction.

The asserted claims are claims 1-4, 8, 11, 12, 14, 17, and 18.

2. Claim Construction of the '063 Patent

The Commission determined to review the ALJ's construction of the terms "mechanically rubbing" / "mechanically rubbed," "a plurality of spacing elements," and "an affixing layer."

a. "Mechanically Rubbed" / "Mechanical Rubbing" of Claims 1 and 11

Before the ALJ, the parties disputed whether the terms "mechanically rubbed" of claim 1 and "mechanically rubbing" of claim 11 require rubbing along the long axis of the spacer. Thomson argued that there is no directional requirement and that the proper construction of the terms is "using a machine or apparatus to apply a moving pressure." Respondents argued on the other hand that the terms require rubbing along the long axis of the spacer. According to AUO, the proper construction of the limitations "mechanically rubbed" of claim 1, "mechanically rubbing" of claim 11, and "rubbed along a first axis" of claim 14 is "the substrate and spacers are mechanically rubbed along the long axis of the spacer in the plane of the substrate."

The ALJ adopted the directional requirement proposed by respondents, but added the term "substantially" to respondents' proposed construction. Specifically, the ALJ construed the term "mechanically rubbed" of claim 1 as "having a moving pressurized friction applied by a machine or apparatus *substantially along the long axis* of the spacing elements" and the term "mechanically rubbing" of claim 11 as "using a machine or apparatus to apply a moving pressurized friction *substantially along the long axis* of the spacing elements formed on the substrate." ID at 58 (emphasis added). The ALJ added the term "substantially" to respondents' proposed directional requirement based on his finding that according to expert testimony, anisotropic spacers do not need to be rubbed precisely along their long axis in order to have strength superior to that of non-anisotropic spacer elements. *Id.* at 61.

We begin our analysis with the language of the claims. Claim 1 requires that the spacing layer of the claimed display cell include, *inter alia*, “a plurality of spacing elements ... said spacing elements being anisotropic in shape” and “the plurality of spacing elements have been mechanically rubbed.” Claim 11 requires that the claimed method of forming a display cell comprise, *inter alia*, “forming a plurality of spacing elements” and “mechanically rubbing over the first substrate having the plurality of spacing elements formed thereon.” The ALJ construed the term “anisotropic in shape,” appearing in both claims 1 and 11, as “having a length dimension that is greater than a width dimension in the plane of the substrates.” ID at 60. We agree with Thomson that nothing in the language of claims 1 and 11 by itself indicates that the mechanical rubbing must occur along the long dimension of the anisotropically-shaped spacer elements, or that the rubbing must occur in any direction at all. The only claim that indicates a rubbing direction is dependent claim 14, which depends from claim 11 via claim 12 and recites “wherein the spacing elements are rubbed along the first axis.”

We find that the broad language in independent claim 11, when compared with the language “rubbed along the first axis” of dependent claim 14, raises a presumption under the doctrine of claim differentiation that the term “mechanically rubbing” of claim 11 does not require rubbing along the “first axis” (*i.e.*, the long axis) of the spacer element. We do not agree with the ALJ that the presumption created by the doctrine of claim differentiation that “mechanically rubbing” of claim 11 is broader than “rubbing along the first axis” of claim 14 is rebutted by the alleged emphasis in the specification that rubbing must occur along the long axis. *See Am. Calcar, Inc. v. Am. Honda Motor Co.*, 651 F.3d 1318, 1337 (Fed. Cir. 2011) (“The doctrine of claim differentiation creates a presumption that each claim in a patent has a different scope. However, the doctrine of claim differentiation is not a conclusive basis for construing

claims.”). We also do not agree with the ALJ that the doctrine of claim differentiation as asserted by Thomson does not apply because the “first axis” of claim 14 refers to the long axis of the *non-active area of the substrate* and not the long axis of the *spacer element*.

We first address the ALJ’s finding that the doctrine of claim differentiation does not apply because the “first axis” of claims 12 and 14 refers to the long axis of the *non-active area of the substrate* and not the long axis of the *spacer element*. In our view, there is no support in the specification for the finding that the “first axis” of claims 12 and 14 refers to the long axis of the *non-active area of the substrate*. The specification discusses a long axis and a short axis only in the context of the length and width of the spacer element and never in the context of the substrate. For example, the description accompanying Figure 7 states that “[s]pacer 54 is anisotropic in shape as it includes first side 56 along an X direction (also known as the long axis) and a second side 58 along a Y direction (not shown in FIG. 7).” ’063 patent at 3:63-66. Similarly, the description accompanying Figure 9 describes that “a LCD rubbing process using a roller 50 that rolls along the X direction (long axis) of the spacers 54”:

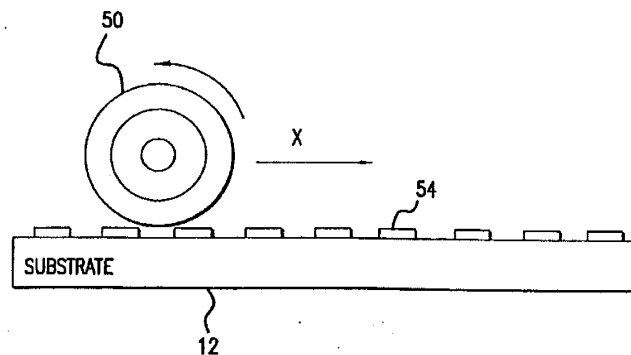


FIG. 9

’063 patent at 4:30-32; Fig. 9. The specification never hints that the non-active area of the substrate even has a long versus short axis.

We also disagree with the ALJ's finding that the terms "a first axis" and "a second axis" of claim 12 cannot describe the length and width of the spacer element itself, allegedly because independent claim 11 already requires that the shape of the spacer element be "anisotropic," *i.e.* "having a length dimension that is greater than a width dimension." We find that the language of dependent claim 12 is a clear example where a dependent claim is redundant to an independent claim. The patentee acted as his own lexicographer and explicitly defined the term "anisotropic" recited in claim 11 as meaning "a longer side along the X direction compared to the shorter side along the Y direction," in which the X direction is "also known as the long axis." *Id.* at 3:66-67. The ALJ essentially adopted this construction by construing "anisotropic" as "having a length dimension that is greater than a width dimension." The parties do not dispute this construction. Thus, the recitation "wherein the spacing elements extend along a first axis and along a second axis shorter than the first axis" of claim 12 is redundant to what is already required by claim 11.

Although we find the language of claim 12 to be redundant to claim 11 and the presumption created by the doctrine of claim differentiation to be rebutted, we do not find that the presumption that claims 11 and 14 have different scopes is rebutted. *Am. Calcar*, 651 F.3d at 1337. In our view, unlike the term "anisotropic," there is no express teaching in the '063 patent specification with respect to the direction of rubbing that would render the scope of claim 14 the same as claim 11. We begin with Figure 9 of the '063 patent, on which the ALJ grounds his construction. As discussed above, the description accompanying Figure 9 states "a conventional LCD rubbing process using a roller 50 that rolls along the X direction (long axis) of the spacers 54." '063 patent at 4:30-32; Fig. 9. The description accompanying Figure 9 goes on to state that "[u]nlike prior art spacers, the spacers of the present invention withstand the rubbing process due to their anisotropic shape." *Id.* at 4:32-34. We find that this language at best shows that the

length-to-width ratio of the spacers allows them to withstand the mechanical rubbing process occurring in the length direction. However, neither this particular description accompanying Figure 9 nor anything else in the specification indicates that rubbing *must* occur in the direction of the long axis of the spacer. Rather, the specification emphasizes that the spacers are anisotropic in shape in order to accomplish two goals: (1) to better withstand rubbing (without specifying the particular direction of rubbing) and (2) so that the spacers can be situated entirely within the non-active areas of the substrate. *Id.* at 2:37-44; 4:42-43. There is no teaching in the specification that would override the presumption that while dependent claim 14 requires rubbing in the direction of the long axis of the spacer element, independent claim 11 does not. Accordingly, we find that the terms “mechanically rubbing” and “mechanically rubbed” do not require rubbing along the long axis of the spacer element or in any particular direction at all.

We turn to the ALJ’s addition of the term “substantially” to his claim construction. It is not clear how the ALJ arrived at his conclusion that the mechanical rubbing needs to occur “*substantially* along the long axis of the spacer.” The only testimony the ALJ cited in support is an equivocal admission by AUO’s expert on cross-examination that when rubbed at [], an anisotropic spacer element had superior strength when compared to a cylindrical spacer element described as prior art in the ’063 patent. *ID* at 61 (citing *Tr.* at 950:19-953:5). According to the ALJ, this particular admission shows that an anisotropic spacer element does not need to be rubbed precisely along the long axis in order to be more effective than prior art spacer elements. However, the ALJ then added in a footnote that the term “substantially” would not include rubbing [], even though the ALJ found that the particular admission indicates that the claimed invention *would* cover rubbing in the [] as long as the spacer elements are anisotropic in shape. *Id.* at n.10. Accordingly, we disagree with the ALJ that the

asserted claims require rubbing “*substantially* along the long axis of the spacer.” As discussed above, the intrinsic evidence is clear that the terms “mechanically rubbing” and “mechanically rubbed” do not contain a requirement of rubbing in any particular direction.

For the foregoing reasons, the Commission reverses the ALJ’s construction of “mechanically rubbed” of claim 1 and “mechanically rubbing” of claim 11. Instead, the proper construction of the terms is “using a machine or apparatus to apply a moving pressure,” which does not contain a directional requirement.

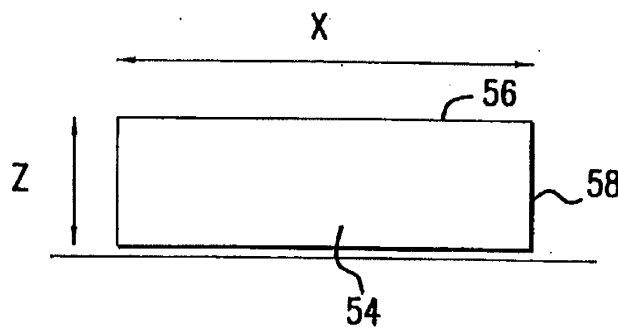
b. “A Plurality of Spacing Elements” of Claims 1 and 11

Before the ALJ, Thomson argued that the proper construction of “a plurality of spacing elements” is “more than one spacing element, where a spacing element is a structure that functions to keep the gap between the two substrates largely uniform.” Respondents argued that the proper construction also requires that the spacing elements contact the second substrate of the display cell. The ALJ adopted respondents’ narrower construction, construing “a plurality of spacing elements” as “two or more structures, not physically connected to one another, which structures serve to substantially uniformly separate two substrates, said structures formed on one of said two substrates and contacting the second substrate.” ID at 48. According to the ALJ, claims 1 and 11 both require that the two substrates are physically attached to one another via the spacing elements, and the specification does not teach away from the concept of requiring the spacing elements to touch both substrates. *Id.* at 45.

We begin our analysis with the language of the claims. In addition to reciting “a plurality of spacing elements separate from one another,” claim 1 recites “wherein said two substrates are affixed to each other after one of the two substrates and the plurality of spacing elements have been mechanically rubbed, the two substrates remaining substantially uniformly separated from each other by said spacing elements.” Similarly, claim 11 recites “attaching a second substrate

on the front surface of said first substrate, said second substrate being kept at a substantially uniform distance from said first substrate by said spacing elements.” The ALJ found that this claim language requires that the spacing elements physically hold the two substrates apart because the language requires that “the two substrates be physically attached to one another and that the spacing elements serve to substantially uniformly separate those attached substrates.” ID at 44-45. We disagree with the ALJ’s reading. Although the claims recite that the second substrate is “affixed” or “attached” to the first substrate, the language does not require that the “spacing elements” play a role in the attachment. The language does not require the spacing elements to physically hold the two substrates apart

Neither do we find this requirement in the ’063 specification. At the outset, we disagree with CMI that the specification discloses forming spacers of a precise thickness Z in order to establish a cell gap of Z height. We find that CMI mischaracterizes the description of the preferred embodiment with respect to Figure 7. Referring to Figure 7, the specification states that “the precise thickness of the spacer 54 in the Z direction can be achieved by spin coating techniques or CVD technology as is well known to one skilled in the art” and that the “the cell gap of Z -height is on the order of $5\text{ }\mu\text{m}$ for LC materials with an optical anisotropy, Δn , of 0.09-0.1,” where “ Z -height strongly depends on the Δn of the LC being used.”



'063 patent at Fig. 7, 4:13-19. Thus, the '063 patent uses the vertical axis as a reference in describing the thickness of the spacer and the distance between the top and bottom substrates. Reference to the Z axis, however, does not mean that the height of the spacer 54 and the distance between the top and bottom substrates are equal. *See* CX-4242C at Q.137. While the thickness between the top and bottom substrates in vertical Z direction is preferably set “on the order of 5 μm ,” the same discussion does not suggest a numerical value for the thickness of the spacers 54, stating only that “the spacers 54 can be made of sufficient size” and that “the precise thickness of the spacer 54 in the Z direction can be achieved by spin coating techniques or CVD technology as is well known to one skilled in the art.” '063 patent at 4:11-18.

The only indication in the specification that the height of the spacer may be equal to the cell gap is in Figures 1-3, depicting prior art spacers 20 between substrates 12 and 14:

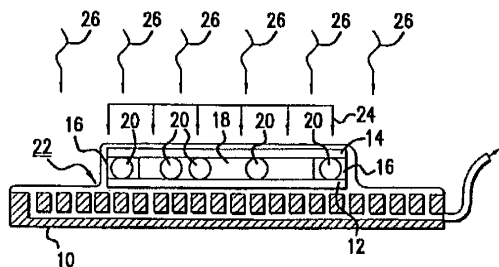


FIG. 1
(PRIOR ART)

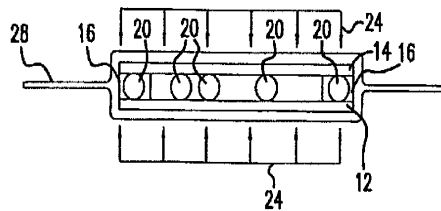


FIG. 2
(PRIOR ART)

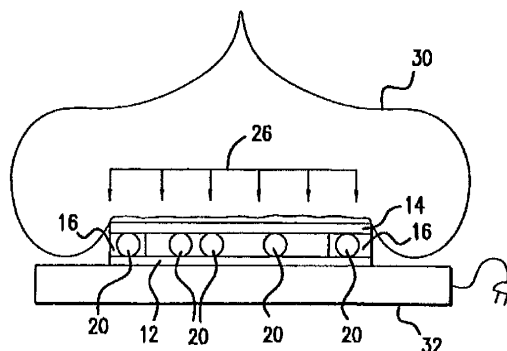


FIG. 3
(PRIOR ART)

'063 patent, Figs. 1-3. However, the accompanying text to Figures 1-3 is silent as to whether contact with the second substrate 14 is an actual requirement for the spacers 20 or whether the spacers 20 are simply depicted as such. Moreover, Figure 12, which also depicts prior art spacer 20, does not seem to show spacer 20 to be in contact with the top substrate:

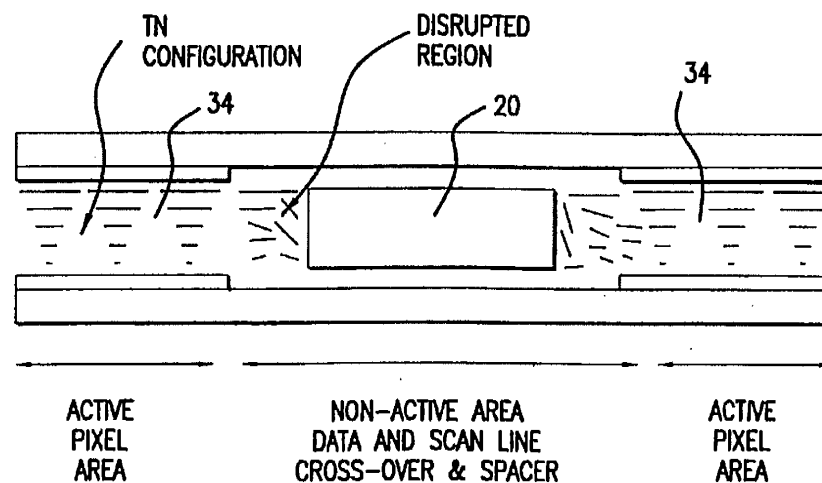


FIG. 12

(PRIOR ART)

Id. at Fig. 12. As for CMI's argument that the contact requirement is present in several references cited on the face of the '063 patent, we do not find the argument sufficient to overcome the silence as to this requirement in the claims and specification of the '063 patent.

See, e.g., Modine Mfg. Co. v. Int'l Trade Comm'n, 75 F.3d 1545, 1553 (Fed. Cir. 1996)

("[I]ncorporation by reference does not convert the invention of the incorporated patent into the invention of the host patent."), *overruled on other grounds by Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co., Ltd.*, 234 F.3d 558 (Fed. Cir. 2000).

As for the extrinsic evidence, neither side has provided persuasive arguments. On one hand, CMI argues that the testimony of Thomson's expert Dr. West actually supports requiring

the “spacing elements” to contact the opposite substrate, because Dr. West allegedly testified that spacers having a fraction of the height of the normal post-like spacers are not able to maintain a uniform cell gap. Written Submission of Respondent CMI In Response to the Commission’s Determination to Review-in-Part a Final Determination (April 9, 2012) (“CMI. Resp.”) at 3-4 (citing CX-4242C at Q.37). We find that CMI mischaracterizes Dr. West’s testimony because the particular portion of Dr. West’s testimony cited by CMI was referring to spacers that are “destroyed or dislodged,” which are different from a spacer deliberately engineered to almost touch the opposing side. CX-4242C at Q.37.

On the other hand, Thomson points to the testimony of Dr. West that a person of ordinary skill in the art would know that spacing elements can cause two substrates to be “substantially uniformly separated from each other,” as required by the claims, without contacting the “second substrate.” Specifically, Dr. West testified that a spacer connected to the bottom substrate that almost contacts the top substrate (and does contact that substrate if the substrate is compressed) can maintain a substantially uniform cell gap because it allows the cell gap to be narrowed only to the point where the second substrate would hit the spacer. CX-4242C at Q.132-33, 200, 566; Tr. at 259:4-17. Dr. West, however, based his testimony on disclosure of a patent that was filed after the invention of the ’063 patent. CX-4242C at Q.133 (citing CX-1127). Thus, we do not find Dr. West’s testimony to be helpful.

Nevertheless, because the claim language does not recite a contact requirement and because the specification is silent on this issue, we believe that Thomson’s construction of “a plurality of spacing elements” is the proper one, *i.e.*, “more than one spacing element, where a spacing element is a structure that functions to keep the gap between the two substrates substantially uniform.” Accordingly, the Commission reverses the ALJ’s construction requiring

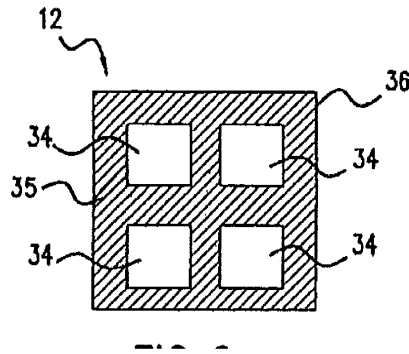
the “plurality of spacing elements” to contact “the second substrate.” We note that although this construction encompasses spacers that do not contact the opposing substrate, for such spacers to meet the “spacing elements” limitation, the spacers must function to keep the cell gap substantially uniform.

c. “Affixing Layer” of Claim 1

Claim 1 recites “a spacing layer, interposed between said two substrates, the spacing layer including an affixing layer . . . the affixing layer covering at least a portion of the non-active area and remaining substantially outside of the active aperture area.” Before the ALJ, Thomson argued that the proper construction of “affixing layer” is “material that attaches the spacing elements to a substrate” and that the “affixing layer” may be part of the material used to form the spacing elements. Respondents argued that the “affixing layer” must be separate and distinct from the spacing elements. The ALJ adopted AUO’s narrower construction, construing “an affixing layer” as “a stratum of material that attaches the spacing elements to a substrate, and which is separate and distinct from said spacing elements.” ID at 34.

We first address Thomson’s argument that because the “affixing layer” and “spacing elements” are recited as part of the same “spacing layer,” the two elements cannot be separate and distinct as construed by the ALJ. While it is clear from the claim language that the two elements are part of the same overall “spacing layer,” it is also clear from the claim language that the “affixing layer” and the “spacing elements” are two distinct (if not necessarily separate) elements within the “spacing layer.” It is a well established principal of claim construction that where a claim lists elements separately, the clear implication of the claim language is that those elements are distinct components of the patented invention. *Becton, Dickinson & Co. v. Tyco Healthcare Grp., LP*, 616 F.3d 1249, 1254 (Fed. Cir. 2010). Thus, the affixing layer should be construed as a distinct component from the spacing elements.

Our construction is supported by the specification. First, Figure 6 of the '063 patent shows spacers 34 as distinct from the affixing layer 35:



'063 patent, Fig. 6. Second, while the specification clearly states that the thickness of the affixing layer should be no more than 1 micron, the specification allows the thickness of the spacing elements to be greater. Specifically, the specification describes an “affixing layer” as “a thin coating of negative photoresist or negative UV curable polyimide” and states that “[t]he thickness of this thin coating should be in the range of approximately 0.1 microns to 0.5 microns and should be at least 0.05 microns, but generally no more than 1 micron and preferably 0.2 microns.” *Id.* at 3:37-42. By contrast, in describing the thickness of the spacing elements, the specification states that “the precise thickness of the spacer 54 in the Z [*i.e.*, vertical] direction can be achieved by spin coating techniques or CVD technology as is well known to one skilled in the art.” '063 patent at 4:13-16. The specification also states that the height of the cell gap between the top and bottom substrates 12 and 14, which contains spacer 54, is preferably set “on the order of 5 μm .” *Id.* at 4:16-19. Even though the spacer 53 may not be exactly the same height as the cell gap, which is disclosed to be around 5 microns, it would not make sense for spacer 54 to be as thin as the affixing layer, which is disclosed to be at most 1 micron. In our view, the statement that “the precise thickness of the spacer 54 in the Z direction can be achieved” implies that an additional layer of negative photoreactive polyimide must be deposited

on top of the affixing layer. Thus, the affixing layer and the spacing elements cannot be exactly the same layer.

It is significant that even Thomson does not argue that the height of the spacers can be the same as the thickness of the affixing layer and that as a result, the height of the spacers can vary greatly from the height of the cell gap. In response to AUO's argument that the specification discloses a disparity in thickness of the affixing layer and the spacing elements, Thomson argues that the cell gap is always larger than the spacer height because the cell gap is measured from the top and bottom substrates in the *active areas*, but the spacers are formed in the *non-active areas*, which have additional layers including electronics. Thomson, however, cites to nothing in the '063 patent to support its definition of "cell gap."

We turn to Thomson's argument that during prosecution of the '063 patent application, the examiner rejected the asserted claims, including claim 1, as anticipated by the Hasegawa reference, which discloses "a negative photosensitive layer disposed on the bottom substrate, wherein portions of the negative photosensitive layer are exposed to light, thereby, forming spacing elements disposed in non-display/active areas." *See* Complainant Thomson's Post-Trial Brief at 19, n.18 (citing JX-6 at THOM3375). We find that even if Thomson's argument were true, the rejection at best shows the views of the examiner and cannot be attributed to the patentee. *Salazar v. Procter & Gamble Co.*, 414 F.3d 1342, 1345 (Fed. Cir. 2005) ("[A]n applicant's silence regarding statements made by the examiner during prosecution, without more, cannot amount to a 'clear and unmistakable disavowal' of claim scope.").

Lastly, AUO's expert Dr. Lowe testified that it was common around the time of the '063 patent invention to use a thin layer, often called an "adhesion promoter," between two materials to promote adhesion of the spacing elements to the substrate. RX-554 at Q.174. Thus, there is

support in the extrinsic evidence that one of ordinary skill in the art at the time of the '063 patent invention would understand that the “affixing layer” is distinct from the “spacing elements.”

In sum, the claims recite the “affixing layer” and the “spacing elements” as two distinct elements, and the specification and prosecution history do not contradict the claim language. The words of a claim are generally given their ordinary and customary meaning as understood by a person of ordinary skill in the art when read in the context of the specification and prosecution history. *Thorner v. Sony Computer Entm't Am. LLC*, 669 F.3d 1362, 1365 (Fed. Cir. 2012). There are only two exceptions to this general rule: 1) when a patentee sets out a definition and acts as his own lexicographer, or 2) when the patentee disavows the full scope of a claim term either in the specification or during prosecution. *Id.* Neither exception applies in this situation. Accordingly, the Commission finds that the proper construction of the term “affixing layer” is “a stratum of material that attaches the spacing elements to a substrate, and which is distinct from said spacing elements.”⁴

3. Infringement of the '063 Patent

The products that allegedly infringe the '063 patent include CMI, Qisda, and BenQ LCD monitors and televisions that contain AUO and CMI LCD modules [

]. The CMI, Qisda, and BenQ LCD displays that use such AUO and CMI modules are identified in CMI's, Qisda's, and BenQ's interrogatory responses. CX-4242C at Q.219; CX-459C; CX-367C; CX-379C.

⁴ Note that we do not require the “affixing layer” to be “separate” from the “spacing elements,” as required by the ALJ's construction. The term “separate” is inaccurate and is not required to resolve the issue of infringement. Two layers of materials deposited on top of one another are not necessarily “separate” layers even though they may be “distinct” layers. Thus, it is sufficient for the issue of infringement to determine whether two layers are “distinct.”

The parties do not dispute that a determination of whether or not the accused modules infringe the asserted independent claims of the '063 patent turns on five issues: (1) whether the accused modules meet the limitation “affixing layer” of claim 1; (2) whether the accused modules meet the limitation “plurality of spacing elements” of claims 1 and 11; (3) whether the accused modules meet the limitation “anisotropic in shape” of claims 1 and 11; (4) whether the accused modules meet the limitation “mechanically rubbed” of claim 1 and the limitation “mechanically rubbing” of claim 11; and (5) whether the accused modules meet the limitation “forming a plurality of spacing elements . . . on the *front surface* . . . of said first substrate” of claim 11. ID at 329, 343.

We address each issue separately.

a. “Affixing Layer” of Claim 1

The ALJ found that the accused AUO and CMI modules do not meet the “affixing layer” limitation of claim 1 under his construction of the term, which requires that the affixing layer be distinct from the spacing elements. *Id.* at 330. The ALJ found that there is no affixing layer between the [] and the bottom substrate in the accused AUO and CMI modules. *Id.* at 331.

As discussed above in our claim construction analysis, we agree with the ALJ that claim 1 requires that the “affixing layer” be distinct from the “plurality of spacing elements.”

Specifically, the “affixing layer” is “a stratum of material that attaches the spacing elements to a substrate, and which is distinct from said spacing elements.” We agree with the ALJ that the accused AUO and CMI modules do not meet this limitation. AUO’s expert Dr. Lowe testified that the photospacers in the accused AUO modules are [

]

[]. RX-554C at Q.164-166. Dr. Lowe points to Figure 4 of Exhibit 28 of Thomson's complaint showing a scanning transmission electron microscope ("TEM") image of []:

[

]

Id. at Q.172. Although Thomson has drawn an arrow pointing to the lower portion of the spacing element and labeled it "Affixing Layer," Dr. Lowe pointed out that the [], and that the [

]. *Id.* CMI's expert Dr.

Wagner testified [] that nothing in the accused CMI module other than the [] themselves adheres to the bottom substrate. RX-636C at Q.118-126.

Lastly, we turn to Thomson's argument that the accused products meet the "affixing layer" limitation even under the ALJ's construction because there is an [].

Thomson Pet. at 40, n.15. In our view, Thomson’s argument is unpersuasive because Thomson has not shown that the alleged [].

Accordingly, we agree with the ALJ that the accused AUO and CMI modules do not meet the “affixing layer” limitation of independent claim 1. As such, the Commission affirms the ALJ’s non-infringement finding with respect to asserted independent claim 1 and asserted dependent claims 2-4 and 8.

b. “Plurality of Spacing Elements” of Claims 1 and 11

With respect to the accused CMI modules, the ALJ found that the [] in the accused CMI modules serve as the “plurality of spacing elements” of claim 1 and 11, but that the [] in the accused CMI modules do not. *Id.* at 333. Under the ALJ’s construction of “plurality of spacing elements,” the spacers in the accused modules must contact the second substrate. *Id.* at 43. According to the ALJ, the [] in the accused CMI modules, but the [] do not because they do not normally contact both substrates between which they are situated. *Id.* at 333.

With respect to the accused AUO modules, the ALJ found that the [] in the accused AUO modules serve as the “plurality of spacing elements” of claims 1 and 11. According to the ALJ, AUO’s expert Dr. Lowe admitted that the [] in the accused AUO modules are []. *Id.* at 332. The ALJ did not address AUO’s argument regarding the [] in AUO’s accused modules.

As discussed above in our claim construction analysis, the proper construction of “a plurality of spacing elements” is “more than one spacing element, where a spacing element is a structure that functions to keep the gap between the two substrates substantially uniform.” The

accused CMI and AUO modules contain [

]. The distinction

between these spacers is [

]. CX-4242C at Q.507; RX-554C at Q.246. While the

[

]:

[

]

CDX-244C; CDX-246C.

With respect to the [

]. RX-554C at Q.246, 263. CMI's expert Dr. Wagner contends,

however, that the [] in the accused CMI modules []
[]. RX-545C at ¶50-52, 87. According to Dr. Wagner, the cell gap in
CMI's accused modules is determined by the [] rather than the []
[]. *Id.* at ¶52. Dr. Wagner does not explain, however, the function of the []
[], if it is the [] that
maintains the cell gap. By contrast, both Dr. West and Dr. Lowe testified that it is impossible for
[] to maintain a uniform cell gap in modern displays, and that []
[]. CX-4242C at Q.547; Tr. at 1350:7-20. Accordingly,
Thomson has shown that the [] in the accused CMI and AUO modules meet the
“spacing elements” limitation of claim 11.

We turn to whether the [] function to maintain a substantially uniform cell
gap and thus meet the “spacing elements” limitation. The expert testimony from both sides is
generally consistent regarding the specific purpose of the []. According to
Thomson's expert Dr. West, the []

[]. CX-4242C at Q.511-513. Similarly, CMI's
expert Dr. Wagner and AUO's expert Dr. Lowe each testified that the []

[]. RX-636C at Q.258; RX-554C at 255-256. In others words, the [] do
not themselves maintain the uniformity of the cell gap, but only []
[]. AUO's and CMI's experts both testified
that [], the
“cell gap tolerance” has been exceeded so that the cell gap is no longer “substantially uniform.”

RX-554C at Q.245-46; RX-636C at Q.263-64. Thus, although we disagree with the ALJ on the proper construction of “spacing elements,” the non-infringement result is the same with respect to the [] in the accused products. Thomson has not shown that the [] in the accused CMI and AUO modules keep the gap between the two substrates substantially uniform and thus meet the “spacing elements” limitation of claim 11.

c. “Anisotropic in Shape” of Claims 1 and 11

Because Thomson has not shown that the [] in the accused products are “spacing elements,” Thomson must show that the [] meet the “anisotropic in shape” limitation of claim 11.

With respect to AUO, we agree with the ALJ that there is no real dispute that the “spacing elements” in the accused AUO modules are “anisotropic in shape.” ID at 330 at n.45. Specifically, AUO’s expert, Dr. Lowe, admitted that [] in the accused AUO modules are []

[], as required by the claims. Tr. at 1340:24-1342:18.

Thus, Thomson has shown that the accused AUO products meet the [] limitation of claim 11.

With respect to CMI, Thomson admits that in some CMI accused products, the [] are not anisotropic. Thom. Resp. at 14. Dr. West admitted on cross-examination, for example, that in the accused [] panel, the [] (shown as the dot in the photograph below) are not anisotropic, and that only the [] (shown as rods in the picture below) are anisotropic:

[

Tr. at 189:2-190:4; CDX-250C. Similarly, CMI’s expert Dr. Wagner testified that while the [

] in the accused CMI modules are anisotropic in shape, *only some* of the [

] in the accused modules are anisotropic in shape. 1385:21-1387:1. Thus, to show infringement, Thomson must identify the specific accused CMI modules that have [

] that are “anisotropic in shape.”

We find Thomson has not satisfied its burden. The testimony of its expert Dr. West with respect to the “anisotropic in shape” limitation did not tease out which accused CMI products have [] that are anisotropic and which products do not. Specifically, Dr. West testified that there are [] spacing elements in the accused CMI products, and that although not “all of the Accused Products have anisotropic [] . . . even panels where just the [] or just the [] are anisotropic meet the claim elements.” CX-4242C at Q.504-507; 604-18. Moreover, Dr. West’s testimony was conclusory at best. While his testimony was based on examination of CMI’s design documents, the particular documents were excluded from evidence for lacking proper translation. *Id.* at Q.616-617. As for the GDS files showing masks used in the specific accused CMI products that

allegedly contain anisotropic [], Dr. West did not testify as to these files. *See* Thom. Resp. at 13-15. Without expert guidance, it is not apparent from any of the mask images shown in these files whether the spacers are [] or [] or whether they are anisotropic. Thus, Thomson has not shown that the accused CMI products meet the “anisotropic in shape” limitation and thus has not shown that they infringe claim 11.

d. “Mechanically Rubbed” / “Mechanically Rubbing” of Claims 1 and 11

The ALJ found that the accused modules (both CMI and AUO modules) do not meet the “mechanically rubbed” / “mechanically rubbing” limitations of claims 1 and 11, either directly or under the doctrine of equivalents, [

] *Id.* at 335-336. As discussed above in our claim construction analysis; however, the term “mechanically rubbing” of claims 1 and 11 means “using a machine or apparatus to apply a moving pressure” and does not contain a directional requirement. As found by the ALJ, it is undisputed that the spacing elements in both the AUO and CMI accused products are rubbed using [

] RX-554C at Q.202-203; RX-636C at Q.227. Because our construction of “mechanically rubbing” of claims 1 and 11 does not contain a directional requirement, we find that Thomson has shown that both the AUO and CMI accused products practice this limitation.

e. “Front Surface” / “Rear Surface” of Claim 11

The ALJ found that the accused modules (both CMI and AUO modules) meet the limitation “forming a plurality of spacing elements . . . on the *front surface* . . . of said first substrate” of claim 11. ID at 344. While the ALJ did not address the terms “front surface” and “rear surface” of claim 11 in his claim construction analysis section of the ID, the ALJ found that claim 11 makes clear that “the front surface” of the first substrate is the surface upon which the

spacing elements are formed, and the surface upon which the second substrate is mounted, thus locating the spacing elements between the two substrates. *Id.* The ALJ rejected AUO's argument that the front and rear surfaces refer to the position of the viewer. *Id.* The ALJ found that the accused products practice the "front surface" limitation of claim 11 because AUO's argument that this limitation is not met is based only on AUO's claim construction, which the ALJ rejected. *Id.*

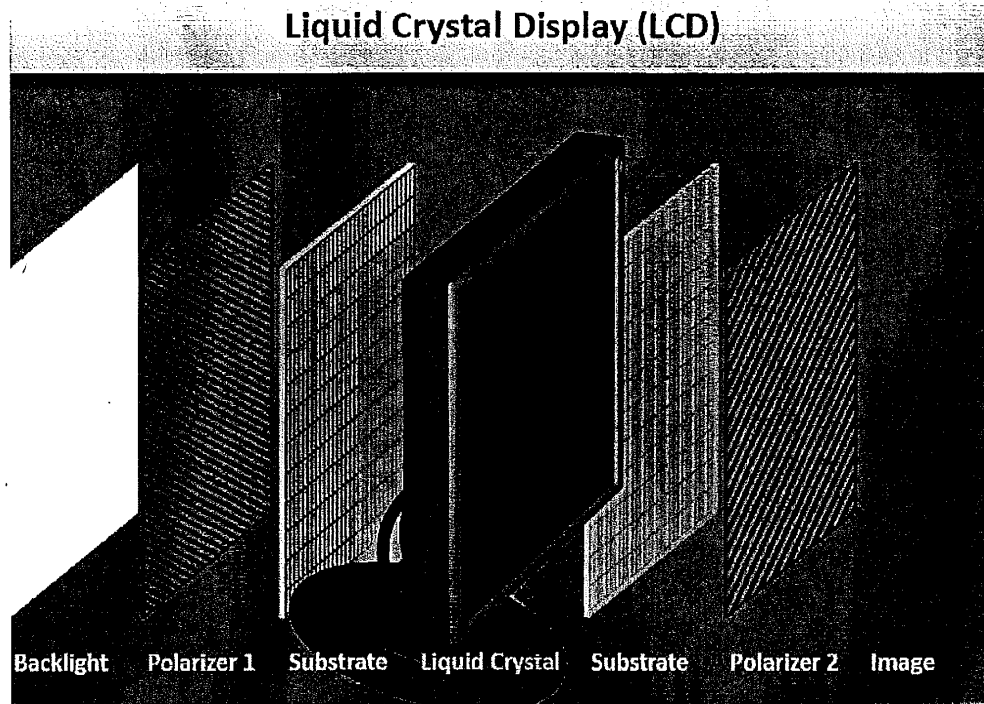
AUO argues that the ALJ erred in failing to address the parties' dispute regarding the proper construction of the terms "front surface" and "rear surface" of claim 11. AUO argues that to the extent that the ALJ construed the "front surface" limitation, he erroneously concluded that the "front surface" is the surface upon which the spacing elements are formed. According to AUO, the proper construction of the terms must take into account the point of reference of the viewer. According to AUO, because the [

].

We disagree with AUO that the ALJ erred in failing to address the proper construction of the terms "front surface" and "rear surface" of claim 11. While the ALJ did not explicitly construe the terms in the claim construction section of his ID, he did make a claim construction finding in addressing infringement of these terms by the accused products. According to the ALJ, "the claim makes clear that *the 'front surface' of the first substrate is the surface upon which the spacing elements are formed, and the surface upon which the second substrate is mounted*, thus locating the spacing elements between the two substrates." *Id.* at 344 (emphasis added). According to the ALJ, "[a]s a result the second substrate is kept at a substantially uniform distance from the first substrate by said spacing elements. *Id.* Thus, based on the claim

language itself, the ALJ construed the “front surface” of the first substrate to mean “the surface upon which the spacing elements are formed, and the surface upon which the second substrate is mounted.” The “rear surface” is the surface opposite the front surface, which the parties agreed to in their proposed constructions. JX-37.

We turn to AUO’s argument that to the extent the ALJ construed “front surface” and “rear surface,” his construction is incorrect and the accused products do not meet the “front surface” limitation under AUO’s construction of the term. AUO’s argument is best explained in reference to the illustration below:



CDX-541. According to AUO, the “front surfaces” of each of the two substrates recited in claim 11 are the surfaces of the two substrates on either side of the liquid crystal that face the viewer (to the right side of the illustration above). AUO argues that [

]

[

]

In our view, the ALJ correctly construed “front surface” as “the surface upon which the spacing elements are formed, and the surface upon which the second substrate is mounted.” In other words, “front surface” is the surface facing the liquid crystal layer of the module, which is away from the viewer. The language of claim 11 makes no reference to the position of the viewer in reciting “a front surface” and “a rear surface,” but refers to the front surface of the first substrate as the surface on which the spacing elements are formed as well as the surface on which the second substrate is attached:

11. A method of forming a display cell comprising:

providing a first substrate which has been partitioned into an active aperture area and a non-active area and has *a front surface* and a rear surface;

forming a plurality of spacing elements separate from one another *on the front surface* and non-active areas of said first substrate, the spacing elements being anisotropic in shape;

mechanically rubbing over the first substrate having the plurality of spacing elements formed thereon; and

attaching a second substrate *on the front surface* of said first substrate, said second substrate being kept at a substantially uniform distance from said first substrate by said spacing elements.

'063 patent at claim 11 (emphasis added). The language further indicates that, as a result, a substantially uniform distance is maintained between the two substrates by placing spacers on the front substrate surface that faces the front surface of the second substrate, *i.e.*, the surface that faces the interior of the display cell.

AUO cites to the introductory section of the '063 patent specification describing the related art to argue that the point of reference for the terms “front” and “rear” is the viewer of the

display. AUO Pet. at 14 (citing '063 patent at 1:13-21). In our view, however, the uses of the term “front” and “rear” at column 1, lines 13-21, of the '063 patent do not concern substrate surfaces — they concern things such as rear projection monitors, and front and rear glass layers.

AUO also cites to the testimony of Dr. Parsons, Thomson's technical expert for the '556 and '674 patents, and argues that Dr. Parsons' testimony directly contradicts the ALJ's conclusion that the term “front surface” and “rear surface” do not relate to the viewer of the display. AUO Pet. at 16 (citing CX-4244C at Q.22-23). We find that AUO mischaracterizes Dr. Parsons' testimony. Dr. Parsons was not testifying about claim terms in the '063 patent. Moreover, Dr. Parsons was testifying about a finished monitor in general and not a substrate in the process of forming a display cell as in claim 11 of the '063 patent.

Lastly, we point out that the ALJ's construction of “front” and “rear” surfaces is consistent with the extrinsic evidence in the form of inventor testimony. The inventor of the '063 patent, Dr. Ho, testified that [

]. Because AUO's argument that this limitation is not met is based only on AUO's claim construction, which the ALJ correctly rejected, we agree with the ALJ's finding that the accused products meet the “front surface” limitation of claim 11.

For the foregoing reasons, we affirm the ALJ's finding that the accused AUO and CMI products do not infringe claim 1. We also affirm the ALJ's finding that the accused CMI products do not infringe claim 11. We reverse the ALJ's finding that the accused AUO products do not infringe claim 11.

f. Dependent Claims 2-4, 8, 12, 14, 17, and 18

The ALJ found that Thomson has shown that the accused modules (both CMI and AUO modules) meet each of the requirements of the dependent claims 2-4, 12, 14, 17, and 18. ID at 339, 346-347. The parties do not challenge these particular findings. The ALJ concluded, however, that because these claims depend from independent claims 1 and 11, which he found not to be infringed, these claims are also not infringed. *Id.*

Because we find that the accused AUO products infringe claim 11, we also find that the accused AUO products infringed claims 12, 14, 17, and 18. With respect to CMI, however, because the accused CMI products do not infringe claim 11, they also do not infringe claims 12, 14, 17, and 18.

4. Obviousness of the Asserted Claims of '063 Patent in View of Sugata and Tsuboyama

The ALJ found that claims 1-4, 8, 11, 12, 14, and 18 are obvious over U.S. Patent No. 4,568,149 to Sugata ("Sugata") in view of U.S. Patent No. 4,775,225 to Tsuboyama ("Tsuboyama"). ID at 202. The ALJ found that Sugata discloses each of the limitations of the above claims except mechanical rubbing of the substrate and spacers *after* the spacers have been formed. *Id.* at 172. The ALJ found that although Sugata discloses forming spacers and the need to apply an orientation controlling treatment (*i.e.*, mechanical rubbing), Sugata does not reveal whether the mechanical rubbing is performed on the spacer members and when the mechanical rubbing occurs. *Id.* at 172-73. The ALJ found the limitation to be disclosed in Tsuboyama and that a person of ordinary skill in the art would be motivated to combine Sugata and Tsuboyama at the time of the invention of the '063 patent. *Id.* at 202-04. The ALJ thus found that respondents have presented a *prima facie* case of obviousness for asserted claims 1-4, 8, 11, 12,

14, and 18. *Id.* at 208. The ALJ found that the secondary considerations offered by Thomson failed to overcome this obviousness showing. *Id.*

With respect to dependent claim 17, the ALJ found that the record lacks clear and convincing evidence that the combination of Sugata and Tsuboyama discloses that the spacing elements are photolithographically formed. *Id.* at 205. Thus, the ALJ concluded that the invention of claim 17 is not obvious.

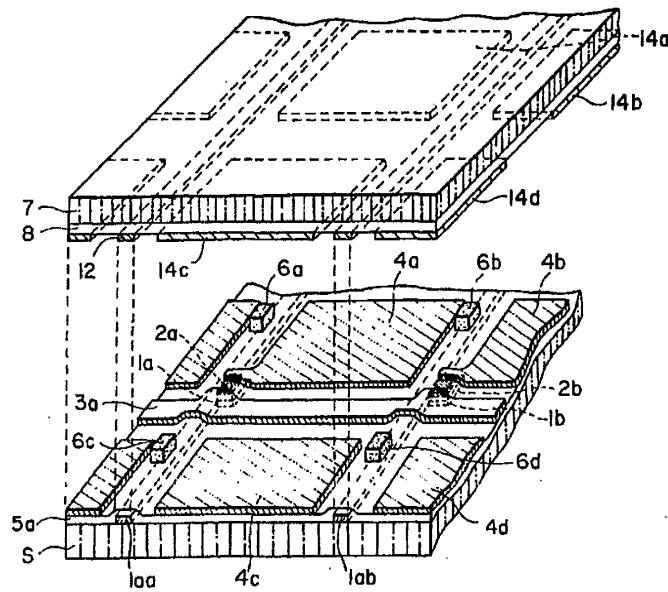
The Sugata patent issued February 4, 1986 and is prior art under 35 U.S.C. § 102(b). RX-15. Sugata teaches an active matrix liquid crystal display panel having spacers for maintaining a uniform gap between two substrates (referred to as electrode plates). *Id.* at Abstract, 2:45-3:2. As found by the ALJ, Sugata discloses almost all of the elements of asserted claims 1-4, 8, 11, 12, 14, and 18 of the '063 patent under either parties' constructions, including anisotropic spacers that remain in the non-active areas of the substrate, an affixing layer that affixes the spacers to one electrode plate, and a cell gap between the two electrode plates that is kept substantially uniform by the spacers. *See* RX-158C at Q.344-373. In addition, Sugata discloses a mechanical rubbing step to control the orientation of the liquid crystal. RX-15 at 4:38-44. The parties dispute, however, whether Sugata's rubbing step is performed *after* the spacers are formed, as required by the claims. Specifically, claim 1 recites "after one of the two substrates and the plurality of spacing elements have been mechanically rubbed," and claim 11 recites the step of "mechanically rubbing over the first substrate having the plurality of spacing elements formed thereon," which occurs after the step of forming the spacing elements. We note that these sequential limitations were added by amendment to distinguish from a prior art reference cited by the Examiner during prosecution of the '063 patent application. JX-6 at THOM3427-31.

The Tsuboyama patent issued on October 4, 1988 and is prior art under 35 U.S.C.

§ 102(b). RX-18. Tsuboyama discloses a liquid crystal device having two substrates (referred to as first and second base plates) and anisotropic spacers between the substrates that are mechanically rubbed along their long axis. *Id.* at Abstract, 2:34-46.

Thomson argues that one of ordinary skill in the art would not be motivated to combine Sugata with the teachings of Tsuboyama so that rubbing occurs *after* the spacers are formed, as required by the claims, because Sugata already discloses rubbing the substrate *prior to* spacer formation. Thus, according to Thomson, one of ordinary skill in the art would have no reason to modify Sugata with the teachings of Tsuboyama. For the reasons explained below, we disagree with Thomson that Sugata discloses rubbing *prior to* spacer formation. Rather, as found by the ALJ, Sugata does not reveal the sequence in which spacer formation and mechanical rubbing occur.

Although spacer formation on a substrate and rubbing of a substrate are discussed in two alternative embodiments in Sugata, it is not clear in either embodiment whether the substrate on which the spacers are formed is the same substrate that is rubbed, *i.e.*, whether rubbing occurs after the spacers are formed. The first embodiment is shown in Figure 3(a):



RX-15 at Fig. 3(a). With respect to Figure 3(a), Sugata states that “in the liquid crystal display panel, spacer members 6a, 6b, 6c, 6d are fixed on row electrodes 1aa, 1ab on the electrode plate S through the insulating layer 5a.” RX-15 at 3:47-51. In other words, spacers 6a-6d are formed on insulating layer 5a. Sugata then states that “each surface of two electrode plates in contact with the liquid crystal may be coated with an *insulating material*.” *Id.* at 4:31-33 (emphasis added). Sugata further states that “*to the insulating film . . . an orientation controlling treatment is applied. . . . As a typical process, the surface of the insulating film is rubbed in one direction with a velvet or cloth.*” *Id.* at 4:38-44 (emphasis added). We find that it is not clear whether the “insulating film” that is rubbed is the “insulating layer 5a” on which the spacers 6a-6d are formed or the “insulating material” that is coated on top of the spacers 6a-6d. On one hand, AUO’s expert Dr. Lowe testified that the insulating film that is rubbed is not the insulating layer 5a on which the spacers 6a-6d are formed, but an additional insulating film (not shown) that is coated on top of the spacers 6a-6d, thereby subjecting spacers 6a-6d to the rubbing treatment after the spacers are formed as required by the claims. RX-158 at Q.335-336. On the other

hand, Thomson's expert Dr. West testified that the orientation insulating film that is rubbed is the same as the insulating layer 5a. Tr. 1586:12-16. We find neither testimony to be persuasive. We find that the first embodiment does not clearly disclose whether rubbing occurs prior to spacer formation, as urged by Thomson, or after spacer formation, as required by the claims.

Neither is the sequence of rubbing and spacer formation clearly disclosed in the second embodiment of Sugata, which is shown in Figure 3(b):

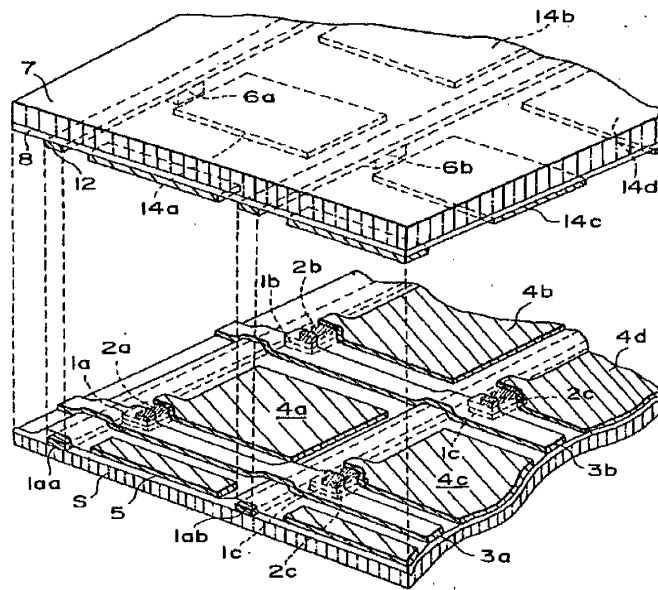


FIG. 3(b)

RX-15 at Fig. 3(b). According to the description accompanying Figure 3(b), spacers 6a-6d are formed on "an insulating layer (not shown)" coated on non-transmissive members 12 and color filters 14a-14d. *Id.* at 5:1-4. The description accompanying Figure 3(b), however, does not mention rubbing of the insulating layer on which the spacers 6a-6d are formed. According to Thomson, this "insulating layer" of Figure 3(b) must be the same "insulating film to which an orientation controlling treatment is applied," as described with respect to Figure 3(a). CX-4304C at Q.319-321. We do not find this argument persuasive. Again, it is not clear whether the "insulating layer" described with respect to Figure 3(b) is the same "insulating film" that is

rubbed. Thus, as with the first embodiment in Sugata, it is not clear whether rubbing occurs prior to spacer formation, as urged by Thomson, or after spacer formation, as required by the claims.

Because we disagree with Thomson that Sugata clearly discloses rubbing the substrate *prior to* forming spacers on the substrate, we also disagree with Thomson's argument that one of ordinary skill in the art would not be motivated to modify Sugata with Tsuboyama's teaching of rubbing *after* spacer formation to arrive at the claimed invention. On the contrary, the evidence indicates that one of ordinary skill in the art would be motivated to combine the references to arrive at the claimed invention.

As found by the ALJ, Sugata and Tsuboyama share a common goal of providing spacing elements that do not disturb the orientation or alignment of the liquid crystal molecules in the active area of the display. Specifically, Sugata states its object as "provid[ing] a liquid crystal display panel in which alignment or orientation of liquid crystal molecules is not disturbed on an image display surface." RX-15 at 2:54-57. Similarly, Tsuboyama states that its invention provides a liquid crystal device "which is free of orientation or alignment defects over the whole area of the device despite spacers which are present" within the liquid crystal. RX-18 at 2:35-38. In Sugata, the goal of avoiding orientation or alignment defects is accomplished by locating the spacers in the light non-transmissive areas of the display panel. RX-15 at 6:49-52. In Tsuboyama, this same goal is accomplished by providing rectangular spacers that are narrow in a direction perpendicular to the rubbing direction, *i.e.*, rubbing in the direction of the long axis, which occurs *after* the spacers are formed. RX-18 at Fig. 3B, 2:46-50, 4:49-51, 4:66-5:6.; Fig. 10, 7:51-58.

The evidence further indicates that a person of ordinary skill in the art would be motivated by the common goal of avoiding orientation defects to combine Sugata's teaching of locating the spacers in the non-active area of the substrate with Tsuboyama's teaching of rubbing along the long axis of the spacers after spacer formation. Specifically, respondents' expert Dr. Lowe testified that a person of ordinary skill in the art who was following the teachings of Sugata would be interested in additional steps that could be taken to avoid defects in the alignment or orientation of the liquid crystal molecules in the image display area in the vicinity of the spacers. RX-158C at Q.542. According to Dr. Lowe, the person of ordinary skill would recognize that, in addition to locating the spacers in the light non-transmissive areas of the display panel, as taught by Sugata, a further improvement could be achieved by forming spacers that are rectangular in shape and rubbing along the long axis of the spacers after the spacers are formed, as taught by Tsuboyama. *Id.*

In addition, the evidence indicates that the combination of Sugata and Tsuboyama is a combination of known elements that yield predictable results. Thomson does not dispute Dr. Lowe's testimony that at the time of the '063 patent, it was known that the mechanical rubbing step could be performed *either before or after* the spacers were formed on the substrate. Tr. 1095:1-8 ("[B]oth processes were known"). Also, the '063 patent itself acknowledges that it was well known to perform rubbing after the spacers have been formed on the substrate, stating that "[p]rior art spacers that are post-like are easily destroyed by the rubbing process." '063 patent at 4:34-35. In addition, Tsuboyama describes in detail the effects of mechanically rubbing anisotropic spacers to the alignment or orientation of the liquid crystals. RX-18 at 4:66-5:6, 10:40-45, cols. 9-12, tbls. 1-4. Thus, we agree with the ALJ that respondents have shown a *prima facie* case of obviousness of claims 1-4, 8, 11, 12, 14, and 18.

We turn to the ALJ's finding that the combination of Sugata and Tsuboyama does not render claim 17 of the '063 patent obvious. Claim 17 recites:

The method of claim 11, wherein the forming step comprises photolithographically forming the spacing elements having the anisotropic shape using a mask.

We find that the evidence shows that Tsuboyama discloses forming spacing elements using photolithography. Specifically, Dr. West discussed in detail that Tsuboyama discloses formation of spacing elements using photoresist, etching, and masking:

The method of forming the spacers of Figure 3A is described in column 8, lines 26-52 of the Tsuboyama patent. First, the two glass substrates covered in stripe ITO electrodes are each coated with a polyimide film. These polyimide films are then heat-cured. Then, a second polyimide film that will result in a film of approximately 1 micron thickness is applied to one of the substrates. *To this polyimide film a photoresist solution is applied, and the materials are pre-baked. Then, the photoresist is exposed through a mask to form a pattern on top of the polyimide film. then the polyimide film is selectively etched using the mask just formed.* At the end of this process the remaining polyimide pattern is heat cured to form the material shown in Figure 3A of Tsuboyama.

CX-4304C at Q.431 (emphasis added).

Moreover, we find that Sugata also discloses forming spacing elements using photolithography. Sugata discloses forming spacing elements by forming a film and then patterning the film through etching:

Spacer members 6a, 6b, 6c, 6d, ..., etc. may be formed by vapor deposition, sputtering and the like with a mask having a predetermined pattern, or by forming a uniform film having a thickness substantially equally to that of the liquid crystal layer by vapor deposition, sputtering[,] coating or the like *and then patterning the film through etching of portions other than those forming spacers.*

RX-15 at 5:44-51 (emphasis added). Dr. Lowe's testimony shows that Sugata's disclosure of "forming a uniform film" and "then patterning the film through etching of portions other than those forming spacers" is a disclosure of photolithographically forming the spacers using a mask.

Q: How are the spacer members formed in Sugata, RX-15?

A: According to Sugata, spacer members 6a, 6b, 6c, 6d may be formed by vapor deposition or sputtering with a mask having a predetermined pattern or by forming a film and then patterning the film through etching of portions other than those forming spacers.

Q: Does Sugata disclose photolithographically forming the anisotropically shaped spacing elements using a mask, as recited in claim 17?

A: Yes. Sugata teaches that spacer members 6a, 6b, 6c, 6d are formed photolithographically using a mask and that they can be anisotropic in shape.

RX-158C at Q.332, 373. As acknowledged by the ALJ, Thomson made no effort to rebut Dr.

Lowe's testimony on this point. ID at 205, n.28 (citing CX-4304C at Q.442-446; CDX-1351).

We disagree with the ALJ that, because Sugata does not contain the word

"photolithographically," this claim element is not disclosed. In addition to the testimony of Dr.

Lowe, the testimony of Thomson's expert Dr. West acknowledges that the act of forming a film and then patterning the film is known as "photolithography":

Q.47. What does it mean that the spacers are "photolithographically formed"?

A. Photolithography is a technique for selectively *patterning* a material on a substrate. CDX-0026 to CDX-0030 illustrates this technique.

CX-4242C at Q.47 (emphasis added). Dr. West also testified that in the '063 patent, "the spacers are first formed when they are *patterned* onto the substrate in only the non-active areas." CX-4304C at Q.295 (emphasis added). Accordingly, we find that respondents have shown a *prima facie* case of obviousness of claim 17.

Lastly, we turn to Thomson's commercial success argument. Once a challenger has presented a *prima facie* case of invalidity, the patentee has the burden of going forward with rebuttal evidence, which may include evidence of secondary considerations of non-obviousness such as commercial success, long felt but unsolved needs, failure of others, and so forth. *Pfizer Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1360 (Fed. Cir. 2007). Commercial success is usually shown

by significant sales in a relevant market and that the successful product is the invention disclosed and claimed in the patent. *Ormco Corp. v. Align Tech. Inc.*, 463 F.3d 1299, 1312 (Fed. Cir. 2006). Evidence of commercial success is only significant, however, if there is a nexus between the claimed invention and the commercial success. *Id.* at 1311-12. If the commercial success is due to an unclaimed feature of the device, or if the feature that creates the commercial success was known in the prior art, the commercial success is irrelevant. *Id.* at 1312.

To show commercial success, Thomson offered the testimony of Dr. West that there is widespread infringement by respondents' accused products and that the photolithographically formed anisotropic spacers of the '063 patent are important to a successful display cell both during the manufacturing and in the field. CX-4304C at Q.449-451. In our view, Dr. West's conclusory testimony that the "['063] spacers are critical to the proper manufacturing and utility in the field of the display cells at issue" is insufficient to demonstrate a nexus between the alleged commercial success and the features that allegedly distinguish the claimed invention from the prior art. *See id.* Dr. West's testimony does not show that the accused products are coextensive with the claimed features of the '063 patent, or that favorable results in the accused products could be achieved only by using the combination of features claimed in the '063 patent. Thus, ALJ correctly found that Thomson's assertions regarding secondary considerations are unsupported by the evidence and fail to rebut respondents' strong showing of obviousness.

Accordingly, we find that respondents have shown by clear and convincing evidence that all asserted claims of the '063 patent, including claim 17, are rendered obvious by Sugata in combination with Tsuboyama. As such, the Commission affirms the ALJ's finding that claims 1-4, 8, 11, 12, 14, and 18 are rendered obvious by Sugata and Tsuboyama and reverses the ALJ's finding that claim 17 is not rendered obvious by the combination.

5. Anticipation of the Asserted Claims of '063 Patent in View of Either Lowe or Miyazaki

U.S. Patent No. 5,801,796 to Lowe ("Lowe") was filed on May 10, 1996, eleven months before the filing date of the '063 patent. U.S. Patent No. 5,978,061 to Miyazaki ("Miyazaki") was filed on September 5, 1996, seven months before the filing date of the '063 patent. As a result, both references are presumptively prior art under 35 U.S.C. § 102(e). Before the ALJ, respondents argued that both references anticipate the '063 patent. Thomson attempted to "swear behind" the references, arguing that the '063 patent invention was conceived and reduced to practice before the filing dates of Lowe and Miyazaki.

The ALJ found that Lowe and Miyazaki are not prior art to asserted claims 11, 12, 14, 17, and 18 of the '063 patent because Thomson has shown that the claims were reduced to practice before the effective dates of Lowe and Miyazaki. *Id.* at 152. The ALJ found, however, that Lowe and Miyazaki are prior art to asserted claims 1- 4 and 8 because Thomson has not shown that the claims were reduced to practice before the effective dates of Lowe and Miyazaki. *Id.*

With respect to Lowe, the ALJ found that even if Lowe is prior art to all asserted claims, Lowe does not anticipate any of the asserted claims. *Id.* at 182. With respect to Miyazaki, the ALJ found that respondents have failed to offer clear and convincing evidence that Miyazaki anticipates claims 1-4 and 8 of the '063 patent. *Id.* at 186. The ALJ found, however, that if Miyazaki is prior art to claim 11 and its dependent claims, it would anticipate claims 11, 12, 14, and 18, but not claim 17. *Id.* at 187.

a. Conception and Reduction to Practice

Priority of invention usually goes to the first party to reduce an invention to practice, unless the other party can show that it was the first to conceive the invention and that it exercised reasonable diligence in later reducing that invention to practice. *Mahurkar v. C.R. Bard Inc.*, 79

F.3d 1572, 1577 (Fed. Cir. 1996). To have conceived of an invention, an inventor must have formed in his or her mind a definite and permanent idea of the complete and operative invention, as it is hereafter to be applied in practice. *Id.* Reduction to practice follows conception. *Id.* at 1577. To show actual reduction to practice, an inventor must demonstrate that the invention is suitable for its intended purpose. *Id.* at 1578. Depending on the character of the invention and the problem it solves, the showing of reduction to practice may require test results. *Id.* Conception and reduction to practice cannot be established solely on the basis of an inventor's testimony but must be corroborated by independent evidence. *Loral Fairchild Corp. v. Matsushita Elec.*, 266 F.3d 1358, 1363 (Fed. Cir. 2001). The sufficiency of such corroborating evidence is evaluated under a "rule of reason," considering all of the pertinent evidence. *Id.*

In an infringement action, when a party offers into evidence a prior art reference challenging the validity of the asserted patent, the burden of production shifts to the patentee to show that the patented invention was conceived and reduced to practice with reasonable diligence. *Stamps.com Inc. v. Edicia Inc.*, 437 Fed. App'x. 897, 908 (Fed. Cir. 2011); *Mahurkar* at 1576. In other words, when a patent's validity is challenged on the basis of a prior art reference, the patentee has the burden of production in antedating (or "swearing behind") the prior art reference. The party that challenges the validity of the asserted patent, however, bears the burden of persuasion by clear and convincing evidence on all issues relating to the status of the reference as prior art. *Id.*

We analyze the evidence presented by Thomson to determine whether Thomson has met its burden of production in showing that the asserted claims of the '063 patent were conceived and reduced to practice before Lowe (filed on May 10, 1996) and Miyazaki (filed on September 5, 1996). Thomson presents the testimony of inventor Dr. Ho, who along with Dr. Crawford,

devised the invention [

]

Thomson presents the following evidence as independent corroboration to Dr. Ho's
testimony: [

]

[

]

We find that [] discusses most of the limitations of the asserted claims. As found by the ALJ, [

], however, does not discuss assembling a second substrate to the first substrate on which the “smart spacers” are fabricated to form a complete display cell, as required by all of the asserted claims. [] also does not discuss whether there is a distinct “affixing layer” between the spacers and the substrate, as required by claims 1-4 and 8. Rather, [

]

With respect to claims 1-4 and 8, which require the presence of an “affixing layer,” we find that Thomson has not met its burden of production in showing that the claimed invention was conceived and reduced to practice before the filing dates of Lowe and Miyazaki. As noted above, [

]

[]. Thus, we agree with the ALJ that Lowe and Miyazaki are prior art to claims 1-4 and 8 of the '063 patent.

We also agree with the ALJ that Lowe and Miyazaki are not prior art to claims 11, 12, 14, 17, and 18. We find that Thomson has met its burden of production with respect to these claims.

[

]. Evaluating the evidence together under a “rule of reason,” we find that Thomson has presented sufficient independent corroboration to Dr. Ho’s testimony that the complete display cells were built, tested, and worked for their intended purpose. “The law does not impose an impossible standard of ‘independence’ on corroborative evidence by requiring that every point of a reduction to practice be corroborated by evidence having a source totally independent of the inventor; indeed, such a standard is the antithesis of the rule of reason.” *Knorr v. Pearson*, 671 F.2d 1368, 1374 (C.C.P.A. 1982). Thus, we find that

Thomson has met its burden of production in showing that claims 11, 12, 14, 17, and 18 of the '063 patent were conceived and reduced to practice before the filing dates of Lowe and Miyazaki, but has not met its burden with respect to claims 1-4 and 8.

We turn to whether respondents have met their ultimate burden of persuasion in establishing Lowe and Miyazaki as prior art with respect to claims 11, 12, 14, 17, and 18. Citing the Federal Circuit's decisions in *Hahn v. Wong*, AUO and CMI argue that documents originating from the inventor cannot serve as independent corroboration. We do not read the Court's case law on reduction-to-practice so narrowly. The Federal Circuit has found that documents made by the inventor can serve as independent corroboration to the inventor's subsequent testimony when the documents show an appreciation of the invention and that the invention was communicated to others. *See, e.g., Purdue Pharma L.P. v. Boehringer Ingelheim GMBH*, 237 F.3d 1359, 1366 (Fed. Cir. 2001) (finding sufficient corroboration in graphs and memos that showed an appreciation of the invention, its benefits, and its practice in the trials, indicating that the invention was communicated to others). [

] and thus shows an appreciation of

the invention. Moreover, the contents of [] were communicated to others.

[

] Although there is no clear author named for [

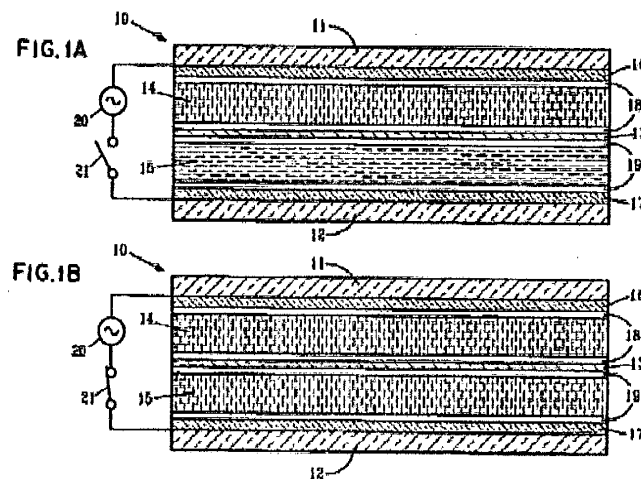
], respondents never asserted that the document originated entirely from the inventors themselves. Thus, we find that respondents have not met their burden, by clear and convincing

evidence, in establishing Lowe and Miyazaki as prior art with respect to claims 11, 12, 14, 17, and 18 of the '063 patent. As discussed above, respondents have, however, established Lowe and Miyazaki as prior art to claims 1-4 and 8.

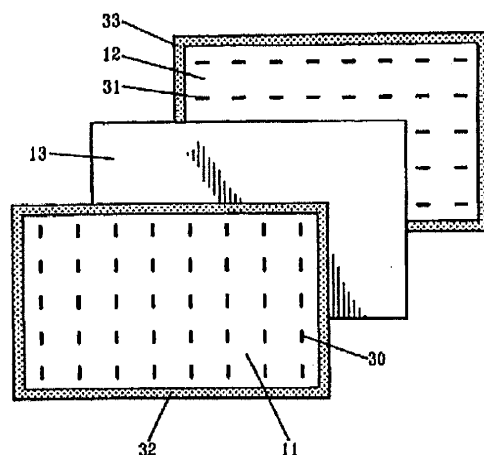
b. Anticipation of the Asserted Claims by Lowe and Miyazaki

We turn to whether claims 1-4 and 8 are anticipated by Lowe and Miyazaki and whether claims 11, 12, 14, 17, and 18 would be anticipated if Lowe and Miyazaki were prior art to those claims.

Lowe discloses a liquid crystal display cell having a stacked double-cell construction:



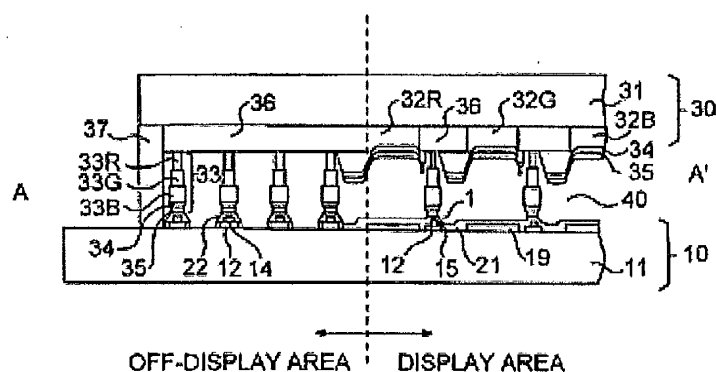
RX-16, Figs. 1A and 1B, Abstract. Thomson does not dispute that Lowe discloses a “transparent front substrate 11,” “a transparent or opaque rear substrate 12,” and an “intermediate substrate 13,” together forming two display cell compartments 14 and 15, each of which is filled with liquid crystal material. RX-16, 3:15-21. Thomson also does not dispute that Lowe discloses anisotropic spacers 30 and 31 that are deposited on the “interpixel” gaps of substrates 11 and 12, respectively, followed by rubbing of the substrates in the direction of the long axis of the spacers.



RX-16, Fig. 2, 4:49-5:9. The parties dispute, however, whether one of the substrates disclosed by Lowe is partitioned into “an active aperture area and a non-active area” as required by claims 1 and 11. Specifically, AUO argues that the rear substrate 12 in Lowe has a multiplicity of transparent pixel electrodes (*i.e.*, “active aperture areas”) separated by interpixel gaps (*i.e.*, non-active areas), in which the spacers are hidden.

The ALJ correctly found that Lowe does not disclose portioning the rear substrate 12 into active aperture areas and non-active areas. Rather, according to Lowe, the “inner surface of the rear substrate is coated with an electrode material 17 which can be transparent or opaque, reflective or light absorbing, depending on the particular liquid crystal effect employed.” *Id.* at 4:22-25. In other words, the rear substrate 12 is either all opaque or all transparent. Although AUO argues that the “interpixel gaps” in which the spacers are formed are the “non-active areas, AUO does not provide any explanation for its assertion. Moreover, as found by the ALJ, Thomson’s expert Dr. West testified that the interstitial gaps are not opaque non-active areas. CX-4304 at Q.350. Thus, we agree with the ALJ that Lowe does not disclose portioning the substrate into active-aperture and non-active areas, as required by the claims. Accordingly, we find that Lowe does not anticipate any of the asserted claims.

We turn to the Miyazaki patent, which discloses a liquid crystal display device having pillar-shaped spacers (shown as 33 in the figure below) for keeping a constant distance between the active matrix substrate (shown as 10 in the figure below) and the color filter substrate of a liquid crystal display cell (shown as 30 in the figure below):



RX-12, Abstract, Fig. 1; RX-158C at Q.423. Respondents challenge the ALJ's finding that Miyazaki does not disclose an "affixing layer" as required claims 1-4 and 8, and the ALJ's finding that Miyazaki does not disclose forming spacers using photolithography as required by claim 17. Thomson, however, does not challenge the ALJ's finding that Miyazaki discloses each of the limitations of claims 11, 12, 14, and 18 of the '063 patent and thus would anticipate if it were prior art.

With respect to claims 1-4 and 8, AUO argues that the ALJ erred in finding that Miyazaki does not disclose an "affixing layer," when Thomson admitted that the red color layer 32R is an "affixing layer" under the ALJ's construction of the term. We do not find AUO's argument persuasive. First, we point out that respondents themselves, who have the ultimate burden of persuasion, never presented their own expert testimony showing that Miyazaki discloses an "affixing layer" under the ALJ's construction of the term, which requires a stratum of material that attaches the spacing elements to a substrate, and which is distinct from said spacing

elements. Rather, Dr. Lowe testified that his anticipation opinion with respect to the “affixing layer” limitation “is based only on Complainants’ proposed construction.” RX-158C at Q.422.

Second, although Dr. West testified that the red color layer 32R functions as a material that attaches the spacer 33 to the substrate, he also testified that red color layer 32R covers one-third of all subpixels and is not substantially outside of the active aperture area. CX-4304C at Q.367-393. Thus, Dr. West’s testimony shows that the red color layer 32R cannot serve as the “affixing layer” of claim 1 even under the ALJ’s construction of the term, because it does not meet claim 1’s requirement that “the affixing layer cover[] at least a portion of the non-active area and remain[] substantially outside the active aperture area.” Accordingly, we agree with the ALJ that respondents have not met their burden by clear and convincing evidence that Miyazaki discloses each of the limitations of claims 1-4 and 8 and thus anticipates those claims. We also agree with the ALJ in light of Thomson’s failure to petition that Miyazaki would anticipate claims 11, 12, 14, and 18 if it were prior art to those claims.

We disagree, however, with the ALJ that Miyazaki does not disclose forming spacers using photolithography as recited in claim 17. The ALJ found that Miyazaki describes forming “stacked” spacers consisting of three color layers of red, blue, and green, and that such a process is much different from the process of photolithography. ID at 186, 188. The ALJ’s finding is contrary to expert testimony from both sides. Thomson’s expert Dr. West testified that the three color layers that form stacked spacers are each formed using a patterning process, which he refers to as “photolithography.” CX-4304C at Q.371, CX-4242C at Q.47. According to Dr. West, “the layer of red color material is *photolithographically* patterned,” and that after the layer of red material has been developed and baked, “a layer of green color material is layered over the entire substrate,” then “a mask is used to pattern the green color layer.” CX-4305C at Q.380

(emphasis added). Dr. West further testified that “a layer of blue material is layered over the entire substrate and patterned using a mask.” *Id.* Also, respondents’ expert Dr. Lowe testified that each of the three color layers are formed using photolithography. RX-158C at Q.433. According to Dr. Lowe, “a red-pigmented photosensitive resist [is] coated on the substrate and then patterned using *photolithography* and a mask to form the red color filters, including the bottom layer of the spacers.” *Id.* (emphasis added). Dr. Lowe further testified that “[t]he green and blue filters are formed in a similar manner, with a portion of the blue and green layers overlapping the red layer in order to form spacers constructed of the stacked color filter layers.” *Id.* Based on the foregoing evidence, we find that respondents have shown by clear and convincing evidence that Miyazaki discloses the limitation of claim 17 of forming spacers using photolithography and thus anticipates the claim.

In sum, we affirm the ALJ’s finding that Lowe and Miyazaki are prior art to claims 1-4 and 8 of the ’063 patent, and that respondents have not shown that either Lowe or Miyazaki anticipates claims 1-4 and 8. Also, we find that if Lowe and Miyazaki were prior art to claims 11, 12, 14, 17, and 18 of the ’063 patent, only Miyazaki would anticipate claims 11, 12, 14, 17, and 18, but respondents have not shown that Lowe or Miyazaki are prior art to those claims.

C. The ’556 Patent

1. The Invention of the ’556 Patent

The ’556 patent is directed to a method for manufacturing an active LCD. An active matrix LCD includes multiple thin-film transistors (“TFTs”), which are switches that control the orientation of liquid crystal, thereby controlling the amount of light passing through the LCD panel. The process for forming TFTs is called the “photo-exposure process” or “PEP,” which involves a series of steps including thin film deposition, photolithography or “masking,” and

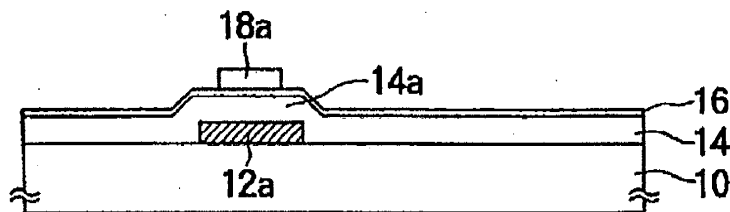
etching. The '556 patent provides a five-mask PEP process for forming a TFT. Dependent claim 3 is the only asserted claim for the '556 patent.

The Commission determined not to review the ALJ's finding that Thomson has not shown that the accused products infringe claim 3.

2. Obviousness of the '556 Patent over Takizawa in view of Possin

Before the ALJ, CMI and AUO argued that U.S. Patent No. 5,483,082 to Takizawa ("Takizawa"), either alone or in combination with U.S. Patent No. 5,041,888 to Possin ("Possin"), renders claim 3 obvious. The ALJ found that respondents have not shown by clear and convincing evidence that Takizawa, either alone or in combination with Possin, renders claim 3 obvious. *Id.* at 241. The ALJ found that Takizawa does not disclose the limitation "forming a plurality of etch stoppers over the plurality of gate electrodes using a second mask," because the testimony of Thomson's expert Dr. Parsons shows that Takizawa does not clearly disclose use of a single mask for forming etch stoppers. *Id.* at 242. The parties did not dispute that Possin does not disclose this limitation. Thus, the ALJ concluded that Takizawa, alone or in combination with Possin, fails to render claim 3 obvious. For the reasons detailed below, we agree with the ALJ's finding.

The Takizawa reference is directed to an improved structure for a TFT matrix device. Takizawa discloses forming a "channel protecting film" or "etch stoppers" 18a over gate electrodes 12a:



RX-45 at Fig. 5B. The ALJ found, however, Takizawa does not disclose, expressly or inherently, whether a mask is used to form the etch stoppers over the gate electrodes. ID at 242. The terms “photolithography” and “mask” are nowhere mentioned in the relevant passages describing formation of the etch stoppers 18a. RX-45 at 15:6-16. Rather, the particular passage in Takizawa simply states that the protecting film 18 is “etched off”:

Then, on the insulating film 14, the non-doped i-type a-Si layer 16, and the protecting film 18 of SiO₂ film or SiN film are formed in the stated order respectively in a 20 nm-thickness and a 150 nm-thickness by plasma CVD (FIGS. 4A to 4D).

Then, the protecting film 18 except a part thereof on the TFT channel unit is etched off using hydrofluoric acid buffer or others. That is, the protecting film 18 is left only above the gate electrode 12a of the TFT unit to form the channel protecting film 18a (FIGS. 5A to 5D).

Id. As pointed out by Thomson’s expert Dr. Parsons, “[w]hile the cited section discloses etching to form a channel protecting film, it does not disclose how many masks were used, if any, in this step.” CX-4306C at Q.100.

Respondents nevertheless argue that it is obvious to a person of ordinary skill in the art to use a single mask to form the etch stoppers over the gate electrodes as described in Takizawa. Respondents rely on the testimony of CMI’s expert Dr. Howard. Specifically, Dr. Howard testified that photolithography was at the time of Takizawa and through today, the predominant way to manufacture TFTs for arrays in LCDs. RX-159C at Q.232. According to Dr. Howard, although Takizawa does not mention using a mask to form “the protecting film 18a” over “gate electrode 12a,” a person of ordinary skill in the art would know that photolithography using a mask would be the most likely and efficient way to accomplish the task. *Id.* at 233. While Dr. Howard’s testimony is not largely disputed, it is nevertheless conclusory and does not explain

how or why a person of ordinary skill in the art would know to use a *single* mask versus multiple masks to form the etch stoppers disclosed in Takizawa. To support the legal conclusion of obviousness, “there must be some articulated reasoning with some rational underpinning.” *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006); *see also Innogenetics, N.V. v. Abbott Labs.*, 512 F.3d 1363, 1373 (Fed. Cir. 2008) (finding that use of the stock phrase “to one of skill in the art it would have been obvious to perform” without more was insufficient to demonstrate obviousness).

Respondents also rely on the admission of Thomson’s expert Dr. Parsons. Specifically, CMI and AUO argue that Dr. Parsons admitted on cross-examination that the cited passage in Takizawa discloses to a person of ordinary skill in the art that *one* or more masks can be used.

Q. But you agree, sir, that based on the description of the etch stopper in the '082 patent, one of ordinary skill would also understand that a mask could be used to form the etch stopper, correct?

A. *One of ordinary skill could presume that one or masks was used.*

* * * * *

Q. Okay. And is it also your understanding, sir, that such persons reading the '082 back in the 1995 time frame would understand that either one or more than one mask could also be used to form etch stoppers?

A. *Yes, I think that's true.*

Tr. at 1621:22-1622:3; 1624:14-19. We find that this particular portion of Dr. Parsons’ testimony, in which he voices presumptions and speculations regarding the understanding of those of skill in the art, is ambiguous at best and cannot provide the missing evidence requisite for a finding of obviousness. *See Star Scientific, Inc. v. R.J. Reynolds Tobacco Co.*, 655 F.3d 1364, 1376 (Fed. Cir. 2011) (“[S]peculative and tentative disclosure of what ‘might’ or ‘may’ [happen] does not sufficiently direct or instruct one of skill in [the] art.”). Dr. Parsons made

these brief statements as part of his overall testimony that one of ordinary skill in the art would understand Takizawa to disclose use of *more than one* mask to form the etch stoppers. Tr. at 1623:14-1624:6, 1698:1-22.

Moreover, Thomson has presented evidence suggesting that Takizawa actually teaches away from using a single mask. As pointed out by Dr. Parsons, the inventors of the Takizawa patent referred to the use of the resist pattern as a mask for the formation of the gate electrode, drain and source electrodes, and the passivation layer, but did not disclose the use of the resist pattern as a mask to form the channel protecting film. *Compare* RX-45, 14:56-63, 15:21-25, 15:46-52 *with* RX-45, 15:11-16. Dr. Parsons testified that this distinction indicates that a single mask was not necessarily used to form the etch stoppers in Takizawa. CX-4306C at Q.104-05.

The ALJ did not err in finding claim 3 not obvious. Respondents bear the burden of proving obviousness by clear and convincing evidence. *See Scanner Techs. Corp. v. ICOS Vision Sys. Corp. N.V.*, 528 F.3d 1365, 1380 (Fed. Cir. 2008) (“We recognize that it is ICOS’s burden to prove invalidity by clear and convincing evidence, and that that burden of proof never shifts to the patentee to prove validity.”). Further, respondents’ evidence must give the fact finder “an abiding conviction that the truth of [the] factual contentions are highly probable.” *Procter & Gamble Co. v. Teva Pharm. USA Inc.*, 566 F.3d 989, 994 (Fed. Cir. 2009). Here, neither the brief testimony of Dr. Howard, nor the ambiguous admission of Dr. Parsons, clearly show that a person of ordinary skill in the art would know to use a single mask to form the etch stoppers in Takizawa. Accordingly, the Commission affirms the ALJ’s finding of non-obviousness of claim 3 of the ’556 patent.

D. The '674 Patent

1. The Invention of the '674 Patent

The '674 patent is directed to an improved structure for the array circuitry of thin-film transistor LCDs. As shown in Figures 1 and 2 below, the improved structure includes (1) a TFT consisting of contact leads 22 and 24 and channel 26, (2) a capacitor electrode 30 and conductive lines 20 formed from a single layer of highly conductive metal (shown in back-slashes), and (3) a conductive element 76:

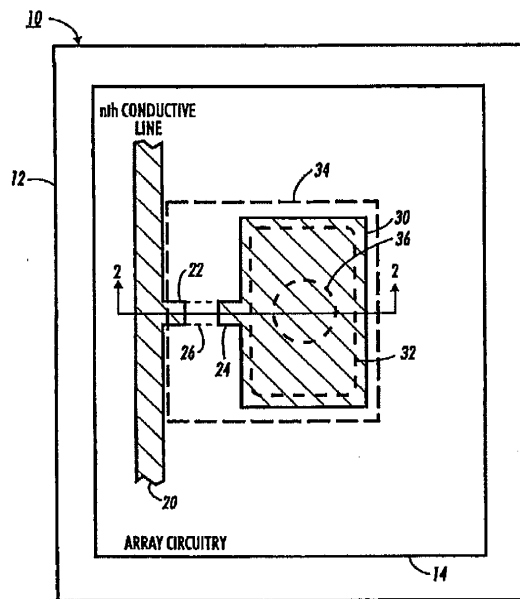


FIG. 1

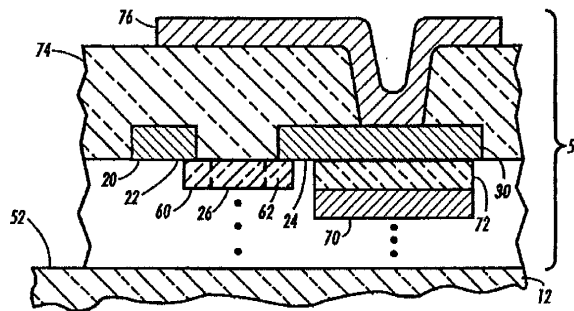


FIG. 2

Contact lead 24 and capacitor electrode 30 are electrically connected, and the capacitor electrode 30 has an exposed part due to an opening 36 in the insulating layer 74 over it. Thus, the overlying conductive element 76 contacts the upper electrode 30 in the exposed part, providing an electrical connection to the switching element via contact lead 24.

The asserted claims are claims 1, 7-9, 11, 13, 14, and 16-18. The Commission determined not to review the ALJ's finding that the asserted claims are infringed by the CMI

accused products including the “Type 2 Array Circuitry” and any Qisda or BenQ accused product incorporating these CMI accused products.

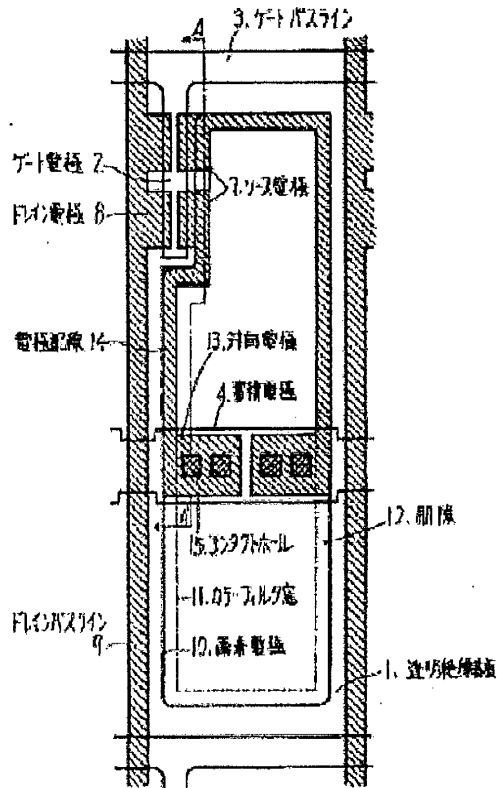
2. Anticipation and Obviousness of the ’674 Patent in view of Fujitsu

Before the ALJ, CMI argued that Japanese Published Application No. JP 06-130415A (“Fujitsu”) anticipates claims 1, 7, 8, 14, 16, 17, and 18 of the ’674 patent. CMI also argued that dependent claims 9, 11, and 13 are obvious over the Fujitsu reference in combination with the knowledge of one of ordinary skill in the art. The ALJ found that CMI has not shown by clear and convincing evidence that any of these claims are anticipated. Specifically, the ALJ found that Fujitsu does not disclose the limitations “the second patterned conductive layer including the N conductive lines and the first and second contact leads and the second electrode of each unit of cell circuitry” of independent claims 1 and “the second patterned conductive layer including the N data lines and the first and second contact leads and the second electrode of each unit of cell circuitry” of claim 16. *Id.* at 254. The ALJ also found that Fujitsu does not disclose the limitation “the second contact lead and the second electrode [are] joined in the second patterned conductive layer.”⁵ *Id.* at 257. The ALJ found that the only explicit disclosure in Fujitsu on whether the recited elements are in a single layer is the diagram of Figure 1 depicting the bus lines, drain and source electrodes, and the opposing electrode in the same shading. *Id.* at 256. The ALJ found that respondents have not met their burden of clear and convincing evidence with this single ambiguous disclosure. The ALJ found that because Fujitsu does not disclose each of the limitations of independent claims 1 or 16, it cannot render obvious dependent claims 9, 11, and 13. *Id.* For the reasons detailed below, we disagree with the ALJ’s finding.

⁵ The ALJ construed the limitation to mean “the second contact lead and the second electrode are electrically connected in the second patterned conductive layer,” which was not challenged by the parties.

Claims 1 and 16 require the “conductive lines” or “data lines” to be in the same layer as the first and second “contact leads” of the “switching element” and the “second electrode” of the “capacitive element.” ’674 patent, claim 1 (“the second patterned conductive layer including the N conductive lines and the first and second contact leads and the second electrode of each unit of cell circuitry”); claim 16 (“the second patterned conductive layer including the N data lines and the first and second contact leads and the second electrode of each unit of cell circuitry”).

Claims 1 and 16 also require that the second contact lead and the second electrode are joined in the same layer. *Id* (“the second contact lead and the second electrode [are] joined in the second patterned conductive layer.”). The parties agree that the “drain bus line” or “data bus line” (labeled 9 in Figure 1 below) of Fujitsu serve as the recited “conductive line” or “data line,” that the source electrode and drain electrode (labeled 7 and 8 in Figure 1 below) serve as the first and second “contact leads” of the “switching element,” and that the opposing electrode (labeled 13 in Figure 1 below) serves as the “second electrode” of the “capacitive element”:



1. Transparent insulating substrate
2. Gate electrode
3. Gate bus line
4. Storage electrode
7. Source electrode
8. Drain electrode
9. Drain bus line
10. Pixel electrode
11. Color filter window
12. Gap
13. Opposing electrode
14. Electrode wiring
15. Contact hole

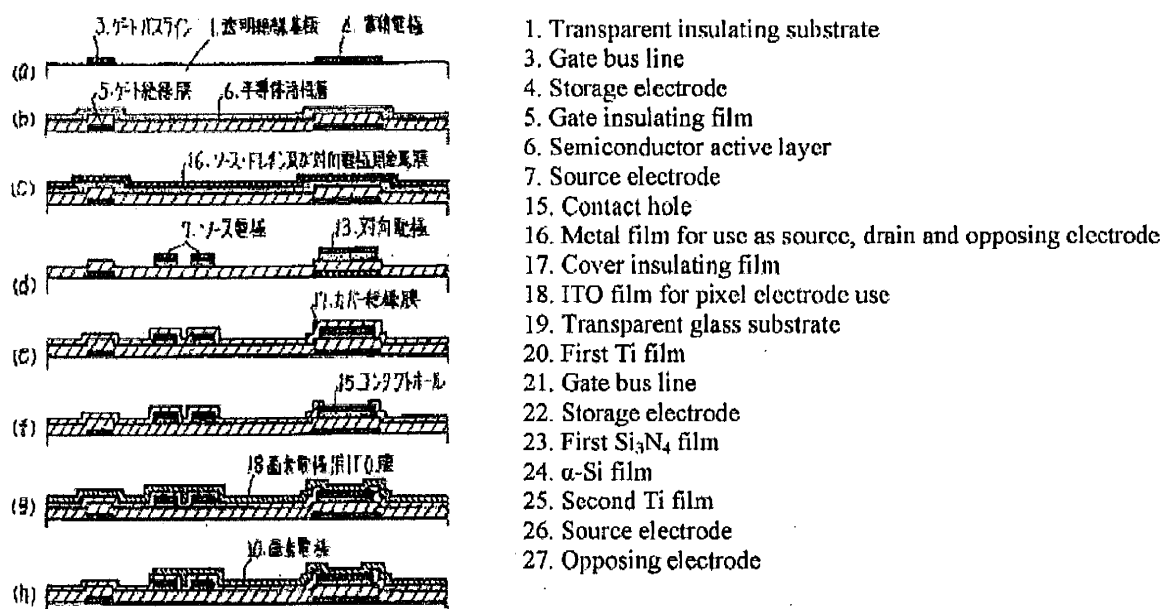
RX-325 at CMI-741-600702. The parties dispute, however, whether one of ordinary skill in the art would understand Fujitsu to also disclose that the data bus line (labeled 9 above) is in the same layer as the source and drain electrodes (labeled 7 and 8 above) and the opposing electrode (labeled 13 above).

We find that when Figure 1 of Fujitsu is viewed in the context of the full disclosure, including Figure 3 and the supporting text, one of ordinary skill in the art would understand that data bus line 9 is formed from the same patterned conductive layer as are the source electrode 7, drain electrode 8, opposing electrode 13, and electrode wiring 14. First, one of ordinary skill in the art would understand that Figures 1 and 2 depict bus line 9, source electrode 7, drain electrode 8, opposing electrode 13, and electrode wiring 14 in the continuous and consistent shading. Dr. Hatalis testified that when elements such as the drain bus lines and drain electrodes are disclosed as being connected to one another, the continuous and consistent shading between

those elements in Figure 1 indicates to one of ordinary skill in the art that they are connected together in the same layer. RX-393C at Q.143.

Figure 3 shows a cross-sectional view of the Fujitsu invention taken along the A-A' line of Figure 1 (and is consistent with Figure 4 which shows a sectional schematic view of the process sequence):

FIG. 3
Description of principles of present invention



RX-325 at Fig. 3. As shown in Figures 3(c) and (d) and the line 16 of the accompanying list, source electrode 7, drain electrode 8, and opposing electrode 13 are formed from the same metal film 16. According to Fujitsu, this metal film 16 (referred to as the second Ti film 25 in Figure 4(d)) is photolithographically patterned to create source electrode 7 (referred to as source electrode 26 in Figure 4(e)), drain electrode 8 (referred to as source electrode 27 in Figure 4(e)), and the opposing electrode 13 (referred to as opposing electrode 27 in Figure 4(e)). *Id.* at ¶ 21. Moreover, with respect to Figure 4(d), Fujitsu discloses that the source electrode 26 and the opposing electrode 27 are connected by electrode wiring 14. *Id.* at ¶ 21.

The only question remaining is whether data bus line 9 is formed from the same metal film 16 (also referred to as Ti film 25) as the drain and source electrodes and the opposing electrode. We acknowledge that Figure 3 does not show data bus line 9 and electrode wiring 14. This is expected because Figure 3 is a cross-sectional view of Figure 1 along the A-A' line, which does not pass through the data bus line 9. We find, however, that the only logical conclusion as to how data bus line 9 is connected to drain electrode 8 is that they are both formed from metal layer 16. Figure 3 and the remaining descriptions in Fujitsu describe a manufacturing process involving three metal layers. Specifically, Fujitsu discloses a first metal layer referred to as Ti film 20 that forms the gate electrode and gate bus lines; a second metal layer referred to as metal film 16 or 25 which contains the source and drain electrodes, opposing electrode, and electrode wiring; and a third metal layer 30 containing the ITO electrode. *Id.* at Fig. 3-4, ¶ 19, 23; RX-393C at Q.145. Thomson has never asserted that the gate metal layer or the ITO layer can be used to form data bus line 9. Thus, the only possible metal layer from which data bus line 9 could be formed is the second metal layer containing the drain electrode 8.

Paragraph 15 of Fujitsu, which describes the sequence in which the components of the Fujitsu invention are deposited, confirms our conclusion:

To explain, in a method for manufacturing a TFT matrix which uses as switching devices bottom gate staggered thin film transistors wherein, as shown in FIG. 1 through FIG. 3, deposited on a transparent insulating substrate 1 in the following sequence are, at least, a gate electrode 2, gate bus line 3 that connects a plurality of said gate electrodes 2, storage electrode 4, gate insulating film 5, semiconductor active layer 6, source electrode 7, drain electrode 8, drain bus line 9 that connects a plurality of said drain electrodes 8 and a pixel electrode 10, the object of the present invention is achieved by using the gap 12 between the pixel electrode 10 and a color filter window 11 that is opened inside said pixel electrode 10 region to form an electrode wiring 14 that connects said source electrode 7 and said opposing electrode 13.

Id. at ¶ 15. Paragraph 15 states, *inter alia*, that “source electrode 7, drain electrode 8, drain bus line 9 that connects a plurality of said drain electrodes 8” are formed after gate insulating film 5 and semiconductive active layer 6.⁶ This disclosure indicates that drain bus line 9 is formed from the same metal layer as the drain electrode 8. Moreover, paragraph 15 mentions only insulating film 5 and makes no mention of another insulating film that is deposited after drain electrode 8 is formed, which would be necessary if drain bus line 9 is formed from a separate layer of metal than drain electrode 8, and connected via a contact hole, as Thomson’s argument suggests.

As explained by CMI’s expert Dr. Hatalis, if the drain bus lines and electrode wiring were in different layers, more complex structures would be required, such as contact holes. RX-393C at Q.146. In other words, if drain bus line 9 is formed from a different metal layer than drain electrode 8, there must be an opening through the insulation layer between the two

⁶ We note the Abstract in Fujitsu provides further support that the source electrode, drain electrode, drain bus line, electrode wiring and opposing electrode are in the same layer. Specifically, the Abstract states: “In a method for manufacturing matrix semiconductor devices . . . wherein deposited on a transparent insulating substrate 1, in succession, are: . . . source electrode 7, drain electrode 8, drain bus line 9 that connects a plurality of drain electrodes 8 and pixel electrode 10, an electrode wiring 14 that connects the source electrode 7 and the opposing electrode 13...”. *See also* claim 1 in Fujitsu.

different metal layers. No such contact holes are disclosed, even though contact holes are disclosed when connecting the other two metal layers, *i.e.*, the gate layer and the ITO layer. *Id.* Furthermore, it was the intent of the Fujitsu reference to eliminate any contact holes on top of the source metal. RX-325 at ¶ 14. Thomson's argument that the drain bus lines and electrode wiring are not in the same layer runs counter to the stated goal of the Fujitsu reference, which is to minimize cost and complexity in the device structures by using fast and easy photolithographic techniques top of the source metal. *Id.* at ¶ 3.

Thomson's expert Dr. Parsons testified that Fujitsu "lists" certain items in the second titanium layer and does not mention the drain bus lines, and that the continuous and consistent shading does not indicate that all of the elements are patterned together in the same layer because Figure 1 is an "overhead view." CX-4307C at Q.57, 60. However, Dr. Parsons does not rebut the substantial textual evidence of Fujitsu relied on by CMI's expert Dr. Hatalis. Only one of Dr. Parsons' statements cited by the ALJ is even related to the question of whether one of ordinary skill in the art would understand the Fujitsu reference as disclosing the electrode wiring and drain bus lines in the second titanium layer. Dr. Parsons states that at the time of the '674 patent, it was "advantageous to form different elements out of different metal layers." CX-4307C at Q.60. Dr. Parsons does not explain, however, why it was advantageous, does not cite to any evidence to support such as statement, and does not testify how such a statement even relates to the Fujitsu reference. We find that this testimony is contrary to the express teachings of the Fujitsu reference and the '674 patent, which both disclose only three metal layers for the purpose of manufacturing devices that "can be formed using photolithographic techniques that are fast and easy to operate." RX-325 at ¶ 003; *see also* JX-2 at 1:31-33 ("[t]he new structure's simplicity and ease of production result from forming several different features in a single layer

of highly conductive metal”). Thus, we find that respondents have shown by clear and convincing evidence that Fujitsu discloses the limitation of the asserted claims requiring that the “conductive lines” or “data lines” be in the same layer as the first and second “contact leads” of the “switching element” and the “second electrode” of the “capacitive element,” and that the second contact lead is joined to the second electrode in the same layer.

Before the ALJ, Thomson also argued that Fujitsu does not disclose the “highly conductive metal” limitation of claims 1 and 16. ID at 252. Thomson, however, does not challenge the ALJ’s finding that respondents have shown by clear and convincing evidence that this limitation is disclosed by Fujitsu. *Id.* at 253-54. Because Thomson does not dispute that Fujitsu discloses the remaining limitations of claims 1, 7, 8, 14, 16, 17, and 18 of the ’674 patent, we find that respondents have shown by clear and convincing evidence that Fujitsu anticipates those claims.

We also find that respondents have shown by clear and convincing evidence that dependent claims 9, 11 and 13 are obvious in light of the Fujitsu reference. Dr. Hatalis testified extensively regarding the knowledge of one of ordinary skill in the art, why one would be motivated to combine the Fujitsu reference with the teachings of several other prior art references, and the likelihood of success in modifying the Fujitsu reference. RX-393 at Q.161-186. For each of claims 9, 11, and 13, Dr. Hatalis analyzed several prior art references that could have been be combined with the Fujitsu reference to render the claims obvious. *Id.*

Regarding claim 9, which recites a second patterned conductive layer of aluminum, Dr. Hatalis testified that aluminum was a well-known metal commonly used in fabricating electrodes and conductive lines since the early 1980s. RX-340; RX-393C at Q.165. Dr. Hatalis testified that aluminum had certain advantages over titanium in the manufacture of TFT-LCDs, such as

the manufacture of larger displays with a higher resolution. RX-338; RX-393C at Q.165. Thus, if one wanted to manufacture a larger display using the structure disclosed in Fujitsu, it would have been natural to select aluminum for the highly conductive metal.

Regarding claim 11, which recites a second patterned conductive layer with sublayers of a highly conductive metal and a refractory metal, Dr. Hatalis testified that the use of a multi-layer metallization using a refractory metal has been well known since the mid-1980s. RX-393C at Q.174; *see also* JX-53C, Yao 6/13/2011 Dep. Tr. 179:23-180:0. Dr. Hatalis testified that the use of a refractory metal sublayer along with a highly conductive metal would have solved the problem of hillocks, which would allow for the manufacture of larger panels at a decreased cost, as taught by the Fujitsu reference. RX-393C at Q.175.

Regarding claim 13 which recites an opening in the second insulating layer with a tapered profile, Dr. Hatalis testified that the tapered profile provided better step coverage at the edge of a tapered via hole. RX-393C at Q.183. In fact, Dr. Hatalis testified that the tapered via hole was a natural result from certain wet or plasma etching processes which were then in use. *Id.*; *see also* JX-53C, Yao 6/13/2011 Dep. Tr. 181:23-182:24. Use of a tapered via hole reduces breakage and results in better contact between the pixel electrode and the second electrode, thereby increasing the manufacturability of the panels and lowering the cost. RX-393C at Q.183.

In rebuttal to respondents' evidence of the obviousness of claims 9, 11, and 13, Thomson relies on Dr. Parsons' conclusory testimony that one of ordinary skill in the art would not have combined elements from two different array circuitry disclosures. *See*, CX-4307C at Q.99, 101, 115, 117, 157, 159, 171, and 189. We disagree with Thomson's argument. "It is not necessary that the inventions of the references be physically combinable to render obvious the invention under review. *In re Sneed*, 710 F.2d 1544, 1550 (Fed. Cir. 1983). The test for obviousness is not

whether the features of a secondary reference may be bodily incorporated into the structure of the primary reference. *In re Keller*, 642 F.2d 413, 424 (CCPA 1981). Rather, the test is what the combined teachings of the references would have suggested to those of ordinary skill in the art. *Id.*; see also *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 418-420 (2007).

Dr. Parsons does not dispute any of CMI's evidence regarding the scope and content of the prior art, the level of ordinary skill, or the similarities between claims 9, 11, and 13 and the prior art. See *Scanner Techs. Corp v. ICOS Vision Sys. Corp N.V.*, 528 F.3d 1365, 1379 (Fed. Cir. 2008) (citing *Graham v. John Deere Co.*, 383 U.S. 1, 17 (1966)). Moreover, Thomson has failed to adduce any reliable secondary evidence of non-obviousness to rebut respondents' prima facie case. *Ashland Oil, Inc. v. Delta Resins & Refractories, Inc.*, 776 F.2d 281, 291-92 (Fed. Cir. 1985). The mere statement that prior art references disclose different array circuitry designs is insufficient evidence to overcome a showing of obviousness. Thus, the clear and convincing evidence demonstrates that the combinations of prior art references described in Dr. Hatalis' testimony disclose all limitations of asserted claims 9, 11, and 13 of the '674 patent.

E. The '941 Patent

1. The Invention of the '941 Patent

The '941 patent, entitled "Method and Device for the Controlling of Matrix Displays," issued on September 19, 2000 to Hirtz. '941 patent. The invention relates to the control of signal processing in an active matrix display consisting of M x N pixels (where M is the number of pixels per line and N is the number of lines). *Id.* at 1:17-19. The control of pixels in an active matrix display is generally carried out line by line, *i.e.*, a video signal containing the information of a picture line is scanned N times. More specifically, the '941 patent is directed to utilizing signals for cathode ray tubes, which include lines or portions thereof that do not contain data

(also known as horizontal blanking, vertical blanking and overwrite periods), for other matrix displays that do not require the same periods having no picture data.

The object of the invention is to reduce the ratio f_t/Z_a , where Z_a represents the number of lines to be displayed and f_t is the clock signal for controlling signal processing and depends on the number M of pixels to be presented per line. *Id.* at 1:30-34; 2:35-38. This reduction in clock frequency is accomplished by expanding the time interval available for executing signal processing algorithms by taking advantage of time periods in which a video signal contains no picture information. *Id.* at 2:38-41.

Figure 2, reproduced below, illustrates the vertical and horizontal blanking periods. In Figure 2, there are Z lines and the duration of each line is T_z . *Id.* at 3:4-10. Z_a is the number of active lines and T_{za} is the active time of each line. *Id.* at 3:1-7.

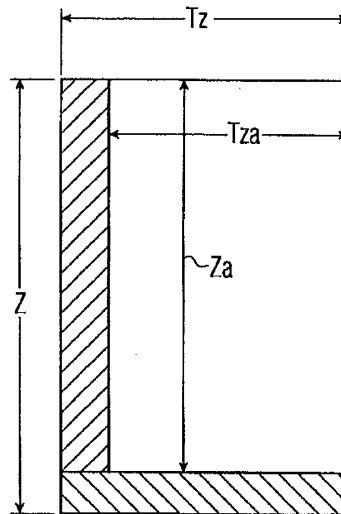


FIG. 2

The first embodiment of the invention is illustrated in Figure 3, reproduced below.

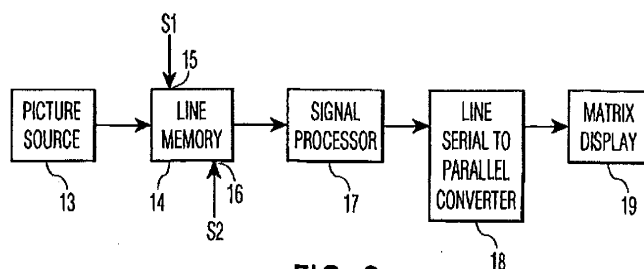


FIG. 3

The picture source processes the signal and outputs the video line by line into memory 14. *Id.* at 3:36-41. S1 is the first input control signal that controls storing the line data. *Id.* at 3:44-46. S2 is the second line control signal that controls reading-out of video line data. *Id.* at 3:46-48. The signal output from line memory is processed by the signal processor and is sent to the line serial to parallel converter. *Id.* at 3:49-52. The output for the line serial to parallel converter results in the video display. *Id.* at 52-53. Unlike a cathode ray tube, the matrix display used in the invention does not need to adjust for “the feedback [for] a beam ... triggering matrix displays” and therefore, the horizontal blanking period, $T_z - T_{za}$, of Figure 2 can be used for executing signal processing algorithms and for triggering the matrix display. *Id.* at 3:54-62. A line initialization time period T_i has to be taken into account when determining the additional available time. *Id.* at 3:64-65. The time for reading out the display is therefore $T_{za}^* \leq T_z - T_i$. *Id.* at 65-4:1. In this embodiment of the invention, the number of picture elements per line M is equal to the number of picture elements per line to be presented M' . *Id.* at 4:3-6. The clock frequency of S1 equals $f_t = M/T_{za}$. *Id.* at 4:7-10. The clock frequency of S2 equals $f_t' = M/T_{za}^*$. *Id.* at 4:12-15. Therefore, the frequency for triggering the display for the same number of lines to be displayed (Z_a) is less than in the prior art. *See id.* at 4:23-25.

Figure 4 illustrates a second embodiment of the invention. The video signal is sent to image memory 20 and S1' controls the storage of the video image data in memory and S2'

controls the reading out of the video image data. In this embodiment both the horizontal and vertical blanking periods are utilized. *Id.* at 4:42-45.

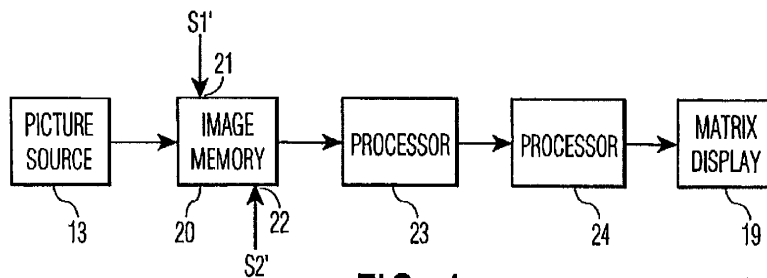


FIG. 4

As in the first embodiment, the clock frequency of S1' is $ft=M/Tza$. *Id.* at 4:53. The line frequency of S1' is $fz=Z/Tb$. *Id.* at 4:55-57. Tb is the picture duration and $Tb = 1/fB$, where fB is the picture frequency. *Id.* The readout of the picture, in this embodiment, includes the vertical blanking period as controlled by S2'. *Id.* at 4:61-63. The time available to display Za lines is designated Tba' and is determined by $Tba \leq Tba' \leq Tb$. *Id.* at 4:64-67. This increases the amount of time to display Za lines, which reduces the line frequency. *Id.* at 5:1-11. The time to display the Za lines results in a prolonged Tz period delineated as Tz'. *Id.* Tz' is determined by $Tz'=Tba'/Za$. *Id.* The line frequency is reduced as illustrated by the equation $fz'=1/Tz'$. *Id.* The resulting reduced clock frequency is equal to $ft'=M/Tza'$. *Id.* at 5:17-20. Figure 5 (reproduced below) illustrates the picture build up in accordance with the second embodiment of the invention. This illustrates that most of the time is now utilized for signal processing. *Id.* at 5:27-31. The time available for signal processing in this embodiment increases over the prior art systems.

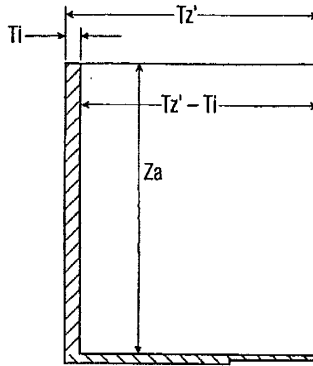


FIG. 5

In the final embodiment described, the video signals are displayed over a larger number of lines, which results in a “vertical upward interpolation [of] the number of lines Za .” *Id.* at 6:17-24.

Thomson asserts independent claims 1 and 4.

2. Claim Construction of the Term “Second Rate” “Determined By” and the ALJ’s Finding of Non-infringement Based on this Term

a. Construction of “Second Rate” “Determined By” Limitations

Thomson challenges the ALJ’s construction of “determined by.” *Thom. Pet.* at 89. Thomson argues that the term “determined by” means “based on,” and asserts that the ALJ re-wrote the claim language by inserting a mathematical formula into the claims when the claims do not call for one. *Thom. Resp.* at 44. MStar argues that “a second rate determined by . . .” and “a second rate which is determined by . . .” means “a frequency equal to the density of picture information to be displayed divided by the time available for display comprising active and inactive parts.” *See MStar Resp.* at 2. MStar contends that Thomson’s construction is so broad that “it would provide no meaningful limitation on the scope of the claim.” *Id.* at 7. Realtek asserts that the ALJ’s construction, which is the construction also advocated by MStar, is the correct construction. *Realtek Resp.* at 4.

The ALJ adopted MStar's proposed construction. *Id.* at 111-13. The ALJ concluded that "[t]he claims make clear that the second rate is 'determined by' two factors: (1) the density of picture information to be displayed; and (2) the time available for display comprising active and inactive parts." *Id.* at 111. However, he found that the specification "provides only one method for determining the second rate," which is "dividing the number of pixel elements by the available time." *Id.* at 111-12. According to the ALJ, "the absurd result that the second rate limitation would be met no matter what mathematical operation is performed, as long as it includes the density of picture information to be displayed and the time available for display comprising active and inactive parts," cannot be correct. *Id.* at 112. The ALJ concluded that "[e]ven if the second rate could be determined in a way that is different than dividing the density of picture information to be displayed by the time available for display comprising active and inactive parts, the intrinsic evidence provides no disclosure of that, and Thomson offers no evidence that such a determination would be within the knowledge of one of ordinary skill in the art." *Id.* at 112. The Commission adopts the ALJ's construction.

We first turn to the language of claims 1 and 4. Both claims recite: "a second rate" "*determined by* the density of picture information to be displayed" and "the time available for" "display." While the language of the claims requires a "rate," the claim language does not explicitly teach how the time available for display and the density of picture information are used to calculate that rate. Based on the claim language the only way to calculate a rate from these factors is to divide the density of picture information must by the available time.

The first column of the specification teaches that "[t]he clock frequency *f* for triggering signal processing devices, which ... control the matrix display, depends on a number *M*' of picture elements to be presented per line." '941 patent at 1:30-34. It further teaches that the

resulting formula is $ft=M'/Tza$, where Tza is the duration of the video signal to be presented within one line. *Id.* at 1:35-38. The first and second embodiments also teach a similar formula. In the first embodiment, the second line control signal $S2$ has a reduced frequency that is calculated by the formula $ft'=M/Tza^*$, where M is the number of picture elements per line to be presented and Tza^* is the time available for reading-out, processing and displaying a line. *Id.* at 3:65-4:15. For the second embodiment, the reduced readout frequency is calculated by the formula $ft'=M/Tza'$, where M is the number of pixels per line and Tza' is the time available to present the active lines (Za). *See id.* at 4:64-5:24. Each of these formulas includes the number of pixels divided by a processing time. Although none of these formulas teach the exact relationship adopted by the ALJ, the same relationship is found in the ALJ's construction when considering the screen in its entirety. The ALJ's construction requires the density of picture information for the whole screen divided by the time available to display the picture information. The formulas discussed in the specification similarly teach a number of pixels divided by the time to display the pixels. Notably, the specification does not teach any other methods for computing the second rate.

Thomson's construction that "determined by" means "based on" allows for any conceivable equation involving the density of picture information and the time available of the claims to be used to calculate the second rate. As noted above, in general terms, the formulas of the specification only teach one way of calculating this, where the numerator relates to a number of pixels that is divided by a denominator that relates to the time available to display those pixels as the methods for calculating the second rate. Thomson's construction goes so far beyond the formulas provided in the specification such that it provides no meaningful limitation on the scope of the claims. Any equation that includes the two variables or includes variables "based on" the

two variables of the claim would meet this claim limitation. Thomson also cites to nothing in the intrinsic record to support its construction. Accordingly, the Commission adopts the ALJ's construction for these limitations.

b. Infringement of the "Second Rate" Limitation

The ALJ found that both MStar and Realtek⁷ products do not meet the "second rate" limitation of the asserted claims and therefore, found that there was no infringement. ID at 401-403, 406-07. Claim 1 is a method claim and claim 4 is an apparatus claim. The parties did not challenge the ALJ's finding of non-infringement under his construction of the "second rate" "determined by" limitations. On review, the Commission affirms the ALJ's finding of non-infringement.

In addition to finding that the accused scaler chips do not meet the "second rate" "determined by" limitation, the ALJ also determined that Thomson failed to provide evidence of direct infringement of claim 1 because the evidence for both MStar and Realtek was directed to the importation, sale for importation, or sale after importation of an apparatus that allegedly performs the claimed process. These actions, the ALJ found, do not constitute direct infringement of claim 1, and therefore, claim 1 is not met by MStar or Realtek. ID at 401, 406. The Commission did not review this finding of the ALJ.

In addition, the Commission concludes that even under Thomson's construction of the "second rate" "determined by" limitations of asserted claims 1 and 4, Respondents MStar and Realtek and Respondents CMU, Qisda, and BenQ, which manufacture products including MStar and Reaktek scaler chips do not infringe.

⁷ The ALJ also determined that Respondents CMU, Qisda and BenQ, which manufacture products including MStar and Reaktek scaler chips, do not infringe claims 1 and 4. ID at 407.

1) *MStar*

MStar explains that there is no dispute about the operation of the Accused MStar Chips or the Accused Monitors. MStar Resp. at 8. Thomson alleges that the calculation of the clock rate, using the following formulas, is the second rate of claims 1 and 4:

[
]

Thom. Resp. at 45. According to MStar, [] is the amount of [

]. MStar Resp. at 9. MStar further explains that the [

] *Id.* The [] variable is the number of [

] *Id.* The [] variable is the number of

[

] *Id.* Finally, the [] is the

number [] *Id.*

Thomson argues that the product of [] and [] is based on the density of pixel information recited in claims 1 and 4. Thom. Resp. at 45-46. MStar argues that it does not infringe the second rate limitation under Thomson's construction because the alleged second rate formula does not include the density of picture information to be displayed. Instead, MStar argues that the formula relied on by Thomson to prove infringement includes a different and larger number that includes [

] MStar Resp. at 9-10. The parties agree that the density of the picture

information to be displayed is the number of pixels on the screen. *See e.g.*, CX-4243C, Q.114; MStar Resp. at 9. During the hearing, Thomson’s expert conceded that the product of [] and [] is not the “density of the picture information to be displayed.” Tr. at 636:1-639:21. MStar therefore argued that Thomson does not and cannot explain how the second rate formula can be based on the number of pixels on the screen when that number is not a factor in the formula. MStar Rep. Thomson Pet. at 11. We agree with MStar.

As noted above, Thomson argues that this calculation is based on the density of the picture information even though the product does not include the variables necessary to determine the density of the picture information. Thomson argues that because the horizontal lines include inactive pixels on the line does not preclude a finding of infringement because “a quantity can be based on one factor without precluding other factors.” Thom. Resp. at 45 n.46. While we agree with this general principle, Thomson does not appear to rely on the claimed factor (*i.e.*, density of picture information) anywhere in the asserted formula. Instead, the product relied upon by Thomson is a larger area than the pixels on the screen. The Commission concludes that MStar and the Respondents manufacture products including the MStar scaler chips that do not infringe this limitation even under Thomson’s construction of the term.

2) *Realtek*

Thomson argues that the accused Realtek products use the following DCLK formulas, depending on whether or not a frame buffer is used:

$$\left[\frac{\text{DCLK}}{\text{Number of Pixels}} \right]$$

$$\left[\frac{\text{DCLK}}{\text{Number of Pixels}} \right]$$

Thomson alleges that the DCLK is based on the density of picture information to be displayed by relying on the product of []

[

] Thom. Resp. at 46. Thomson relies on the product of

[

] for the buffered products. *Id.*

However, Realtek argues that products relied upon by Thomson include a number greater than the number of pixels in each horizontal line because they include non-active or blanking pixels.⁸ Realtek Resp. at 13. Therefore, [] cannot be used to calculate the density because it includes the blanking periods. Thomson argues that the fact that

[

] includes additional inactive pixels does not change the fact that it is determined by the number of picture elements to be controlled. Thom. Resp. at 46 n.47.

Realtek argues that Thomson has failed to provide any evidence that [] is a combination of the pixels in a horizontal line and the number of blanking pixels in the line such that it would be “based on” the density of picture information. Realtek Resp. at 6. As with the MStar formulas, Thomson does not appear to even rely on the claimed factor (i.e., density of picture information) anywhere in the asserted formula. Instead, the product relied upon by Thomson includes a larger number of horizontal pixels. For the reasons noted above with respect to MStar, we also conclude that Realtek does not meet the “second rate” “determined by” limitations even under Thomson’s construction of the term.

3. Whether Claim 4 Requires an “Input Video Signal” for a Finding of Infringement

Realtek challenges the ALJ’s determination that an “input video signal” is not required to find infringement of claim 4 of the ’941 patent. Realtek Pet. at 4.

⁸ [

] Realtek Resp. Thom. Pet. at 4 n.1.

Thomson asserts that claim 4 requires an apparatus “capable of processing an input signal,” and does not require that the apparatus include the input video signal. Thom. Resp. at 47. According to Thomson, “Realtek’s argument is contrary to Federal Circuit law that courts must take care to avoid reading process limitations into an apparatus claim.” ID at 399. Realtek argues that the Realtek-Based products generate an input video signal. Realtek Resp. at 15. Realtek contended that the input video signal “can only be supplied by an end user when the Realtek-based Products are connected to a third-party video source.” *Id.*

The ALJ adopted Thomson’s position with respect to claim 4 of the ’941 patent. ID at 402. The ALJ concluded that claim 4 is “directed to ‘an apparatus for controlling a matrix display.’” *Id.* The apparatus “does not include an input video signal, but performs certain actions when presented with an input video signal.” *Id.* Therefore, the ALJ held that “the inquiry is whether or not the accused Realtek products include all of the elements of the claimed apparatus; the fact that Realtek does not provide the input video signal is not relevant.” *Id.* The ALJ cites *Hewlett-Packard Co. v. Bausch & Lomb Inc.*, 909 F.2d 1464, 1468 (Fed. Cir. 1990), for support for the proposition, “[a]pparatus claims cover what a device is, not what a device does.” *Id.*

The Commission affirms the ALJ’s finding that claim 4 does not require an input signal. Infringement of an apparatus claim is based on whether the alleged infringing device is the same device as disclosed by the claims. In this instance, claim 4 requires that the memory be controlled by a first control signal so that active portions of an input video signal are stored at a first rate. The parties dispute whether or not this is a functional or structural limitation. Realtek argues that the claim does not require that the memory be “*capable* of being controlled by a first control signal,” but instead requires that it be “controlled by” the input video signal, and

therefore, this limitation of the apparatus claim must be met in order to find infringement.

Realtek Resp. at 16. In contrast, Thomson argues that the preamble provides an apparatus “for controlling a matrix display” and that this language indicates that the claim is directed to an apparatus that can be used to control a matrix display. Thom. Rep. Realtek Pet. at 3-4. Thomson also argues that the claim language does not require the presence of a video signal, but requires the capability of processing the input signal. *Id.* at 4; Thom. Resp. at 47.

The Federal Circuit has stated that method claims and apparatus claims are different, and method claims require performance of steps for infringement. *See generally Hewlett-Packard Co. v. Bausch & Lomb Inc.*, 909 F.2d 1464, 1468 (Fed. Cir. 1990); *Baldwin Graphic Sys., Inc. v. Siebert, Inc.*, 512 F.3d 1338, 1344-45 (Fed. Cir. 2008). “[A]pparatus claims cover what a device is, not what a device does.” *Hewlett-Packard Co.*, 909 F.2d at 1468.

We first look to the language of the claim which requires “a memory controlled by a first control signal so that active portions of an input video signal having active and inactive portions ... are stored.” ’941 patent at 8:7-10. We find that the language is functional language and that the input signal is not a limitation of the claims but describes what the device must be capable of doing when presented with an input video signal. Moreover, the specification teaches that: “The invention in question concerns a method for controlling a matrix display according to the preamble of the main claim as well as a device, suitable for executing the method according to the invention, according to the preamble of the first device claim.” ’941 patent at 1:7-11. While the claim language itself is the primary focus, this paragraph illustrates that the inventors intended the invention be a device that is “suitable for” (*i.e.*, capable of) executing the method.

Realtek argues that claim 5 bolsters its position that the input signal must be met because claim 5 refers to “transmitted video signal” which therefore, requires that claim 4 have an input

video signal. Realtek Rep. Resp. at 7. However, Thomson argues that this refers to a video signal, which must be transmitted by its nature. We agree with Thomson and believe that this language is functional. Accordingly, the Commission affirms the ALJ's finding that claim 4 does not require an input video signal to be infringed.

4. Anticipation of the Asserted Claims of the '941 Patent by Baba

Respondents challenge the ALJ's finding that the asserted claims of the '941 patent are not anticipated by Japanese Patent Publication No. H2-70186 ("Baba").

The Baba reference discloses a method and apparatus for converting an interlaced signal for a CRT display into a progressive signal for an LCD display. RX-168. Interlacing is a technique for transmitting video signals that is still widely used with many CRT displays. CX-4308C at Q.24; RX-160 at Q.42-44. Interlacing reduces the data transfer rate of a video signal in half by splitting each image or "frame" to be displayed into two fields, each field containing half the lines of the full frame: one field containing the odd lines and the other field containing the even lines. *Id.* The cathode ray gun of a CRT display first paints the odd field on the phosphor on the inside of the screen, and then – before the image can fade – paints the even field. CX-4308C at Q.24. Thus, for interlaced signals, only half of the lines of a frame are updated during a single "field period," generally 1/60 second. Tr. at 1758:4-1759:18; 1760:3-25. A non-interlaced video signal is known as a progressive video signal. In a progressive video signal, all the lines of an input frame are updated at the same time. *Id.* at 1753:20-1754:1.

The ALJ found that Baba does not anticipate the asserted claims of the '941 patent because it does not disclose the limitation "a ratio f_t/z_a is reduced from the ratio required for a cathode ray tube, where f_t is a clock frequency for signal processing and for controlling the display, and z_a represents the number of lines to be displayed." ID at 283. Before the ALJ, the

parties agreed that in Baba the input clock frequency (*i.e.*, the frequency associated with the CRT) is 6.72 MHz and the output clock frequency (*i.e.*, the frequency associated with the LCD) is 10.06 MHz. RX-160 at Q.190; CX-4308C at Q.197. The parties also agreed that in Baba, the value z_a associated with the LCD is 200. The parties disagreed, however, over the value of z_a associated with the CRT. The value of z_a , *i.e.*, “number of lines to be displayed,” associated with the CRT affects whether the ratio f_t/z_a for the LCD display in Baba is “reduced from the ratio f_t/z_a required for a cathode ray tube,” as required by the claims. According to MStar, a standard 200-line CRT display receiving an interlaced video signal, only 100 lines are updated on the CRT display in a given “field period” and thus the value of z_a for a CRT display is 100. RX-160 at Q.190. According to Thomson, the value of z_a in a standard 200-line CRT display receiving an “interlaced” video signal, the value of z_a is still 200 because all 200 lines are visible to the user at any given time. CX-4308C at Q.197.

The ALJ found Thomson’s argument to be more persuasive. ID at 284. The ALJ found that the parties both relied on a textbook authored by Thomson’s expert Dr. Ferraro for an understanding of “interlaced” signals in a CRT. *Id.* According to the ALJ, the relevant passage in the textbook indicates that even though only half of the lines in an interlaced display are updated during a specific time period, the number of lines to be displayed includes both the odd and even lines because the set of lines that is not updated during the specific period is still shown to the user. *Id.* at 285. Accordingly, the ALJ found that the proper z_a value for the ratio required for a CRT in the Baba reference is 200, and that using this z_a value, Baba fails to disclose the reduced f_t/z_a limitation of claims 1 and 4. *Id.*

We find that the parties’ arguments can be boiled down to this: whether the limitation “the number of lines to be displayed” means the number of lines updated in a given field period,

as alleged by MStar, or whether the limitation means the number of lines actually displayed at any given time, as urged by Thomson and adopted by the ALJ. For interlaced signals, a field period refers to the time in which half of the lines (even or odd) are updated in a CRT, which usually has a duration of 1/60 second. We find that neither the intrinsic evidence nor extrinsic evidence supports MStar's construction that the value of z_a associated with a CRT display is the number of lines that are *updated* for a given field period.

First, we disagree with MStar that by reading the claim language "the number of lines to be displayed" as meaning the number of visible lines at any given time, the ALJ read out the words "to be." In our view, the words "to be" do not provide a temporal limitation. Rather, the key word here is "displayed," which does not mean "updated," as urged by MStar.

The words "to be displayed" are used in the specification to refer to all visible lines at any given time. The specification discusses upscaling all 482 interlaced lines (including odd and even lines) of a CRT display to 560 lines on a matrix display. '941 patent at 6:40-43. Specifically, the specification states: "[w]hen using a matrix display, with, for example, 560 lines, and when processing a video image according to the M standard (US standard) using approx. 482 active lines, it is possible to expand the picture to be displayed to 560 lines." The M standard, as MStar's expert Dr. Drabik admitted under cross-examination, refers to the U.S. standard which uses an interlaced video signal displayed on a CRT with 482 active lines in each frame (and therefore, 482 lines that are visible to the viewer at any point in time). CX-4308C at Q.91-92; Tr. at 1232:18-1233:7-15. If the number of lines "to be displayed" did not mean the number of lines that are visible to the user at a given point for both the CRT display and the matrix display, the specification would have referred to upscaling from 291 lines, rather than 482 lines, to 560 lines. *Id.*

We disagree with MStar's interpretation of the written description accompanying Figure 2 of the '941 patent, which states that in interlaced signals, z_a represents the number of lines that are active, "*i.e.*, contain picture information." '941 patent at 3:4-7. We find that this statement merely defines z_a as the number of lines in the image with respect to the number of total lines available on the physical display, as opposed to the number of odd or even lines in the image that are updated in a given field period. As shown by Figure 2, z_a denotes the entire height of an image of an interlaced signal in comparison to the total available lines z of the display:

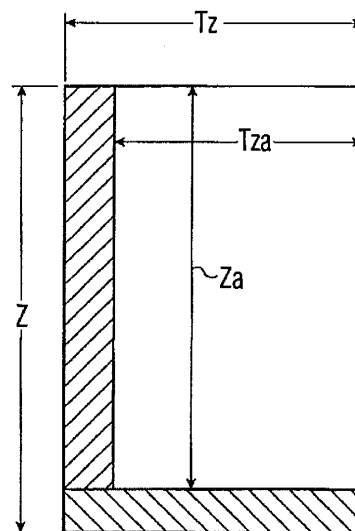


FIG. 2

Id. at Fig. 2. The description does not suggest that z_a refers to half of the lines of the image updated for a given field.

We turn to MStar's argument that because Dr. Ferraro's textbook states that for an interlaced signal, only half of the lines are "displayed" at any give time while the other half of the lines are not "displayed," the limitation "the number of lines to be *displayed*" for an interlaced signal in a 200-line CRT display is only 100. MStar Resp. at 19. We believe that MStar mischaracterizes the evidence. Dr. Ferraro explained at the hearing that the term "displayed" stated in the textbook does not mean actual image displayed, but rather the number

of even or odd lines that are updated in a given field period. Tr. at 1771:18-23. Dr. Ferraro's testimony shows that at any given time, all lines of an interlaced signal are displayed on a CRT display, not just half. *Id.* at 1769-1771.

Nothing in the intrinsic or extrinsic evidence indicates that *za* should be interpreted as anything but the total number of lines of an image, regardless of whether the image is interlaced or progressive. Applying this construction of *za*, Baba would not disclose a reduced *ft/za* ratio as required by the claims. In Baba, the input ratio required for a 200-line CRT would be 6.72MHz/200 and the output ratio for the LCD of Baba would be 10.06 MHz/200. Thus, we agree with the ALJ that Baba does not disclose the limitation requiring a reduction of the ratio *ft/za*. Accordingly, the Commission affirms the ALJ's finding that Baba does not anticipate the asserted claims of the '941 patent.

5. Exclusion of the ViewFrame II+2 LCD Panel

At the hearing, the ALJ granted Thomson's motion *in limine* to exclude physical exhibit RPX-12, also known as ViewFrame II+2, which is allegedly a prior art device to the '941 patent. He excluded it on the ground that the date of the device was insufficiently corroborated:

Motion in limine number 7, regarding the nView ViewFrame II + 2, this motion is granted. What this essentially amounts to is inventor's testimony regarding an alleged prior art invention and the date of it. The ViewFrame II + 2 has not been shown to be – contain any indication of when it was made and the testimony of the inventor that he thinks it is the same product or he is certain it is the same product is still not going to get past the date issue. That must be corroborated independently by some sort of documentary evidence. This motion is granted.

Tr. at 31:23-32:11. In response to the ALJ's ruling, MStar's counsel pointed out during the hearing that the physical device itself (RPX-12) contains a serial number, indicating when the device was sold. Tr. at 33:14-20. MStar's counsel also pointed out to the ALJ that MStar's

response to Thomson's motion *in limine* cites a list of documents showing that the ViewFrame II+2 was created and sold before the priority date of the '941 patent. *Id.* The ALJ did not consider MStar's arguments, stating that "I am going to let you just go ahead and appeal this." *Id.* at 33:24-25. The ALJ affirmed his evidentiary ruling in the final ID. ID at 281 n.32.

We disagree with the ALJ's ruling and find that MStar has presented sufficient evidence showing that the ViewFrame II+2 was sold before the priority date and that the ALJ should have allowed the evidence in to consider whether the ViewFrame II+2 anticipates the asserted claims of the '941 patent.

The evidence shows that counsel for MStar purchased the device designated RPX-12 through eBay. MStar Resp. at 22. Mr. Vogeley, the inventor of ViewFrame II+2, testified in a deposition that he inspected the exhibit and testified that RPX-12 is a true, accurate, and unmodified version of the ViewFrame II+2 device that he designed, and that his company manufactured and sold before the priority date. RX-161 at Q.260-63. Mr. Vogeley testified that the particular exhibit of RPX-12 was sold roughly in the late 1980's, as indicated by the date on the circuit board. *Id.* at Q.258-60. The circuit board of ViewFrame II+2 (RPX-12) is stamped with "COPYRIGHT N VIEW CORPORATION 1989," and the microcontroller is also stamped with a 1989 copyright. In addition, several articles and documents dated before the priority date mention sales of the ViewFrame II+2. RX-217 at 18-19, 24; RX-219C at MS0208735, MS0208740; RX-220; JX-41; RX-211; RX-218C at MS0208360-63; MStar Resp. at 23-24. Accordingly, Mr. Vogeley's testimony that the ViewFrame II+2 was sold prior to the priority date is sufficiently corroborated by other evidence.

Thomson argues that MStar cannot provide any chain of title evidence to describe where the device of RPX-12 came from and how it has been handled before it came into MStar's

possession. Thom. Resp. at 50; Thom. Rep.; MStar Pet. at 12. We disagree with Thomson's argument that the proponent of a physical exhibit must establish such a strict chain-of-custody to justify admission of the exhibit into evidence. *See Certain Ceramic Capacitors and Products Containing Same*, Inv. No. 337-TA-692, Order No. 46 (July 19, 2010) (denying motion *in limine* and rejecting a strict chain-of-custody requirement for the admissibility of evidence).

Accordingly, we find that RPX-12 and corresponding evidence was improperly excluded.

We also find that there is clear and convincing evidence that ViewFrame II+2 is prior art under 35 U.S.C. § 102(b) because the evidence supports the conclusion that it was sold more than one year prior to the priority date. RX-161 at Q.259; RX-217 at 18, 19, 24; RX-219C at MS0208735, MS0208740; RX-220; JX-41; RX-211; RX-218C at MS0208360-63. Although the Commission is cognizant of the fact that the '941 patent will expire on August 26, 2012, in view of the reversal of this evidentiary ruling, a remand is required to address the issue of invalidity in light of ViewFrame II+2.⁹ The ALJ should determine whether the '941 patent is invalid in light of RPX-12 and the related evidence.

F. Economic Prong of the Domestic Industry Requirement

In a patent-based action, a complainant must demonstrate that a domestic industry either exists in the United States or is in the process of being established to prove a violation of section 337. *See* 19 U.S.C. § 1337(a)(2). Sections 337(a)(2) and (3) set forth the domestic industry requirement:

(2) Subparagraphs (B), (C), (D), and (E) of paragraph (1) apply only if an industry in the United States, relating to the articles protected by the patent, copyright, trademark, mask work, or design concerned, exists or is in the process of being established.

⁹ Commissioner Dean Pinkert takes no position on whether RPX-12 anticipates the asserted claims of the '941 patent.

(3) For purposes of paragraph (2), an industry in the United States shall be considered to exist if there is in the United States, with respect to the articles protected by the patent, copyright, trademark, mask work, or design concerned—

- (A) significant investment in plant and equipment;
- (B) significant employment of labor or capital; or
- (C) substantial investment in its exploitation, including engineering, research and development, or licensing.

19 U.S.C. §§ 1337(a)(2) and (3).

When a complainant seeks to satisfy the economic prong of the domestic industry requirement specifically through its investments in licensing under section 337(a)(3)(C), the complainant must show that it has made a substantial investment in the exploitation of the asserted patent through licensing. 19 U.S.C. § 1337(a)(3)(C); *Multimedia Display and Navigation Devices and Sys., Components Thereof, and Prods. Containing Same*, Inv. No. 337-TA-694, Comm’n Op. at 7-8 (Aug. 8, 2011) (“*Navigation Devices*”). To be considered “exploitation” though licensing within the meaning of the statute, the complainant must demonstrate that a particular activity: (1) relates to the asserted patent; (2) relates to licensing; and (3) occurred in the United States.¹⁰ *Id.* Activities that meet these three requirements can be considered in our evaluation of whether a complainant has satisfied the domestic industry requirement, but our inquiry does not end there. *Id.* A complainant must also demonstrate that the amount of its investment in these activities is substantial. In the portfolio licensing context, the Commission has indicated that it considers the relative importance of the asserted patent to the licensing investment to determine to what extent the claimed licensing investments can be attributed to the asserted patent. *Navigation Devices*, Comm’n Op. at 8 (“Because Pioneer’s activities are associated both with the asserted patents and unasserted patents, a key issue

¹⁰ Because the statute requires that investment activities satisfy all three of these requirements, the absence of a connection to any one of them will defeat complainant’s attempt to rely on those activities to satisfy the domestic industry requirement. *Navigation Devices*, Comm’n Op. at 15 n.12.

presented is the strength of the nexus between the activities and the asserted patents.”). Finally, a complainant must establish that its investment in licensing the asserted patent is substantial. *Id.*

1. Which of Thomson’s Activities Constitute Exploitation Through Licensing

As an initial matter, the Commission considers which of Thomson’s activities constitute exploitation through licensing for purposes of section 337(a)(3)(C).¹¹ *Id.* Thomson identifies four categories of activities that it alleges are licensing: (1) activities relating to Thomson’s LCD licensing program, including employee time, facility use, travel, and product acquisition; (2) the purchase of the Xerox/PARC patent portfolio; (3) litigation of the underlying section 337 investigations and the parallel, stayed district court action; and (4) the reexamination of the ’006, ’674, and ’556 patents.

Respondents do not dispute that the first category of activities relates to licensing. *See* CMI Pet. at 74-80. As discussed below, the Commission holds based on the evidence in this investigation, that the latter three categories of activities do not constitute exploitation through licensing under section 337(a)(3)(C).

With respect to the second category, Thomson’s acquisition of the Xerox/PARC patent portfolio, Thomson argues that it paid [] to acquire the portfolio of patents, including the ’063, ’674, and ’556 patents, as well as Xerox’s licensing work product. Thomson asserts that this investment should be considered part of its own licensing program because [

] and because Xerox’s

licensing work product could be used to incorporate Xerox’s licensing program into Thomson’s

¹¹ Respondents do not challenge the ALJ’s conclusion that Thomson’s activities occurred in the United States, nor do they challenge that these activities sufficiently relate to the asserted patents to be considered in our domestic industry analysis. Thus, we begin by determining which of complainants’ activities relate to licensing and then turn to the strength of the relationship to the asserted patents.

future LCD Licensing Programs. Thom. Resp. at 65-66. Thomson proposes that, as a general matter, investments by a prior patent owner should be attributed to a complainant when the “Complainant acquired an existing licensing program with the intent of continuing that program and did in fact continue that program.” Thom. Resp. at 67. Respondents counter that Thomson had no involvement with any licensing activities associated with these patents before buying them from Xerox, so Thomson’s purchase of the portfolio is solely related to ownership, not licensing. CMI Resp. at 71.

The ALJ found that Thomson did not show that its expenditures for acquiring the Xerox/PARC portfolio actually relate to licensing. ID at 424. We agree. Congress clearly stated that it did not intend mere patent ownership to constitute a domestic industry and as such cannot constitute exploitation through licensing:

The mere ownership of a patent or other form of intellectual property rights would not be sufficient to satisfy this test. The owner of the property right must be actively engaged in steps leading to the exploitation of the intellectual property, including application engineering, design work, or other such activities.

S. Rep. No. 100-71, at 130 (1987). Thomson’s investment in Xerox/PARC’s patent portfolio simply reflects a transfer of legal title of the patents in return for monetary consideration. In other words, Thomson merely assumed the status of a patent owner through its investment in Xerox/PARC’s patents. And while Thomson argues that it purchased Xerox/PARC’s portfolio because of its licensing prospects, we agree with the ALJ that “Thomson’s motivation is similar to most patent owners, who acquire patents, either through prosecution or purchase, for the purpose of exploiting them for financial gain.” ID at 424. Therefore, we find that Thomson has not presented evidence sufficient to establish that the acquisition of the Xerox/PARC portfolio constitutes an investment in the exploitation of the patents through licensing. Nor do we believe that Xerox’s work product should be attributed to Thomson. There is no evidence that Thomson

had any involvement with Xerox/PARC's licensing activities before purchasing the patent portfolio, and Thomson has failed to present sufficient evidence as to why these expenses should be attributed to Thomson. Nor has Thomson shown that its purchase of Xerox's patent portfolio and work product constitutes the type of active engagement envisioned by Congress to be considered exploitation of the patent through licensing. Thus, we hold that under the facts presented here, Thomson's acquisition costs are patent ownership costs that do not warrant consideration in our evaluation of whether Thomson satisfies the domestic industry requirement.

Turning to the third category, Thomson's litigation of the underlying section 337 investigations (337-TA-741 and 337-TA-749) and the parallel, stayed Delaware district court action, Thomson argues that these actions relate to licensing because Thomson attempted to license the patents prior to filing the actions and because litigation is part of its overall licensing strategy. Thomson Pet. at 95-96; Thom. Resp. at 70. Thomson also argues that the Commission should consider the litigation expenses for its parallel district court action, regardless of whether or not it is stayed, because "there is a nexus between those expenses and licensing the asserted patents." *Id.* at 70-71. Respondents respond that allowing legal expenses for the instant investigations and corresponding district court action would nullify the nexus to licensing requirement. CMI Resp. at 72 n.23. Respondents further argue that the Commission has held that only activities that occurred before the filing of a complaint are relevant to whether or not a domestic industry exists and that Thomson incurred these expenses after the complaint was filed. *See id.* at 73.

The ALJ, relying on Commission precedent holding that only costs incurred before the filing of the complaint generally can be considered in assessing a complainant's domestic industry, held that Thomson did not provide evidence about which litigation costs were incurred

before the filing of the complaint. ID at 425. The Commission agrees with the ALJ that, as a general matter, “only activities that occurred before the filing of a complaint with the Commission are relevant to whether a domestic industry exists or is in the processing of being established,” and that Thomson failed to distinguish between costs incurred prior to and after filing its complaint. *See Video Game Sys. and Controllers*, Inv. No. 337-TA-743, Comm’n Op. at 5 (Jan. 20, 2012) (“*Video Game Systems II*”) (quoting *Coaxial Cable Connectors Components Thereof and Prods. Containing Same*, Inv. No. 337-TA-650, Comm’n Op. at 51 n.17 (Apr. 14, 2010) (“*Coaxial Cables*”)); ID at 425.

Moreover, the Commission holds that, regardless, litigation expenses for an underlying section 337 investigation may not establish a domestic industry under section 337(a)(3)(C). In *John Mezzalingua Associates, Inc. v. International Trade Commission*, the Federal Circuit held that litigation expenses do not automatically constitute evidence of the existence of a domestic industry. 660 F.3d 1322, 1328 (Fed. Cir. 2011) (“We agree with the Commission that expenditures on patent litigation do not automatically constitute evidence of the existence of an industry in the United States established by substantial investment in the exploitation of a patent.”). In *Coaxial Cable Connectors*, Inv. No. 337-TA-650, the Commission further held that litigation costs, if directly related to licensing, may be considered in determining whether a domestic industry exists. *Coaxial Cables*, Comm’n Op. at 50-51. The Commission did not consider, however, the propriety of allowing its own investigations to support the domestic industry requirement. We hold that underlying section 337 investigations and parallel, stayed district court actions should not be considered in our domestic industry analysis. Holding to the contrary would essentially eliminate the domestic industry requirement. Every complainant that comes before the Commission invests resources when bringing a section 337 complaint and

supporting its allegations during the investigation. Permitting complainants to rely on these activities and investments to establish a domestic industry, would be inconsistent with the statute and legislative history which imposes an affirmative requirement of demonstrating the domestic industry, one which cannot be automatically fulfilled by the filing of a Section 337 complaint. Thomson has not shown that Congress intended the filing of a section 337 complaint to support the very allegations a complainant must prove, even when the investment is intended to further a licensing agenda. In addition, Thomson failed to show that the district court action that was filed to parallel the section 337 investigation and was stayed pursuant to 28 U.S.C. § 1659 had a clear relationship to licensing. Therefore, Thomson's litigation of the underlying section 337 investigations (337-TA-741 and 337-TA-749) and the parallel, stayed Delaware district court action do not warrant consideration in our evaluation of whether Thomson satisfies the domestic industry requirement.

Finally, with respect to the fourth category, Thomson's reexamination of the '006, '674, and '556 patents, Thomson argues that its ongoing reexamination expenses have a strong nexus to licensing because the proceedings were initiated by respondent Qisda Corp. during the course of licensing negotiations and Thomson incurred these expenses as part of its continuing efforts to maintain and license the asserted patents to respondents and other potential licensees. Thom. Resp. at 71. Respondents argue that the reexamination expenses were incurred in connection with the instant investigation and corresponding district court action and that there is no nexus between the reexamination of three of the asserted patents and licensing of those patents to warrant consideration under section 337(a)(3)(C). CMI Resp. at 74-75.

The ALJ again relied on the fact that Thomson did not provide evidence about which costs were incurred before the filing of the complaint to conclude that the reexamination costs

should not be considered. ID at 425. We agree with the ALJ, and further note that, in general, reexaminations are simply a continuation of prosecution that reaffirm or modify the boundaries of the patentee's ownership. Thomson has not established evidence to show that these activities and investments are encompassed within the statute. Nor has Thomson shown that these activities and expenditures constitute more than mere patent ownership. S. Rep. No. 100-71 at 130; *Video Game Sys. and Controllers*, Inv. No. 337-TA-743, Comm'n Op. at 8 (Apr. 14, 2011) ("*Video Game Systems I*") (holding that "patent prosecution activities alone would be insufficient to establish the domestic industry requirement under section 337(a)(3)(C)"). Accordingly, on the facts presented here, the Commission finds that these activities and investments do not fall within section 337(a)(3)(C).

Thus, only Thomson's activities relating to its LCD licensing program, including employee time, facility use, travel, and product acquisition, constitute exploitation through licensing and may be considered to determine the level of investment Thomson made in the asserted patents.

2. What Did Thomson Invest in Licensing the Asserted Patents

Having identified the activities that constitute exploitation through licensing, Thomson must establish what amount it invested in those activities and ultimately what portion can be attributed to the patents at issue. *See Navigation Devices*, Comm'n Op. at 12-13. Oftentimes, a complainant can only provide the Commission with the total amount it invested in licensing the entire patent portfolio. It may be challenging, if not impossible, to allocate a particular dollar amount to each asserted patent. This does not preclude a complainant from establishing a domestic industry. *See Navigation Devices*, Comm'n Op. at 11-12 n.8. Indeed, Congress, by using the word "substantial," indicated that no mathematical precision is required when assessing the amount a complainant has invested in each patent. *See* 19 U.S.C. § 1337(a)(3)(C); *Certain*

Stringed Musical Instruments and Components Thereof, Inv. No. 337-TA-586, Comm’n Op. at 25-26 (May 16, 2008) (“[T]here is no need to define or quantify the industry itself in absolute mathematical terms.”). In addition, we see no reason to believe that Congress intended the domestic industry to be established only on the basis of licenses covering individual patents. However, a complainant must provide additional evidence to establish the extent that it was investing in licensing the asserted patent. *Navigation Devices*, Comm’n Op. at 9 (“Where the complainant’s licensing activities and investments involve a group of patents or a patent portfolio, the complainant must present evidence that demonstrates the extent of the nexus between the asserted patent and the complainant’s licensing activities and investments.”). In this regard, the Commission will consider all evidence demonstrating the importance of the asserted patent to the licensing expenses incurred. *Id.* at 9-11. In *Navigation Devices*, the Commission indicated it will consider, among other things: evidence showing that complainants’ licensing activities are particularly focused on the asserted patent among the group of patents in the portfolio, including evidence that the patent was discussed during the licensing negotiation process, it has been successfully litigated before by the complainant, it relates to a technology industry standard, it is a base patent or a pioneering patent, it is infringed or practiced in the United States, or the market recognizes its value in some other way; whether the licensee’s efforts relate to “an article protected by” the asserted patent; the number of patents in the portfolio; the relative value contributed by the asserted patent to the portfolio; the prominence of the asserted patent in licensing discussions, negotiations and any resulting license agreement; and the scope of technology covered by the portfolio compared to the scope of the asserted patent. *Id.*

Thomson argues that it invested a total of [] in its LCD licensing program, including [] in employee costs, [] in facility costs, [] in travel expenses, and [] in product acquisition for analyzing potentially infringing products. Thom. Resp. at 53. While the ALJ attributed this entire amount to each of the asserted patents, ID at 420, we decline to do so. Thomson's investment relates to the entire LCD patent portfolio, not just the asserted patents. In *Navigation Devices*, the Commission declined to adopt a policy whereby every investment in a patent portfolio is automatically allocated in its entirety to each individual patent in the portfolio. *Navigation Devices*, Comm'n Op. at 13. Instead, we adopted a fact-focused and case-specific inquiry to determine whether a substantial investment in the exploitation of the asserted patent has been made by the complainant. *Id.* Thus, of the total amount invested in licensing the LCD patent portfolio, Thomson must establish that a substantial investment can be attributed to the patents at issue. *See id.*

The Commission finds that Thomson has shown that a substantial portion of the [] invested in licensing the LCD patent portfolio can be attributed to the asserted patents because Thomson's licensing activities were heavily focused on the asserted patents among the group of patents in the portfolio. Thomson Licensing LLC is a U.S. company with offices in Princeton, New Jersey, whose primary business is licensing patents []. In this regard, Thomson has multiple ongoing licensing programs, including one directed to its LCD patent portfolio. The ALJ found that as of December 31, 2009, Thomson owned [] patents and patent applications worldwide, but that only a very small subset [] is part of its LCD licensing program. *Id.* []

[

] The ALJ also found that “[w]ith respect to the LCD licensing program, Thomson has executed [] licenses [] that cover the ’006 and ’941 patents” and “[] of those licenses [] cover the ’063, ’556, and ’674 patents.” ID at 420. “Thomson has also executed [] release agreements with fields of use that cover the ’006 and ’941 patents.” *Id.* This evidence shows that Thomson has negotiated a number of licenses that cover the asserted patents.

In addition, as the ALJ found, Thomson prominently asserted the five patents in many of its licensing negotiations. [

] During licensing discussions, Thomson actively relied on the five asserted patents [

]

[

] The ALJ also found that “[o]ne or more of the patents-in-suit were asserted in each of the Thomson patent books identified.” *Id.* He also relied on five exemplary patent books which included claim chart proofs [] For example, the patent book for [] provides claim charts []. In the patent book for [] claim charts are for asserted patents; in the patent book for [] claim charts are for asserted patents; in the patent book for [] claim charts are for asserted patents; and finally, a supplemental patent book was provided to [] including claim charts for two of the asserted patents. *ID* at 419-20. The ALJ concluded, and we agree, that “while Thomson’s patent books may mention more than [] patents . . . and Thomson’s LCD licensing program includes approximately [] patents . . . , Thomson typically provides claim charts for a small subset of patents, often including the asserted patents, from its LCD licensing program.” *Id.* at 420. The parties did not challenge these findings by the ALJ in their petitions for review.

The claim charts signify that particular patents were important to Thomson for purposes of licensing negotiations. They also establish that Thomson believed the claims of those patents actually cover commercially-available products of the targeted company. We conclude that Thomson was specifically investing in these particular patents through licensing. Also, the evidence indicates that Thomson has executed licenses with or is in active negotiations with much of the industry [

] and suggests that the industry agreed that the risk of infringing these patents was

sufficiently high to warrant taking a license and that the patents are important to the industry. *See* ID at 420 (citing CX-4245C at Q.203).

Thus, we agree with the ALJ that the evidence shows that Thomson's licensing activities for licensing the LCD patent portfolio are particularly focused on the asserted patents and conclude that a substantial portion of the [] invested in procuring the LCD portfolio licenses can be attributed to the asserted patents. We find further support for this conclusion in Thomson's estimate that [] relates directly to the asserted patents. To reach this estimate, Thomson's witness Ms. Coto determined what percentage of time each employee spent on the asserted patents and based on these percentages, she calculated the costs associated with those employees. She also reviewed the travel expense reports and verified with the relevant employees whether or not the expenditures related to licensing the asserted patents, and finally, she attributed the percentage of the employee time to the square footage of office space, insurance, and overhead to determine the total facility costs attributable to the asserted patents. CX-4247C at Q.69-98; *see generally* ID at 427-28.

Respondents argue, however, that the [] is itself inflated because it includes costs incurred prior to the acquisition of the Xerox/PARC patent portfolio.¹² *See* CMI Resp. at 63. Thomson acquired the Xerox/PARC patent portfolio, which included three of the five

asserted patents, in [] Respondents argue that none of the costs

¹² Respondents also argue that the [] investment Thomson identified as specifically related to the asserted patents "remains overstated given that this amount still encompasses expenses related to a host of other patents, including Thomson's numerous other licensing programs, in addition to the five asserted patents that are among the patents involved in the LCD licensing program." CMI Resp. at 63; *see also* CMI Resp. at 61 nn.13-14. However, Respondents did not make this argument in their Petitions for Review. Respondents only asserted that Thomson's [] investment is overstated because of the timing issue addressed above. Respondents also did not assert that the [] investment they identify was overstated in their Petitions for Review nor did they provide a lower number for the Commission to consider. Therefore, we believe that Respondents' arguments have been waived before the Commission.

incurred between [] can be attributed to the asserted patents and that without those employee and facility costs, Thomson's total investment is only [

]. We disagree that the investment made between [] cannot be attributed to the two patents owned by Thomson during that time period. [

] Thus, in our view, a portion of the amount invested from [] can be attributed to the after-acquired patents as well.

3. Whether Thomson's Investments in Licensing the Asserted Patents Are Substantial

Finally, Thomson must establish that the amount invested in the exploitation of each patent through licensing is substantial. 19 U.S.C. § 1337(a)(3)(C). The Commission has indicated that whether an investment is substantial may depend on the industry and the size of the complainant. *Navigation Devices*, Comm'n Op. at 15. The Commission has also indicated that it will consider, among other things, the existence of other types of "exploitation" of the asserted patent such as research, development, or engineering; the existence of license-related ancillary activities such as ensuring compliance with license agreements and providing training or technical support to its licensees; whether the complainant's licensing activities are continuing; and whether the complainant's licensing activities are those that are referenced favorably in the legislative history of section 337(a)(3)(C). *Id.* at 16. Finally, the Commission has stated that "there is no minimum monetary expenditure that a Complainant must demonstrate to qualify as a domestic industry under the 'substantial investment' requirement of this section." *Stringed Musical Instruments*, Comm'n Op. at 25.

Although we cannot credit Thomson's entire investment in licensing the LCD patent portfolio to the five asserted patents, we conclude that Thomson's investment in licensing the asserted patents is nevertheless substantial. The close relationship between the licensing activities and the asserted patents in combination with the amount invested in those activities leads us to this conclusion. Moreover, because we find that, at a minimum, Thomson has invested [] in licensing the asserted patents and Thomson focused heavily on each of the asserted patents during licensing, we conclude that the amount invested in each patent is substantial.

As in *Navigation Devices*, this is not an instance where the complainant is an individual, a university, or other entity with limited resources. The ALJ found, and we agree, that Thomson Licensing LLC is a U.S. company with offices in Princeton, New Jersey. ID at 418. Thomson's primary business is licensing patents [] Thomson Licensing LLC and Thomson Licensing SAS are both subsidiaries of the French Company Technicolor. *Id.* As of December 31, 2009, Thomson owned [] patents and patent applications worldwide and approximately [] patents are part of its LCD licensing program. *Id.* Moreover, the ALJ noted that "[f]rom January 1, 2008 through June 30, 2010, Thomson's worldwide expenditures on its licensing programs totaled over [] and its United States expenditures on its licensing programs totaled approximately []

[] Thus, Thomson has significant resources and a vast international patent portfolio at its disposal. Nevertheless, Thomson's [] dollar investment in licensing its LCD portfolio represents [] of its total U.S. licensing expenditures and [] of its worldwide licensing expenditures, while the patents included in its LCD licensing program make up only [] of Thomson's approximately [] patent and patent applications. In addition,

Thomson's [] dollar investment in the asserted patents represents [] of its total U.S. licensing expenditures and [] of its worldwide licensing expenditures. This suggests that Thomson was investing more heavily in its LCD patent portfolio and in the five asserted patents than in its other patents.

We also believe that Thomson's investment is substantial in relation to the industry. Unlike in *Navigation Devices*, where Pioneer identified a number of potential licensees but executed a few licenses, one of which was a cross license resulting from litigation, Thomson has successfully licensed its LCD portfolio, including the asserted patents, to a large portion of the industry. *Navigation Devices*, Comm'n Op. at 24-25. Indeed, by late 2010, Thomson had licensed or was in renewal talks with licensees representing [] of the LCD monitor market and Thomson had licensed or was in active negotiations with over [] of the LCD digital television market. CX-4245C at Q.203. Thomson has shown that it was able to license or was in licensing negotiations with a majority of the industry, thereby demonstrating the significance of its investment in licensing the asserted patents in view of the particular industry at issue.

In addition, we agree with the ALJ, for the reasons he indicated, that the substantial nature of Thomson's investment is bolstered by the fact that Thomson invests in license-related ancillary activities and the fact that its licensing activities are ongoing. We also note, as the ALJ did, that Thomson has generated [] in revenue from its licenses covering the patents-in-suit. CX-4247C, Q.100; CX-4246C, Q.128-129. Of the [] in revenue, [] was collected from [] and over [] was collected between []. While the ALJ found that this was strong evidence that Thomson's investment is substantial, we caution that royalties do not constitute the investment itself. *Navigation Devices*, Comm'n Op. at 24. Rather, they are circumstantial

evidence that an investment was made and are consistent with our conclusion that Thomson's investment in the asserted patents was substantial.

Further, we note that the fact that complainant's licensing activities are revenue-driven and target existing production detracts somewhat from the substantiality of Thomson's investment. *See Navigation Devices*, Comm'n Op. at 25. Thomson argues that its activities fall into the category of production-driven licensing because it developed and practiced two of the five patents (the '006 and '941 patents) until it was forced from the market, and it supported PARC's research and development by purchasing Xerox/PARC's patents. Thom. Resp. at 63. We find however that, like complainant in *Navigation Devices*, Thomson's current business is based on a revenue-driven licensing model, so its investments are entitled to less weight.¹³

Finally, Thomson argues that it has invested in other types of "exploitation" of the asserted patents such as research, development, or engineering. Specifically, Thomson argues that it developed the technology covered by two of the five patents and that it has a history of developing, innovating, manufacturing, and licensing. Thom. Resp. at 61-62. In addition, Thomson argues that it manufactured radios, moved into the turntable business, and then into televisions, though Thomson does not allege that it currently manufactures or performs research and development or engineering in the United States. CX-4646C, Q.24. We do not believe that this supports Thomson's assertion that its investment in the exploitation of the asserted patents is substantial. Thomson provides no information about the amount it invested in research, development, or engineering for us to assess whether this contributes to its investment in the asserted patents' exploitation. Rather, we find that, taken as a whole, Thomson's investment in

¹³ In *Coaxial Cables*, the Commission held that section 337(a)(3)(C), by using the broad term "licensing," "does not limit the types of licensing activities that the Commission can consider," including revenue-driven licensing, even when viewed in light of the legislative history drawn to production-based licensing. *Coaxial Cables*, Comm'n Op. at 49-50.

licensing the asserted patents is substantial. We, therefore, affirm the ALJ's conclusion that Thomson has established the economic prong of the domestic industry requirement under section 337(a)(3)(C).

VI. CONCLUSION

For the foregoing reasons, the Commission reverses the ALJ's determination of violation of section 337 with respect to the asserted claims of the '674 patent. The Commission affirms, with modifications, the ALJ's determination of no violation of Section 337 with respect to the asserted claims of the '006, the '063, and the '556 patents, as well as the ALJ's determination that the economic prong of the domestic industry requirement has been met. With respect to the '941 patent, we remand to the ALJ for a determination of whether the ViewFrame II+2 prior art invalidates the '941 patent.

By order of the Commission.



Lisa R. Barton
Acting Secretary to the Commission

Issued; July 6, 2012

**CERTAIN LIQUID CRYSTAL DISPLAY DEVICES,
INCLUDING MONITORS, TELEVISIONS, AND MODULES,
AND COMPONENTS THEREOF**

337-TA-749

CERTIFICATE OF SERVICE

I, Lisa R. Barton, hereby certify that the attached **COMMISSION OPINION** has been served by hand upon the Commission Investigative Attorney, Daniel L. Girdwood, Esq., and the following parties as indicated, on **July 6, 2012**



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